

Quantitative sparse sum-of-squares hierarchies and large private convex recourse

Abstract

Sparse moment-sum-of-squares hierarchies exchange finitely many separator moments between the bags of a junction tree. We quantify how this restricted information affects convergence. For box problems, common polynomial kernels and a density correction turn every feasible moment family into probability laws with exactly consistent separator marginals. For the ordinary sparse quadratic module, the order- r gap is bounded by the sum of the absolute nonconstant local Chebyshev coefficients times $O(\log^3 r/r^2)$, with constants depending on width and local degree and no additional factor in the number of bags. For the full sparse preordering, we give an explicit rounding proof of the inverse-square rate previously stated in lecture slides. A fixed quadratic with two bags has a preordering gap of exact order r^{-2} even with exact local measures, showing a limitation caused solely by finite separator information.

We also study large private continuous vectors entering each bag objective as a convex quadratic. Their degree remains two while the shared degree grows; at fixed shared width, SDP block sizes are polynomial in private dimension and relaxation order. Nonempty fixed private polytopes give $O(r^{-2})$ gaps. Affine fibers with a fixed constraint matrix and complete recourse give the sharp general rate $O(1/r)$. The inverse-square bound returns when projected optimal multipliers have summable weighted Chebyshev coefficients, or each component is a Lipschitz function of its corresponding shared coordinate. Extensions cover polynomial constraints under a global error bound, finite states with common continuous boxes, and exact rational box certificates with positive slack and size polynomial in expanded SDP dimensions and rational input length.

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1 Introduction

Global solvers for nonconvex polynomial and mixed-integer nonlinear problems need lower bounds that are both strong and affordable. Moment-sum-of-squares (SOS) hierarchies [18, 28] give a monotone sequence of semidefinite programming (SDP) lower bounds that converges to the global minimum under standard Archimedean assumptions, which ensure boundedness through the polynomial generators. The size of a dense relaxation nevertheless grows quickly with the number of variables. When the problem has a sparse coupling structure, *correlative sparsity* replaces one dense moment matrix by one local moment matrix per bag of a junction tree, and local moments are required to agree on the variables that adjacent bags share [19, 38]. This keeps the SDP blocks small when the bags are small. Under running intersection and the corresponding local Archimedean assumptions, the sparse hierarchy still converges [10, 19].

How fast it converges is less well understood. For dense problems on a box, a series of works based on polynomial kernels established error $O(r^{-2})$ at relaxation order r for the box preordering (Schmüdgen-type certificates) [22] and, very recently, $O(\log^3 r/r^2)$ for the ordinary box quadratic module (Putinar-type certificates) [9]. For sparse hierarchies the published quantitative theorem gives the slower exponent $2/(w+3)$ for the box preordering, where w is the largest bag size [17]. The inverse-square rate for the sparse preordering was subsequently stated in lecture slides [23, 25]; we discuss this below.

This paper studies the rate of sparse hierarchies, and in particular the role played by the agreement of separator moments. We consider two settings. The first is polynomial optimization on a box with a junction-tree decomposition of the objective. The second, which motivates much of the paper, attaches to each bag a large vector of private continuous variables that enter the objective as a convex quadratic and are constrained by a polytope that may depend affinely on the shared variables.

1.1 Private convex recourse

Many nonconvex models have a small number of coupling decisions and many continuous decisions that are convex once the coupling decisions are fixed. As an illustration, consider a network of regions arranged along a tree. Each region b has a few shared design or price variables x_{B_b} , some of which it shares with neighboring regions, and a dispatch problem in p_b continuous variables y_b

(flows, production levels, storage), with

$$\min_{y_b} c_b(x) + q_b(x)^\top y_b + y_b^\top Q_b(x) y_b \quad \text{subject to} \quad A_b y_b \leq a_b + E_b x_{B_b}, \quad y_b \in [-1, 1]^{p_b}.$$

The dependence on x can be an arbitrary polynomial (for example, prices multiplying quantities, or a nonconvex investment cost c_b), while for each fixed x the dispatch problem is a convex quadratic program. If $E_b \neq 0$, the design changes capacities and other right-hand sides; this is affine recourse.

A standard total-degree SOS hierarchy treats the $v_b + p_b$ variables of a bag alike, so its moment matrices have $\binom{r+v_b+p_b}{r}$ rows; this is prohibitive even for moderate p_b . Convexity suggests a cheaper relaxation: let the degree grow only in the shared variables and keep private moments of degree at most two. The resulting *private-degree-two hierarchy* (Definition 2.14) has local blocks with $(p_b + 1) \binom{r-|I|+v_b}{v_b}$ rows. For two shared variables per bag, fifty private variables, and $r = 6$, the largest block has 1,428 rows instead of about 4×10^7 . The question is what accuracy such a relaxation guarantees and on what that accuracy depends. Kahl and Henrion use partial lifting of a limited set of nonlinear variables in geometric reconstruction [15]. Guo and Wang keep an SOS-convex decision-variable module at fixed degree inside a robust polynomial matrix inequality hierarchy whose order grows in the uncertainty variables [11]. Here we quantify the accuracy of a sparse private-degree-two hierarchy, with separate rates for fixed domains, affine recourse, and regular projected multipliers.

1.2 How separator information can limit convergence

In a sparse hierarchy, each bag communicates with its neighbors through moments of the shared variables up to degree $2r$. Two actual local probability measures can therefore use different separator distributions while agreeing on all the communicated moments. The sharp examples in Sections 5 and 6 isolate this effect in two bags whose local minima are $-h(x)$ and $h(x)$ on a one-dimensional separator. In these paired models the exact local-measure relaxation has gap $2E_{2r}(h)$, where $E_{2r}(h)$ is the best uniform approximation error by polynomials of degree at most $2r$ [12]. The local measures also give feasible finite-order SDP points, so this identity supplies a lower bound for their SDP gaps. It need not equal the finite SDP gap or describe a general sparse model.

For $h(x) = |x|$ the approximation error has exact order $1/r$. A Lipschitz derivative gives an inverse-square approximation upper bound; the piecewise quadratic in Theorem 5.4 attains that order. A smoother function, or a polynomial, can be approximated faster. These examples distinguish the information lost at a separator from errors in local positivity.

The upper bounds use a common kernel to turn arbitrary feasible truncated functionals into local laws with exactly matching separator marginals. For the preordering the kernel is positive under the local cone and preserves mass. For the ordinary module we first obtain signed exactly normalized densities, then correct them by one common reference density. Junction-tree gluing produces a global box law. With private convex quadratics on fixed polytopes, conditional private means are feasible and preserve the inverse-square upper bound. Affine recourse makes those means feasible at a conditional source point rather than the kernel output; repair at the output costs $O(1/r)$ in general. An active private constraint can create a corner, but need not do so. Regularity of the projected multipliers controls the repair more precisely and restores the faster bounds under the hypotheses of Section 7.

1.3 Contributions

Table 1 summarizes the results. Throughout, r is the relaxation order, w is the bag width, and the finite-order bounds display their dependence on the polynomial data, geometry, and regularity

Model	Hierarchy	Upper bound	Lower bound / remark
Box, sparse objective	preordering	$O(r^{-2})$ (Theorem 3.4)	$\Omega(r^{-2})$ (Theorem 5.4); rate stated earlier [23]
Box, sparse objective	ordinary module	$C_f O(\log^3 r/r^2)$ (Theorem 4.6)	$\Omega(r^{-2})$ (Theorem 5.4)
Fixed private polytope, convex QP	private-degree-two	$O(r^{-2})$ (Theorem 6.10)	needs private quadratic bounds and convexity
Affine complete recourse	private-degree-two	$O(r^{-1})$ (Theorem 6.15)	$\Omega(r^{-1})$, also total-degree (Theorem 6.17)
Affine recourse, weighted Chebyshev or coordinatewise Lipschitz multipliers	private-degree-two	$O(r^{-2})$ (Theorem 7.2; Theorem 7.3 with $\beta = 1$)	$\Omega(r^{-2})$ (Example 7.4)
Polynomial constraints, global Hölder error bound with exponent α	preordering; ordinary module	$O(r^{-\alpha})$; $O((\log r/r)^\alpha)$ (Theorems 8.5 and 8.8)	$\Omega(r^{-1})$ possible for $\alpha = 1$
Finite-state labels with box costs	preordering; ordinary module	as for the box (Theorems 8.12 and 8.13)	labels never rounded

Table 1: Summary of the rate results. Upper bounds hold for every feasible moment family through an explicit rounding construction; lower bounds are for fixed instances and hold for actual local measures.

parameters. The box hierarchies use total degree at most $2r$; the recourse hierarchy uses shared degree at most $2r$ and private degree at most two.

Exactly consistent rounding for the box preordering. Theorem 3.4 shows that every feasible family of the sparse box preordering of order r can be rounded to a probability law on the box whose expected objective exceeds the relaxation objective by at most $3A/(2m^2 + 1)$, where $m = \lfloor r/w \rfloor + 1$ and A is a weighted sum of the local Chebyshev coefficients. Hence the gap is at most $3w^2A/(2r^2)$. We do not claim this rate as new: it was stated for two overlapping bags in lecture slides of Magron [23, 25], with attribution to [17], whose published Theorem 6 has the exponent $2/(w + 3)$; we have not been able to resolve this difference. The squared Fejér (Jackson-type) kernel is classical [7, 17, 22]. Our proof is self-contained and does not pass through an approximation of separator value functions. It gives explicit constants, a rounding to a finite Chebyshev grid (Corollary 3.7), and the template for every later upper bound. A real SOS certificate with the same error follows from SDP duality (Corollary 3.8).

The ordinary sparse quadratic module. The ordinary module uses $v + 1$ PSD blocks for a bag of size v instead of up to 2^v . The normalized interval-positive kernel used for the preordering has a tensor certificate involving products of box generators, which are absent from the ordinary module. Products of globally SOS kernels are available and provide the starting point for the ordinary construction. Theorem 4.6 transfers the recent dense bound of Gribling, de Klerk and Vera [9] to the sparse ordinary module: the gap is at most $C_f O_{w,d}(\log^3 r/r^2)$, where C_f is the sum over bags of the Chebyshev coefficient norms of the nonconstant parts of the local objectives and the

implicit constant depends only on the width w and the local degree d . After this normalization, the bound does not depend on the number of bags or on the shape of the tree; the size of the SDP still grows with the number of bags. The kernel and its SOS normalization are from [9]. A polynomial kernel that is nonnegative on the whole source line for every output and has mass identically one cannot depend on the source variable (Remark 4.3). Thus this source-dependent SOS kernel has a mass defect, and its local densities need not have matching separator marginals. We repair this with an exactly normalized signed kernel, a pseudomoment Cauchy–Schwarz bound on the negative parts of the resulting densities, and one common correction in all bags, which preserves exact consistency. No assumption on an attained polynomial separator dual is needed, in contrast with the conditional transfer principle of Gamertsfelder and Mourrain [8]. The theorem-specific contribution is this exactly consistent finite-order transfer, normalized by C_f and uniform in bag count and tree shape at fixed width and degree. The slides mention a sparse Putinar rate without stating its exponent, so we keep the claim separate from priority for a general sparse rate.

Separator information is the obstruction. Theorem 5.4 gives a fixed quadratic in three variables, with two bags of size two, whose sparse preordering gap is of exact order r^{-2} , while the dense preordering is exact at order two. The lower bound holds even if every local functional is an actual probability measure, so no strengthening of local positivity removes it while separators exchange only moments of degree $2r$. The same witnesses give $\Omega(r^{-2})$ for the ordinary module. Proposition 5.5 computes the first two relaxation values exactly for both cones; at order two the whole gap is due to separator information. Qualitative failure of finite sparse exactness with a nonpolynomial separator is known [27]; the example adds a sharp rate for a quadratic objective.

Large private convex blocks. For a fixed private polytope, Theorem 6.10 proves gap $O(r^{-2})$ for the private-degree-two hierarchy, with constants that depend on the shared width and the coefficients but whose dependence on private dimension enters only through these coefficients. The kernel smooths only shared variables; private decisions are recovered as conditional means, which are feasible because the polytope is fixed, and a matrix Jensen inequality controls the convex cost. Two safeguards are necessary: without the redundant private quadratic bounds the relaxation can be unbounded below at every sufficiently high order (Proposition 6.11), and without private convexity the rounding fails even when the objective depends only on private variables (Proposition 6.12). For this hierarchy the moment value equals the supremum of certificate values, and every level strictly below it is certified (Proposition 6.9); we do not claim attainment at the boundary.

Affine recourse and its sharp rate. With complete recourse and a fixed private constraint matrix, Theorem 6.15 proves gap $O(r^{-1})$: the conditional private mean is guaranteed to be feasible at the conditional *source* mean of the shared variables, but need not be feasible at the output point. A Hoffman error bound [14] repairs it at a cost proportional to the kernel’s transport distance. Theorem 6.17 shows that $1/r$ is the correct order in general: a two-bag example with one shared variable x , private variables y_1 and y_2 , objective $-xy_1 + y_2$, and private constraints $y_2 \geq x$, $y_2 \geq -x$ has gap of exact order $1/r$, for the private-degree-two hierarchy and, separately, for the total-degree sparse preordering. Merging the two bags, or splitting at the active-set change, gives an exact full-preordering certificate at low order.

Regular multipliers restore the inverse-square rate. The sharp affine example has a corner in its separator value function and a discontinuity in its projected multipliers. Theorems 7.2 and 7.3 show that the rate r^{-2} returns, for the same unchanged hierarchy, if the *projected* optimal

multipliers—the part of the dual solution that acts on the shared variables—have absolutely summable weighted tensor Chebyshev coefficients, or each component is a Lipschitz function of its corresponding shared coordinate, the case $\beta = 1$ of the Hölder theorem. Here the i th component has the form $\ell_{b,i}(x_i)$, with $\ell_{b,i}$ univariate. More generally, coordinatewise Hölder dependence of order $0 < \beta \leq 1$ gives $O(r^{-(1+\beta)})$. A key step bounds pseudomoments of Chebyshev polynomials whose total degree exceeds r (Lemma 7.1). Strict convexity of the private problem does not imply the required regularity (Example 7.5); classical parametric quadratic programming gives sufficient conditions [2, 35]. We do not cover arbitrary multivariate Lipschitz projected multipliers.

Extensions. Section 8 adds polynomial constraints under a *global* Hölder error bound with exponent $\alpha \in (0, 1]$. The rounded law has small expected squared constraint violation and is then repaired onto the feasible set, giving $O(r^{-\alpha})$ for box-preordering products times single constraints and $O((\log r/r)^\alpha)$ for the ordinary constrained module (Theorems 8.5 and 8.8). The ordinary-module proof bounds expected squared violation at scale $O(\log^2 r/r^2)$ through a polynomial displacement certificate, then uses global geometric repair. The same rates hold for real certificates with arbitrarily small positive slack, without strict feasibility (Proposition 8.9); boundary attainment is not established. A global error bound can be much worse than separate local bounds. Finite-state variables with bag-local label restrictions are handled without rounding any label, for both cones (Theorems 8.12 and 8.13). Finally, Section 9 turns the box bounds into exact rational certificates with any rational positive slack, with encoding length polynomial in the expanded SDP size and rational input length, including the slack. Positive slack beyond the proved finite-order error supplies the real cone membership needed for this specialization of established rational SOS methods [6, 30].

1.4 Discretization and computational interpretation

The rounding theorems bound the increase in objective when a feasible moment family is converted into a feasible law, and hence bound the error of the specified relaxation. In the private convex setting this comparison is one-sided, because taking conditional means can improve the objective. The accuracy bounds alone do not give numerical running times: SDP size grows with bag count, and geometric and coefficient constants can be large. Discretization also exploits bounded interaction width: Bienstock and Muñoz develop approximate linear programming formulations for polynomial optimization with bounded intersection treewidth [4]. For our box problems, evaluating the objective on a Chebyshev grid and minimizing by dynamic programming over the junction tree gives error $O(r^{-2})$ with $O(r)$ nodes per coordinate (Corollary 3.7). With exact local quadratic programs, the analogous grid bound is $O(r^{-2})$ for fixed private domains and $O(1/r)$ for general affine complete recourse (Remark 6.20). If a proved grid error is ε_r and the grid optimum is G_r , then $G_r - \varepsilon_r \leq f^* \leq G_r$ also provides a lower bound. The hierarchy results instead control the specified sparse SDP, every feasible moment point, and the associated algebraic certificate cone. Their accuracy bounds do not establish a generic runtime or practical speedup.

1.5 Organization

Section 2 fixes notation, the three hierarchies, and standard tools. Section 3 proves the preordering rounding theorem and introduces the construction used throughout. Section 4 treats the ordinary module, and Section 5 gives the matching lower bound and the exact small-order values. Section 6 studies fixed and affine private recourse, and Section 7 the regular-multiplier rates. Section 8 covers polynomial constraints and finite-state variables, and Section 9 rational certificates. Section 10

discusses limitations and open problems. The appendix proves the failure result when private quadratic bounds are omitted.

2 Setting, hierarchies, and standard tools

This section fixes the sparse structure, the Chebyshev notation, the three sparse hierarchies studied in the paper, and the standard facts that later proofs use. We record each fact with the degree bounds used later. The lemmas in [Sections 2.3](#) and [2.4](#) are standard; the private-degree-two hierarchy of [Section 2.5](#) is the specific relaxation analyzed in [Sections 6](#) and [7](#).

2.1 Sparse structure

Definition 2.1 (Junction tree). Let $\mathcal{T}$ be a finite tree with nodes $b = 1, \dots, t$, and let each node carry a *bag* $B_b \subseteq \{1, \dots, n\}$. The bags form a *junction tree* if they cover $\{1, \dots, n\}$ and satisfy the *running-intersection property*: for every coordinate i , the nodes b with $i \in B_b$ induce a connected subtree of $\mathcal{T}$. For an edge $e = bc \in E(\mathcal{T})$, the *separator* is $S_e = B_b \cap B_c$. We write $v_b = |B_b|$ and call $w = \max_b v_b$ the *width*.

In [Sections 2](#) to [7](#) and in the box parts of [Sections 8](#) and [9](#), the width satisfies $w \geq 1$ and the relaxation order satisfies $r \geq 1$. The finite-state models of [Section 8](#) also allow bags with no continuous coordinates, and models with no continuous coordinates at all; the conventions for those cases (empty kernel products, order zero) are stated there and override the standing assumptions $w \geq 1$ and $r \geq 1$.

An example whose data involve only private variables can be placed within the standing positive-width convention by adding one unused shared coordinate in $[-1, 1]$ to its bag. This leaves the original optimum unchanged. A private functional extends to the rectangular hierarchy by evaluating that shared coordinate at zero, as in [Proposition 6.12](#). Individual bags with no shared coordinates are also allowed when the overall width is positive; their shared polynomial space consists of constants and their kernel product is one.

Throughout, $x \in [-1, 1]^n$ and $x_B = (x_i)_{i \in B}$. The basic problem is

$$f^* = \min_{x \in [-1, 1]^n} f(x), \quad f(x) = \sum_{b=1}^t f_b(x_{B_b}), \quad (2.1)$$

where every f_b is a real polynomial in the variables of its bag. The decomposition of f into bag polynomials is part of the data: constants in our bounds depend on it, and different decompositions of the same f can give different constants. Any product of nondegenerate compact intervals can be mapped to $[-1, 1]^n$ by an invertible affine change of each coordinate (fixed coordinates are first eliminated); this preserves bags, total degrees, and agreement of separator moments, so we state all results on $[-1, 1]^n$. When a model is stated on another box we say which affine change is used.

Running intersection has a convenient consequence for polynomial identities.

Lemma 2.2 (Decompositions of zero). *Let $p_b \in \mathbb{R}[x_{B_b}]_{\leq k}$ for $b = 1, \dots, t$ satisfy $\sum_b p_b = 0$ as a polynomial in x . Then there are polynomials $\phi_e \in \mathbb{R}[x_{S_e}]_{\leq k}$, one for each edge, and signs fixed by an orientation of $\mathcal{T}$, such that each p_b is the signed sum of the ϕ_e over the edges incident to b .*

Proof. Induct on t ; the case $t = 1$ is trivial. Let ℓ be a leaf with neighbor c and edge $e = \ell c$. By running intersection, a coordinate of $B_\ell \setminus S_e$ belongs to no other bag. Hence every monomial of p_ℓ that involves such a coordinate appears in no other p_b , so its coefficient in $\sum_b p_b = 0$ is its coefficient

in p_ℓ , which must be zero. Thus $p_\ell =: \phi_e \in \mathbb{R}[x_{S_e}]_{\leq k}$. Replace p_c by $p_c + \phi_e$, delete the leaf, and apply the induction hypothesis to the remaining bags; they still satisfy running intersection, and coordinates that occurred only in the deleted leaf no longer occur and can be dropped. $\square$

2.2 Chebyshev calculus

Let T_k and U_k be the Chebyshev polynomials of the first and second kind, $T_k(\cos \theta) = \cos(k\theta)$ and $U_{k-1}(\cos \theta) = \sin(k\theta)/\sin \theta$. They satisfy $|T_k| \leq 1$ on $[-1, 1]$,

$$T_j T_k = \frac{1}{2}(T_{j+k} + T_{|j-k|}), \quad 1 - T_k(x)^2 = (1 - x^2) U_{k-1}(x)^2 \quad (k \geq 1). \quad (2.2)$$

Let μ be the normalized arcsine probability measure $d\mu(x) = dx/(\pi\sqrt{1-x^2})$ on $[-1, 1]$. Then $\int T_j T_k d\mu = 0$ for $j \neq k$, $\int T_0^2 d\mu = 1$, and $\int T_k^2 d\mu = \frac{1}{2}$ for $k \geq 1$. For a bag B we write μ^B for the product measure and, for a multi-index $\alpha \in \mathbb{N}^B$ (here $\mathbb{N} = \{0, 1, 2, \dots\}$, so zero indices are allowed), $T_\alpha(x) = \prod_{i \in B} T_{\alpha_i}(x_i)$.

Every polynomial p in the variables of B has a unique expansion $p = \sum_\alpha c_\alpha T_\alpha$. We use two coefficient functionals:

$$\|p\|_{\text{ch}} = \sum_\alpha |c_\alpha|, \quad \mathbf{A}(p) = \sum_\alpha |c_\alpha| \sum_{i \in B} \alpha_i^2. \quad (2.3)$$

Since $|T_\alpha| \leq 1$ on the box, $\|p\|_\infty \leq \|p\|_{\text{ch}}$. The weighted budget $\mathbf{A}(p)$ ignores the constant coefficient and is the quantity controlled by a kernel whose multipliers satisfy $1 - \gamma_k = O(k^2/m^2)$. For the decomposed objective (2.1) we write

$$C_b = \|f_b - c_{b,0}\|_{\text{ch}}, \quad C_f = \sum_b C_b, \quad \mathbf{A} = \sum_b \mathbf{A}(f_b), \quad (2.4)$$

where $c_{b,0}$ is the constant Chebyshev coefficient of f_b . Thus C_b is the Chebyshev norm of the nonconstant part. Both C_f and $\mathbf{A}$ vanish exactly when every f_b is constant, and $\mathbf{A} \leq d^2 C_f$ when every f_b has total degree at most d . The oscillation $\text{osc}(f_b) = \max f_b - \min f_b$ over the bag box satisfies $\text{osc}(f_b) \leq 2C_b$.

Lemma 2.3 (Products). *For polynomials p, q in the variables of a bag, $\|pq\|_{\text{ch}} \leq \|p\|_{\text{ch}} \|q\|_{\text{ch}}$ and $\mathbf{A}(pq) \leq \|p\|_{\text{ch}} \mathbf{A}(q) + \mathbf{A}(p) \|q\|_{\text{ch}}$.*

Proof. By (2.2), $T_\alpha T_\beta$ is the average, over the $2^{|B|}$ sign choices, of T_γ with $\gamma_i \in \{\alpha_i + \beta_i, |\alpha_i - \beta_i|\}$. The average has coefficient norm at most one, and the average of $\sum_i \gamma_i^2$ equals $\sum_i (\alpha_i^2 + \beta_i^2)$. Expanding pq and using the triangle inequality proves both bounds. $\square$

2.3 Truncated cones and the sparse box hierarchies

For a bag B and an integer $k \geq 0$, a sum of squares of degree at most $2k$ means a finite sum of squares of polynomials of degree at most k . We write $\Sigma_{2k}(B)$ for these polynomials in the variables x_B , and set $\Sigma_{2k}(B) = \{0\}$ for $k < 0$; terms whose degree allowance is negative are therefore absent, also when counting matrices.

Definition 2.4 (Truncated box cones). The *truncated ordinary box quadratic module* of order r on B is

$$\mathbf{M}_r(B) = \left\{ \sigma_0 + \sum_{i \in B} (1 - x_i^2) \sigma_i : \sigma_0 \in \Sigma_{2r}(B), \sigma_i \in \Sigma_{2r-2}(B) \right\}.$$

The *truncated full box preordering* of order r on B is

$$P_r(B) = \left\{ \sum_{I \subseteq B} \sigma_I \prod_{i \in I} (1 - x_i^2) : \sigma_I \in \Sigma_{2r-2|I|}(B) \right\},$$

where terms with $|I| > r$ are absent. Every summand in either cone has total degree at most $2r$, and $M_r(B) \subseteq P_r(B)$.

The ordinary module uses one positive semidefinite (PSD) matrix for the squares and one for each box generator: $v + 1$ matrices for a bag of size v . The preordering adds the products of distinct generators, up to 2^v matrices. The difference matters both for the size of the relaxation and, as [Section 4](#) shows, for the proof technique.

A linear functional $L : \mathbb{R}[x_B]_{\leq 2r} \rightarrow \mathbb{R}$ is M_r -positive if $L(p) \geq 0$ for all $p \in M_r(B)$, and similarly for P_r . Equivalently, its moment matrix and the localizing matrices of the relevant generator products are positive semidefinite. Such an L is a *truncated functional*; it need not be the integral against any measure.

Definition 2.5 (Sparse box hierarchies). Let $r \geq 1$ with $\deg f_b \leq 2r$ for all b . A *feasible ordinary-module family* of order r is a family $(L_b)_{b=1}^t$ of linear functionals $L_b : \mathbb{R}[x_{B_b}]_{\leq 2r} \rightarrow \mathbb{R}$ such that

- (i) $L_b(1) = 1$ for every b ;
 - (ii) L_b is $M_r(B_b)$ -positive for every b ;
 - (iii) (*exact separator consistency*) $L_b(p) = L_c(p)$ for every edge $e = bc$ and every $p \in \mathbb{R}[x_{S_e}]_{\leq 2r}$.
- A *feasible preordering family* replaces (ii) by $P_r(B_b)$ -positivity. The *moment values* are

$$\rho_r^{\text{mod}} = \inf \left\{ \sum_b L_b(f_b) \right\}, \quad \rho_r^{\text{pre}} = \inf \left\{ \sum_b L_b(f_b) \right\},$$

the infima over feasible ordinary-module and preordering families. The *certificate values* are

$$\lambda_r^{\text{mod}} = \sup \left\{ \lambda \in \mathbb{R} : f - \lambda = \sum_b q_b, q_b \in M_r(B_b) \right\}, \quad (2.5)$$

and λ_r^{pre} is defined in the same way with $P_r(B_b)$. The identity $f - \lambda = \sum_b q_b$ is an identity of polynomials in x .

Consistency is imposed only along tree edges. By running intersection, two bags B_b, B_c then agree on every polynomial in $x_{B_b \cap B_c}$ of degree at most $2r$, because every bag on the tree path from b to c contains $B_b \cap B_c$.

Lemma 2.6 (Weak duality). *For every order r ,*

$$\lambda_r^{\text{mod}} \leq \rho_r^{\text{mod}} \leq \rho_r^{\text{pre}} \leq f^*, \quad \lambda_r^{\text{mod}} \leq \lambda_r^{\text{pre}} \leq \rho_r^{\text{pre}}. \quad (2.6)$$

Proof. Evaluation at a minimizer x^* of (2.1), $L_b(p) = p(x_{B_b}^*)$, is a feasible preordering family with objective f^* , so $\rho_r^{\text{pre}} \leq f^*$. Every preordering family is an ordinary-module family, so $\rho_r^{\text{mod}} \leq \rho_r^{\text{pre}}$; the inclusion of cones gives $\lambda_r^{\text{mod}} \leq \lambda_r^{\text{pre}}$. If $f - \lambda = \sum_b q_b$ with $q_b \in M_r(B_b)$, then $\sum_b (f_b - q_b) - \lambda = 0$, where the constant $-\lambda$ is assigned to bag 1. Apply [Lemma 2.2](#) with $k = 2r$: the left side is a signed sum of separator polynomials, on which adjacent functionals agree. Hence $\sum_b L_b(f_b) - \lambda = \sum_b L_b(q_b) \geq 0$ for every feasible family. The same argument applies to P_r . $\square$

Two different statements recur in the paper and should not be confused. A *rounding statement* applies to every feasible family (L_b) : it constructs a probability law ν on the feasible set with $\int f d\nu \leq \sum_b L_b(f_b) + E$. Taking the infimum gives $f^* - \rho_r \leq E$. The objective value of an arbitrary feasible family is *not* a lower bound on f^* ; the moment value ρ_r and every certificate value are. A *certificate statement* asserts that $f - f^* + E$ lies in $\sum_b M_r(B_b)$ or $\sum_b P_r(B_b)$, with real coefficients. For the box hierarchies the two are linked by [Lemma 2.10](#) below.

Lemma 2.7 (Moment Cauchy–Schwarz). *Let V be a finite-dimensional space of polynomials, G a polynomial, and L a linear functional with $L(Gp^2) \geq 0$ for all $p \in V$. Then $L(Gpq)^2 \leq L(Gp^2)L(Gq^2)$ for all $p, q \in V$. In particular, if $L(Gp^2) = 0$ then $L(Gpq) = 0$ for all $q \in V$.*

Proof. The bilinear form $(p, q) \mapsto L(Gpq)$ on V is positive semidefinite. $\square$

Lemma 2.8 (Chebyshev moments). *Let $L : \mathbb{R}[x_B]_{\leq 2r} \rightarrow \mathbb{R}$ be $M_r(B)$ -positive. For every multi-index α with $|\alpha| \leq r$,*

$$0 \leq L(T_\alpha^2) \leq L(1), \quad |L(T_\alpha)| \leq L(1).$$

Proof. Order the coordinates of B . By (2.2) and telescoping,

$$1 - T_\alpha(x)^2 = \sum_{i \in B} (1 - x_i^2) U_{\alpha_i-1}(x_i)^2 \prod_{j < i} T_{\alpha_j}(x_j)^2, \quad (2.7)$$

where terms with $\alpha_i = 0$ are zero. The i -th term is $(1 - x_i^2)$ times the square of a polynomial of degree at most $|\alpha| - 1 \leq r - 1$. Hence $1 - T_\alpha^2 \in M_r(B)$, which gives $L(T_\alpha^2) \leq L(1)$, and $T_\alpha^2 \in M_r(B)$ gives $L(T_\alpha^2) \geq 0$. [Lemma 2.7](#) with $G = 1$, $p = 1$, $q = T_\alpha$ completes the proof. $\square$

The lemma is used only for $|\alpha| \leq r$, half the degree available to L . Bounds on Chebyshev moments of higher total degree require separate arguments; one such argument, for the preordering, appears in [Lemma 7.1](#).

Lemma 2.9 (Compactness). *Let (L_b) be a feasible ordinary-module family of order r . Then $|L_b(x^\beta)| \leq 1$ for every bag and every monomial with $|\beta| \leq 2r$. Consequently the feasible sets of both sparse box hierarchies are nonempty and compact, and ρ_r^{mod} and ρ_r^{pre} are attained.*

Proof. Let $|\alpha| \leq r$ and $\alpha_i > 0$. With $q = x^{\alpha - e_i}$, the generator test gives $L_b((1 - x_i^2)q^2) \geq 0$, that is, $L_b(x^{2\alpha}) \leq L_b(x^{2(\alpha - e_i)})$. Iterating down to $\alpha = 0$ gives $0 \leq L_b(x^{2\alpha}) \leq 1$. Write any β with $|\beta| \leq 2r$ as $\beta = \alpha + \alpha'$ with $|\alpha|, |\alpha'| \leq r$; [Lemma 2.7](#) gives $L_b(x^\beta)^2 \leq L_b(x^{2\alpha})L_b(x^{2\alpha'}) \leq 1$. The feasible sets are defined by finitely many closed conditions on finitely many bounded coordinates, and they contain point evaluations. $\square$

Lemma 2.10 (Strict feasibility and certificates). *For each bag, let L_b be integration against μ^{B_b} . This family is feasible for both sparse box hierarchies, and every moment and localizing matrix in [Definition 2.4](#) is positive definite. Consequently $\lambda_r^{\text{mod}} = \rho_r^{\text{mod}}$ and $\lambda_r^{\text{pre}} = \rho_r^{\text{pre}}$, and both suprema in (2.5) are attained: $f - \rho_r^{\text{mod}} \in \sum_b M_r(B_b)$ and $f - \rho_r^{\text{pre}} \in \sum_b P_r(B_b)$, each summand of total degree at most $2r$, with real coefficients.*

Proof. Product arcsine measures have the same marginals on every separator, so the family is feasible. If $q \neq 0$ is a polynomial and $I \subseteq B_b$, then $q^2 \prod_{i \in I} (1 - x_i^2)$ is nonnegative and not almost everywhere zero on the open box, where μ^{B_b} has a positive density; its integral is positive. Hence all moment and localizing matrices are positive definite.

Write the moment problem as a finite conic program in the local moment vectors, with the PSD constraints of [Definition 2.4](#), the normalizations, and the separator equalities. By [Lemma 2.9](#)

its value is finite. Its conic dual has a free scalar for the normalization of each bag, a polynomial $\phi_e \in \mathbb{R}[x_{S_e}]_{\leq 2r}$ for each separator equality, and asks that each f_b minus its scalar minus the signed separator polynomials lie in the local cone. Summing these local memberships cancels the ϕ_e , so dual feasibility implies a decomposition as in (2.5); conversely, Lemma 2.2 turns every decomposition $f - \lambda = \sum_b q_b$ into a dual feasible point with value λ . The dual value is therefore λ_r^{mod} (respectively λ_r^{pre}). The conic duality theorem for a strictly feasible primal with finite value [3] gives equality of the values and attainment of the dual. $\square$

Finite SDP dual attainment, as in Lemma 2.10, is different from the existence of exact polynomial separator functions in an infinite-dimensional marginal problem. The latter can fail even when every finite SDP is well posed; Section 5 gives a quadratic example.

2.4 Univariate positivity and gluing

Lemma 2.11 (Interval positivity with degree bounds). *Let $k \geq 0$, and let $p \in \mathbb{R}[x]$ have degree at most $2k$ and satisfy $p \geq 0$ on $[-1, 1]$. Then $p = \sigma_0 + (1 - x^2)\sigma_1$ with $\sigma_0 \in \Sigma_{2k}$ and $\sigma_1 \in \Sigma_{2k-2}$.*

Proof. If $\deg p = 2j$ is even, $j \leq k$, the Markov–Lukács theorem [32] gives $p = \sigma_0 + (1 - x^2)\sigma_1$ with $\sigma_0 \in \Sigma_{2j} \subseteq \Sigma_{2k}$ and $\sigma_1 \in \Sigma_{2j-2} \subseteq \Sigma_{2k-2}$; a nonnegative constant is the case $j = 0$. If $\deg p = 2j - 1$ is odd, $1 \leq j \leq k$, the same theorem gives $p = (1 + x)\tau_+ + (1 - x)\tau_-$ with $\tau_{\pm} \in \Sigma_{2j-2}$. The identity $1 \pm x = \frac{1}{2}(1 \pm x)^2 + \frac{1}{2}(1 - x^2)$ converts this into the claimed form, with $\sigma_0 = \frac{1}{2}[(1 + x)^2\tau_+ + (1 - x)^2\tau_-] \in \Sigma_{2j} \subseteq \Sigma_{2k}$ and $\sigma_1 = \frac{1}{2}(\tau_+ + \tau_-) \in \Sigma_{2j-2} \subseteq \Sigma_{2k-2}$. $\square$

Zeros of p on the interval are allowed. Multiplying univariate representations in distinct coordinates gives elements of the preordering, not of the ordinary module, because products $(1 - x_i^2)(1 - x_j^2)$ appear. This is why the full preordering is the natural cone for product kernels (Section 3) and why the ordinary module needs a different kernel (Section 4).

All the laws we construct live on products of compact intervals and finite sets; these are standard Borel spaces, so regular conditional distributions exist [16].

Lemma 2.12 (Gluing on a junction tree). *Let each coordinate i range over a standard Borel space X_i , and let ν_b be a probability measure on $X_{B_b} = \prod_{i \in B_b} X_i$ for each bag. Suppose that for every edge $e = bc$ the marginals of ν_b and ν_c on X_{S_e} coincide. Then there is a probability measure ν on $\prod_{i=1}^n X_i$ whose marginal on X_{B_b} is ν_b for every b . If each ν_b is concentrated on a measurable set $F_b \subseteq X_{B_b}$, then ν is concentrated on $\{x : x_{B_b} \in F_b \text{ for all } b\}$.*

Proof. Root $\mathcal{T}$ and list the bags so that every bag follows its parent. Let $U_k = B_1 \cup \dots \cup B_k$. Suppose $\nu^{(k)}$ is a probability measure on X_{U_k} with marginals $\nu_1, \dots, \nu_k$. Let $b = k + 1$ have parent c and separator $S = B_b \cap B_c$. If $i \in B_b \cap B_j$ with $j \leq k$, the tree path from b to j passes through c , so $i \in B_c$ by running intersection; hence $B_b \cap U_k = S$. Disintegrate $\nu_b(dx_S, dx_{B_b \setminus S}) = \nu_{b,S}(dx_S) \kappa(x_S, dx_{B_b \setminus S})$, and define $\nu^{(k+1)} = \nu^{(k)}(dx_{U_k}) \kappa(x_S, dx_{B_b \setminus S})$. Its marginal on X_{U_k} is $\nu^{(k)}$. Its marginal on X_S is that of ν_c , which equals $\nu_{b,S}$ by assumption; hence its marginal on X_{B_b} is ν_b . The kernel κ is determined only $\nu_{b,S}$ -almost everywhere, and any version gives the same marginals; in particular an empty separator gives a product. For the last claim, $\nu(\{x_{B_b} \notin F_b\}) = \nu_b(X_{B_b} \setminus F_b) = 0$ for each of the finitely many bags. $\square$

Gluing of consistent local measures under running intersection is classical in sparse polynomial optimization [19] and for finite distributions [37]. The issue addressed in this paper is how to obtain consistent local *measures* from truncated functionals of finite order, with quantitative control of the objective.

2.5 Recourse models and the private-degree-two hierarchy

Sections 6 and 7 study bags that also carry large private continuous vectors entering the objective as convex quadratics.

Definition 2.13 (Recourse model). Let $(\mathcal{T}, (B_b))$ be a junction tree on the *shared* coordinates $x \in [-1, 1]^n$. Each bag b also carries a *private* vector $y_b \in \mathbb{R}^{p_b}$, $p_b \geq 0$; private vectors of different bags are distinct variables. Its *fiber* is

$$P_b(x_{B_b}) = \{y \in [-1, 1]^{p_b} : A_b y \leq a_b + E_b x_{B_b}\}, \quad (2.8)$$

with real $A_b \in \mathbb{R}^{J_b \times p_b}$, $a_b \in \mathbb{R}^{J_b}$, and $E_b \in \mathbb{R}^{J_b \times v_b}$. The local objective is

$$f_b(x, y) = c_b(x) + q_b(x)^\top y + y^\top Q_b(x) y = \mathbf{z}^\top H_b(x) \mathbf{z}, \quad \mathbf{z} = \begin{pmatrix} 1 \\ y \end{pmatrix}, \quad H_b = \begin{pmatrix} c_b & q_b^\top/2 \\ q_b/2 & Q_b \end{pmatrix}, \quad (2.9)$$

with polynomial coefficients in x_{B_b} and symmetric Q_b . We assume *private convexity*: $Q_b(x) \succeq 0$ for every x in the shared bag box. When its feasible set is nonempty, the problem is

$$f^* = \min \left\{ \sum_b f_b(x_{B_b}, y_b) : x \in [-1, 1]^n, y_b \in P_b(x_{B_b}) \text{ for all } b \right\}.$$

The model has a *fixed private domain* if $E_b = 0$ for all b , and *complete recourse* if every fiber is nonempty at every shared box point. The quantitative theorems of Sections 6 and 7 assume nonempty fixed polytopes or complete recourse, respectively.

The shared variables carry an arbitrary nonconvex polynomial structure. The private variables are continuous, belong to one bag, and appear convexly; only the shared coordinates are coupled across bags. Private convexity is a pointwise assumption, not an SOS condition on Q_b ; verifying it may itself be hard in general, and we use it only where an application supplies it.

For $I \subseteq B_b$ write $w_I(x) = \prod_{i \in I} (1 - x_i^2)$. Let $\ell_{bj}(x, y) = (a_b + E_b x - A_b y)_j$ be the j -th row of the fiber constraints, and let $\epsilon_{bj} = 1$ if row j of E_b is nonzero and $\epsilon_{bj} = 0$ otherwise.

Definition 2.14 (Private-degree-two hierarchy). Let $r \geq 1$ with every entry of H_b of degree at most $2r$. A feasible family of order r consists of linear functionals

$$L_b : \mathbb{R}[x_{B_b}]_{\leq 2r} \otimes \mathbb{R}[y_b]_{\leq 2} \longrightarrow \mathbb{R},$$

defined on polynomials of shared total degree at most $2r$ and private total degree at most two, such that, for every $I \subseteq B_b$:

- (R0) $L_b(1) = 1$;
- (R1) $L_b(w_I(s_0(x) + s(x)^\top y_b)^2) \geq 0$ for all polynomials $s_0, s_1, \dots, s_{p_b}$ of degree at most $r - |I|$;
- (R2) $L_b(w_I s(x)^2 \ell_{bj}(x, y_b)) \geq 0$ for every row j and every polynomial s of degree at most $r - |I| - \epsilon_{bj}$;
- (R3) $L_b(w_I s(x)^2 (1 - y_{b,k}^2)) \geq 0$ for $k = 1, \dots, p_b$ and every s of degree at most $r - |I|$;
- (R4) $L_b(p) = L_c(p)$ for every edge $e = bc$ and every $p \in \mathbb{R}[x_{S_e}]_{\leq 2r}$.

Conditions with a negative degree allowance are absent. The moment value is $\rho_r^{\text{rec}} = \inf \sum_b L_b(f_b)$ over feasible families.

Every polynomial tested in (R1)–(R3) has shared degree at most $2r$ and private degree at most two. Evaluation at a feasible point is a feasible family, so $\rho_r^{\text{rec}} \leq f^*$. The affine bounds $1 \pm y_{b,k} \geq 0$ are implied, since $1 \pm y = \frac{1}{2}(1 \pm y)^2 + \frac{1}{2}(1 - y^2)$. The quadratic bounds (R3) are redundant for the original problem but essential for the relaxation: without them the relaxation can be unbounded

below at every sufficiently high order ([Proposition 6.11](#)). Shared positivity uses the full shared box preordering. The rows with $\epsilon_{bj} = 1$ have their degree allowance reduced by one so that the products in (R2) stay within shared degree $2r$.

The point of the hierarchy is its size. The PSD matrix in (R1) indexed by I has

$$(p_b + 1) \binom{r - |I| + v_b}{v_b} \quad (2.10)$$

rows, the localizing matrices in (R2)–(R3) have at most $\binom{r - |I| + v_b}{v_b}$ rows, there are 2^{v_b} choices of I , and the number of local moments is at most $\binom{p_b + 2}{2} \binom{2r + v_b}{v_b}$. For fixed width these sizes are polynomial in p_b , J_b , and r , and the exponent of r depends only on the number v_b of shared coordinates in the bag. By contrast, a standard total-degree hierarchy on all $v_b + p_b$ variables of the bag has a moment matrix with $\binom{r + v_b + p_b}{r}$ rows. For $v_b = 2$, $p_b = 50$, and $r = 6$ these numbers are 1,428 and 40,475,358. The constants in our error bounds, however, can grow with the private dimension through the coefficients of H_b and the fiber data; the hierarchy is not dimension-free in accuracy.

Partial lifting of selected nonlinear variables is an established way to reduce relaxation size [[15](#)]. Holding an SOS-convex decision-variable module at fixed degree while the uncertainty-variable order grows is also used in robust polynomial matrix inequality hierarchies [[11](#)]. Here we give a quantitative analysis of the specified sparse private-degree-two hierarchy.

The private-degree-two hierarchy keeps the private degree at two while retaining shared degree up to $2r$ in mixed moments, so its moments have joint degree up to $2r + 2$. A total-degree hierarchy of order r retains all joint moments of degree up to $2r$, including higher private degrees; if it uses a full preordering, it also contains products of distinct private constraints. We do not compare the two formulations by inclusion. Lower bounds that we state for both are proved separately for each, by actual local measures, which are feasible for every hierarchy considered here.

3 Exactly consistent rounding for the full preordering

This section proves that every feasible family of the sparse box preordering of order r can be rounded to a probability law on the box whose expected objective exceeds the relaxation objective by at most $3A/(2m^2 + 1)$, where $m = \lfloor r/w \rfloor + 1$. Consequently $f^* - \rho_r^{\text{pre}} \leq 3w^2A/(2r^2)$. The inverse-square rate itself is not new ([Remark 3.9](#)). We give a complete proof because its construction is the template for every upper bound in the paper, and because it yields explicit constants, a finite-grid rounding, and a real certificate.

The construction applies one fixed univariate polynomial kernel $K_m(x, u)$ in every coordinate of every bag and evaluates the local functional on the resulting tensor kernel:

$$h_b(u) = L_b \left(\prod_{i \in B_b} K_m(x_i, u_i) \right), \quad u \in [-1, 1]^{B_b}.$$

Four properties of the kernel do four different jobs.

- (1) *Finite degree.* The kernel has degree $2(m - 1)$ in x_i , so the tensor kernel lies in the domain of L_b .
- (2) *Certified positivity.* For each fixed output u_i , the polynomial $K_m(\cdot, u_i)$ is nonnegative on $[-1, 1]$, and its interval certificate has the right degree. Multiplying the certificates over a bag gives an element of the truncated preordering, so $h_b \geq 0$ although L_b need not be a measure.
- (3) *Exact mass.* $\int K_m(x, u) d\mu(u) = 1$ identically in x . Integrating h_b over the coordinates outside a separator therefore removes their kernel factors exactly.

- (4) *One kernel everywhere.* The separator marginal of h_b is L_b applied to a polynomial in the separator variables of degree at most $2r$. Adjacent functionals agree on such polynomials, so adjacent separator marginals are *equal as functions*, not merely close.

Running intersection then glues the local laws into one law on the box (Lemma 2.12). The kernel acts diagonally on Chebyshev polynomials, multiplying T_k by γ_k with $1 - \gamma_k \leq 3k^2/D_m$, and Lemma 2.8 bounds the Chebyshev moments of L_b . Together these control the change in objective.

3.1 The Jackson-type kernel

Fix an integer $m \geq 1$. Let $\tau_j = (m - |j|)_+$ for $j \in \mathbb{Z}$, and let

$$a_k = \sum_{j \in \mathbb{Z}} \tau_j \tau_{j-k}, \quad \gamma_k = \frac{a_k}{a_0} \quad (k \geq 0), \quad D_m = 2m^2 + 1.$$

The sums are finite, $a_k = a_{-k}$, and $a_k = 0$ for $|k| > 2m - 2$. Define the trigonometric polynomial and the bivariate polynomial

$$J_m(t) = \frac{1}{a_0} \left| \sum_{j=0}^{m-1} e^{ijt} \right|^4, \quad K_m(x, u) = 1 + 2 \sum_{k=1}^{2m-2} \gamma_k T_k(x) T_k(u). \quad (3.1)$$

Lemma 3.1 (Jackson-type kernel). *The following hold.*

- (a) $J_m(t) = 1 + 2 \sum_{k=1}^{2m-2} \gamma_k \cos(kt)$, and $J_m \geq 0$.
- (b) $a_0 = (2m^3 + m)/3$, the coefficients a_k are nonnegative integers, and $0 \leq \gamma_k \leq 1$ for all $k \geq 0$, with $\gamma_0 = 1$ and $\gamma_k = 0$ for $k > 2m - 2$.
- (c) For every integer $k \geq 0$,

$$0 \leq 1 - \gamma_k \leq k^2(1 - \gamma_1) = \frac{3k^2}{D_m}. \quad (3.2)$$

- (d) $K_m(\cos \theta, \cos \phi) = \frac{1}{2}(J_m(\theta - \phi) + J_m(\theta + \phi))$. Hence $K_m \geq 0$ on $[-1, 1]^2$.
- (e) For every integer $k \geq 0$ and every $x \in \mathbb{R}$,

$$\int_{[-1, 1]} T_k(u) K_m(x, u) d\mu(u) = \gamma_k T_k(x). \quad (3.3)$$

In particular $\int K_m(x, u) d\mu(u) = 1$ identically in x , and the same holds with the roles of x and u exchanged.

- (f) K_m has rational coefficients and degree $2m - 2$ in each variable.

Proof. (a) Counting the pairs (j, j') in $\{0, \dots, m-1\}^2$ with $j - j' = \ell$ gives $|\sum_{j=0}^{m-1} e^{ijt}|^2 = \sum_{\ell} \tau_{\ell} e^{i\ell t}$. Squaring, the coefficient of e^{ikt} is $\sum_j \tau_j \tau_{j-k} = a_k$, because τ is even. Dividing by a_0 and using $a_k = a_{-k}$ gives the cosine series. Nonnegativity is clear from the definition.

(b) Directly, $a_0 = \sum_j \tau_j^2 = m^2 + 2 \sum_{j=1}^{m-1} j^2 = (2m^3 + m)/3$. The a_k are sums of products of nonnegative integers, and $a_k \leq a_0$ by the Cauchy–Schwarz inequality. The support statement was noted above.

(c) For $k \geq 1$, the cosine series in (a) gives $\gamma_k = \int_{-\pi}^{\pi} \cos(kt) J_m(t) dt / (2\pi)$; this also holds for $k > 2m - 2$, where both sides vanish. Since $J_m \geq 0$ has integral one, $1 - \gamma_k = \int (1 - \cos kt) J_m(t) dt / (2\pi)$. The inequality $|\sin(k\theta)| \leq k|\sin \theta|$ follows by induction from $\sin((k+1)\theta) = \sin(k\theta)\cos \theta + \cos(k\theta)\sin \theta$. With $\theta = t/2$ it gives $1 - \cos(kt) = 2\sin^2(kt/2) \leq k^2(1 - \cos t)$, and integration proves the upper bound in (3.2). For the equality, $a_0 - a_1 = \frac{1}{2} \sum_j (\tau_j - \tau_{j-1})^2$. The differences $\tau_j - \tau_{j-1}$ equal $+1$ for $-m+1 \leq j \leq 0$, equal -1 for $1 \leq j \leq m$, and vanish otherwise, so $a_0 - a_1 = m$ and $1 - \gamma_1 = m/a_0 = 3/(2m^2 + 1)$.

- (d) This follows from (a) and $\cos(k(\theta - \phi)) + \cos(k(\theta + \phi)) = 2 \cos(k\theta) \cos(k\phi)$.
(e) Orthogonality of Chebyshev polynomials under μ gives $\int T_k T_j d\mu = \delta_{jk}/2$ for $k \geq 1$ and $\int T_0^2 d\mu = 1$. Inserting the definition of K_m proves (3.3) for $k \leq 2m - 2$, and both sides vanish for $k > 2m - 2$. The kernel is symmetric.
(f) The γ_k are ratios of integers. $\square$

This kernel is the classical squared Fejér, or Jackson-type, construction used for dense box hierarchies [7, 22] and, in tensor form on bags, by Korda, Magron and Ríos-Zertuche [17]. We derived the exact coefficients and estimates used below; we do not claim the kernel as new. The bound (3.2) holds for every k , including frequencies above the kernel degree, so the objective degree need not be smaller than the kernel degree.

3.2 Positive local densities with identical separator marginals

For a bag B and $u \in [-1, 1]^B$ write $K_{m,B}(x, u) = \prod_{i \in B} K_m(x_i, u_i)$. As a polynomial in x it has total degree at most $2|B|(m - 1)$. Its coefficients are polynomials in u , so $L(K_{m,B}(\cdot, u))$, obtained by applying L in the variable x , is a polynomial in u .

Lemma 3.2 (Local densities). *Let $m \geq 1$ and let (L_b) be a feasible preordering family of order r with $v_b(m - 1) \leq r$ for every bag. For $u \in [-1, 1]^{B_b}$ define*

$$h_b(u) = L_b(K_{m,B_b}(\cdot, u)), \quad h_S^{(b)}(u_S) = L_b(K_{m,S}(\cdot, u_S)) \quad (S \subseteq B_b), \quad (3.4)$$

with $h_\emptyset^{(b)} = 1$. Then:

- (a) h_b is a polynomial of degree at most $2m - 2$ in each coordinate, and $h_b(u) \geq 0$ for every $u \in [-1, 1]^{B_b}$;
- (b) for every $S \subseteq B_b$, $\int h_b d\mu^{B_b \setminus S} = h_S^{(b)}$ identically; in particular $\int h_b d\mu^{B_b} = 1$;
- (c) for every edge $e = bc$, $h_{S_e}^{(b)} = h_{S_e}^{(c)}$;
- (d) for every $\alpha \in \mathbb{N}^{B_b}$ with $|\alpha| \leq 2r$,

$$\int T_\alpha h_b d\mu^{B_b} = \left(\prod_{i \in B_b} \gamma_{\alpha_i} \right) L_b(T_\alpha). \quad (3.5)$$

Proof. (a) The degree statement is clear. Fix u . By Lemma 3.1(d) the polynomial $K_m(\cdot, u_i)$ is nonnegative on $[-1, 1]$ and has degree at most $2(m - 1)$. Lemma 2.11 with $k = m - 1$ gives $K_m(x_i, u_i) = \sigma_{0,i}(x_i) + (1 - x_i^2)\sigma_{1,i}(x_i)$ with $\sigma_{0,i} \in \Sigma_{2m-2}$ and $\sigma_{1,i} \in \Sigma_{2m-4}$ (for $m = 1$, $K_m = 1$ and $\sigma_{1,i} = 0$). Multiplying over $i \in B_b$,

$$K_{m,B_b}(x, u) = \sum_{I \subseteq B_b} \left(\prod_{i \notin I} \sigma_{0,i}(x_i) \prod_{i \in I} \sigma_{1,i}(x_i) \right) \prod_{i \in I} (1 - x_i^2).$$

The bracket is a sum of squares of polynomials of degree at most $v_b(m - 1) - |I| \leq r - |I|$, so every term belongs to $\text{Pr}(B_b)$, and $h_b(u) \geq 0$. The representation may depend on u in an arbitrary way; it is used only at the fixed point u .

(b) Every coefficient of $K_{m,B_b}(x, u)$ as a polynomial in x is a polynomial in u , and L_b is linear, so integration in u and application of L_b commute. By Lemma 3.1(e), integrating $K_m(x_i, u_i)$ in u_i gives the constant one. Integrating out the coordinates of $B_b \setminus S$ therefore leaves $L_b(K_{m,S}(\cdot, u_S))$.

(c) $K_{m,S_e}(x, u)$ is a polynomial in x_{S_e} of total degree at most $2|S_e|(m - 1) \leq 2r$. Exact separator consistency (Definition 2.5(iii)) applied coefficientwise in u gives the claim.

(d) Expanding the definition, $K_{m,B_b}(x,u) = \sum_{\beta} (\prod_i c_{\beta_i} \gamma_{\beta_i}) T_{\beta}(x) T_{\beta}(u)$, where β ranges over multi-indices with $\beta_i \leq 2m-2$, $c_0 = 1$ and $c_k = 2$ for $k \geq 1$. Since $\int T_{\alpha_i} T_{\beta_i} d\mu = \delta_{\alpha_i \beta_i} / c_{\alpha_i}$, integration against $T_{\alpha}(u)$ gives $(\prod_i \gamma_{\alpha_i}) L_b(T_{\alpha})$ if $\alpha_i \leq 2m-2$ for all i , and zero otherwise; in the latter case some γ_{α_i} vanishes. The value $L_b(T_{\alpha})$ is defined because $|\alpha| \leq 2r$. $\square$

Remark 3.3 (Degree accounting and the role of the preordering). Two features of [Lemma 3.2](#) cannot be weakened casually.

First, positivity is certified through the univariate certificates, not through the total degree of the tensor kernel. A product of interval nonnegative polynomials whose total degree is at most $2r$ can be negative under a feasible functional. Take two variables, $r = 1$, and $L(1) = 1$, $L(x) = L(y) = \frac{1}{2}$, $L(x^2) = L(y^2) = 1$, $L(xy) = -\frac{1}{2}$. The moment matrix on $(1, x, y)$ has eigenvalues $0, \frac{3}{2}, \frac{3}{2}$, and $L(1-x^2) = L(1-y^2) = 0$, so L is P_1 -positive (at order one the preordering has no products of generators). Yet $L((1-x)(1-y)) = -\frac{1}{2}$, although $(1-x)(1-y) \geq 0$ on the box and has degree two. A univariate factor of degree q needs interval certificate degree $2\lceil q/2 \rceil$; the even kernel degree $2(m-1)$ makes the count in [Lemma 3.2\(a\)](#) exact.

Second, the expansion of K_{m,B_b} contains products of distinct box generators. The ordinary module does not contain these products, and a normalized kernel that is merely nonnegative on the interval does not in general have an ordinary-module representation of the tensor product. This is why [Section 4](#) uses a different kernel.

Finally, the same kernel must be used for a given coordinate in all bags containing it, with the same parameter m . With different kernels in two bags, equal input moments would not give equal output marginals.

3.3 The rounding theorem

Recall that d is the largest total degree of the local polynomials f_b and that $A = \sum_b A(f_b)$ is the weighted Chebyshev budget of [\(2.4\)](#); it is zero for constant local terms and is not affected by additive constants.

Theorem 3.4 (Rounding for the sparse box preordering). *Let $r \geq \max\{w, d\}$ and $m = \lfloor r/w \rfloor + 1$. For every feasible preordering family (L_b) of order r there is a Borel probability measure ν on $[-1, 1]^n$ such that the marginal of ν on $[-1, 1]^{B_b}$ is $h_b d\mu^{B_b}$ for every bag, with h_b from [\(3.4\)](#), and*

$$\left| \int f d\nu - \sum_b L_b(f_b) \right| \leq \frac{3A}{D_m}. \quad (3.6)$$

Consequently

$$0 \leq f^* - \rho_r^{\text{pre}} \leq \frac{3A}{2m^2 + 1} \leq \frac{3w^2 A}{2r^2}, \quad (3.7)$$

and for every feasible family there is a point $x \in [-1, 1]^n$ with $f(x) \leq \sum_b L_b(f_b) + 3A/D_m$.

Proof. Since $m-1 = \lfloor r/w \rfloor$, we have $v_b(m-1) \leq w \lfloor r/w \rfloor \leq r$ for every bag, so [Lemma 3.2](#) applies. By parts (a) and (b), each $\nu_b = h_b d\mu^{B_b}$ is a probability measure on $[-1, 1]^{B_b}$. By parts (b) and (c), adjacent laws have the same separator marginal, namely $h_{S_e}^{(b)} d\mu^{S_e}$. [Lemma 2.12](#) gives a probability measure ν on $[-1, 1]^n$ with marginals ν_b , and $\int f d\nu = \sum_b \int f_b d\nu_b$.

Fix a bag and expand $f_b = \sum_{\alpha} c_{b,\alpha} T_{\alpha}$; every α with $c_{b,\alpha} \neq 0$ has $|\alpha| \leq d \leq r$. By [\(3.5\)](#),

$$\int f_b d\nu_b - L_b(f_b) = \sum_{\alpha} c_{b,\alpha} \left(\prod_i \gamma_{\alpha_i} - 1 \right) L_b(T_{\alpha}).$$

Lemma 2.8 gives $|L_b(T_\alpha)| \leq 1$. For numbers $\gamma_i \in [0, 1]$ we have $0 \leq 1 - \prod_i \gamma_i \leq \sum_i (1 - \gamma_i)$ by telescoping, so (3.2) gives $|1 - \prod_i \gamma_{\alpha_i}| \leq 3 \sum_i \alpha_i^2 / D_m$. Hence $|\int f_b d\nu_b - L_b(f_b)| \leq 3A(f_b)/D_m$. Summing over bags proves (3.6).

Since ν is supported on the box, $f^* \leq \int f d\nu \leq \sum_b L_b(f_b) + 3A/D_m$, and some point of the box has value at most $\int f d\nu$. Taking the infimum over feasible families gives the first inequality in (3.7) together with Lemma 2.6. Finally $m > r/w$, so $2m^2 + 1 > 2r^2/w^2$. $\square$

The theorem is a statement about *every* feasible family, not only an optimal one. By Lemma 2.9 an optimal family exists; rounding it gives a law ν with $\int f d\nu \leq \rho_r^{\text{pre}} + 3A/D_m$. The objective value of an arbitrary feasible family is not a lower bound on f^* ; the moment value ρ_r^{pre} is. The constant depends on the chosen decomposition of f through A , and A can grow with the number of bags. The rounded law is supported on the box, but it need not satisfy any further constraint, and it need not respect prescribed integer values.

Remark 3.5 (Explicit gluing for polynomial densities). For the laws in Theorem 3.4, the conditional laws used in Lemma 2.12 can be written down, so no general disintegration theorem is needed. Root $\mathcal{T}$ and let bag b have parent separator S and new coordinates $C = B_b \setminus S$. Let $h_S = h_S^{(b)}$. Define the kernel $\kappa(u_S, du_C) = h_b(u_S, u_C) h_S(u_S)^{-1} d\mu^C(u_C)$ where $h_S(u_S) > 0$, and $\kappa(u_S, \cdot) = \mu^C$ where $h_S(u_S) = 0$. For measurable $A \subseteq [-1, 1]^S$ and $F \subseteq [-1, 1]^C$, the parent law gives A the mass $\int_A h_S d\mu^S$, so the glued law gives $A \times F$ the mass $\int_{A \cap \{h_S > 0\}} \int_F h_b d\mu^C d\mu^S$. Since $h_b \geq 0$ and $\int h_b d\mu^C = h_S$, the inner integral vanishes where $h_S = 0$, so this mass equals $\nu_b(A \times F)$. Rectangles determine ν_b , and the induction of Lemma 2.12 goes through. The same construction with sums applies to the finite laws in Corollary 3.7, and with h_b^+ in place of h_b to the laws of Section 4.

3.4 Finite grid rounding and certificates

Because each density h_b is a polynomial of known coordinate degree, the integrals in Theorem 3.4 can be replaced by an exact quadrature on a fixed Chebyshev grid. The rounded law then lives on a finite grid, and the best grid point can be found by dynamic programming over the junction tree.

Lemma 3.6 (Chebyshev quadrature). *Let $N \geq 1$ and $\xi_j = \cos((2j-1)\pi/(2N))$ for $j = 1, \dots, N$. For every univariate polynomial q of degree at most $2N-1$, $\frac{1}{N} \sum_{j=1}^N q(\xi_j) = \int q d\mu$.*

Proof. It suffices to take $q = T_k$, $0 \leq k \leq 2N-1$. For $k = 0$ both sides equal one. For $1 \leq k \leq 2N-1$, put $\theta = k\pi/(2N) \in (0, \pi)$. Then $\sum_{j=1}^N e^{i(2j-1)\theta} = e^{i\theta}(1 - e^{2iN\theta})/(1 - e^{2i\theta})$, and the denominator is nonzero. If k is even, $e^{2iN\theta} = e^{ik\pi} = 1$ and the sum vanishes. If k is odd, the sum equals $2e^{i\theta}/(1 - e^{2i\theta}) = i/\sin \theta$, whose real part is zero. Hence $\sum_j T_k(\xi_j) = 0 = N \int T_k d\mu$. $\square$

Let d_∞ be the largest coordinate degree of the local polynomials, that is, the largest α_i with $c_{b,\alpha} \neq 0$ (zero if all f_b are constant).

Corollary 3.7 (Rounding to a fixed grid). *Let r and m be as in Theorem 3.4, let $N_g = m + \lfloor d_\infty/2 \rfloor$, and let $\Xi = \{\xi_1, \dots, \xi_{N_g}\}$ be the nodes of Lemma 3.6 with $N = N_g$. For every feasible preordering family (L_b) of order r ,*

$$\min_{x \in \Xi^n} f(x) \leq \sum_b L_b(f_b) + \frac{3A}{D_m}. \quad (3.8)$$

In particular $f^ \leq \min_{\Xi^n} f \leq \rho_r^{\text{pre}} + 3A/D_m \leq f^* + 3A/D_m$. Given the tables of local values f_b on Ξ^{B_b} , a grid minimizer can be computed by dynamic programming over $\mathcal{T}$ with $\sum_b N_g^{v_b}$ table entries and at most $2tN_g^w$ additions and comparisons.*

Proof. Since $2\lfloor d_\infty/2 \rfloor \geq d_\infty - 1$, we have $2N_g - 1 \geq 2(m-1) + d_\infty$. The density h_b has degree at most $2m - 2$ in each coordinate, and $f_b h_b$ has degree at most $2m - 2 + d_\infty$ in each coordinate. Applying Lemma 3.6 coordinate by coordinate, the tensor rule with weights $N_g^{-v_b}$ integrates h_b , $f_b h_b$, and h_b with respect to any subset of its coordinates exactly. Define finite bag laws $p_b(\xi_j) = N_g^{-v_b} h_b(\xi_j)$ for $\xi_j \in \Xi^{B_b}$. They are nonnegative by Lemma 3.2(a) and have total mass one. Summing p_b over the coordinates of $B_b \setminus S$ gives $N_g^{-|S|} h_S^{(b)}$ at the remaining nodes, by exact quadrature and Lemma 3.2(b); by part (c) adjacent bags have equal separator marginals. Lemma 2.12 for finite sets (or Remark 3.5 with sums) gives a law on Ξ^n with these bag marginals. Its expected objective is $\sum_b \sum_j p_b(\xi_j) f_b(\xi_j) = \sum_b \int f_b h_b d\mu^{B_b}$, which is the quantity bounded in the proof of Theorem 3.4. Some grid point is no worse than the mean, proving (3.8). The chain of inequalities follows from Lemma 2.6 and (3.7), applied to an optimal family (Lemma 2.9).

For the dynamic program, root $\mathcal{T}$ and process bags from the leaves up. For a bag b with parent separator S , form the table of f_b plus the messages received from its children, each a function of a child separator and hence a table lookup, and minimize over the coordinates of $B_b \setminus S$ to obtain the message to the parent. This uses $(1 + \#\text{children}(b))N_g^{v_b}$ additions and comparisons at bag b , and $\sum_b (1 + \#\text{children}(b)) = 2t - 1$. Storing the minimizing choices recovers a grid minimizer. $\square$

The operation count is in exact real arithmetic; the nodes ξ_j are algebraic numbers, and no bit-complexity claim is made. The grid bound $\min_{\Xi^n} f - f^* \leq 3A/D_m$ on its own is a polynomial approximation statement of the type established by Piazzon and Vianello [31]. The proof describes the dynamic programming algorithm that minimizes the grid objective on a junction tree. Grid optimization with dynamic programming therefore attains the same exponent without solving any SDP. The content of Corollary 3.7 is the comparison: the grid optimum is at most the objective of every feasible point of the SDP plus $3A/D_m$. For an optimal SDP point, the grid optimum and the moment value differ by at most this error. Bounded-treewidth polynomial optimization also admits LP approximation schemes [4], so no algorithmic advance is claimed.

Corollary 3.8 (Sparse preordering certificate). *Let r and m be as in Theorem 3.4. Then*

$$f - f^* + \frac{3A}{D_m} \in \sum_{b=1}^t P_r(B_b),$$

with real coefficients and every summand of total degree at most $2r$. Equivalently, $\lambda_r^{\text{pre}} = \rho_r^{\text{pre}} \geq f^ - 3A/D_m$.*

Proof. By Lemma 2.10, $f - \rho_r^{\text{pre}} = \sum_b q_b$ with $q_b \in P_r(B_b)$. By Theorem 3.4, the constant $\rho_r^{\text{pre}} - f^* + 3A/D_m$ is nonnegative; add it to q_1 , where it is a square of a constant. $\square$

The certificate has real coefficients. Rational certificates and their size are treated in Section 9. Finite SDP dual attainment, used here, is different from the existence of exact polynomial separator functions; Theorem 5.4 shows that the latter can fail for a fixed quadratic objective.

3.5 Relation to prior work

Remark 3.9 (What is prior in Theorem 3.4). The rate is not new. Magron’s lecture slides of 7 July 2025 [23] (slide 23 of 44) and of 16 February 2026 [25] (slide 35 of 90) state an $O(d^{-2})$ gap for two overlapping bags with local full preorderings and total-degree truncation $2d$, attributed to Korda, Ríos-Zertuche and Magron. The published article of these authors [17], Theorem 6, gives for the sparse box preordering a degree bound equivalent to the rate $O(r^{-2/(w+3)})$ for fixed data, using coordinatewise degree bounds; the difference from total degree costs only width-dependent

factors. We could not resolve the difference between the slides and the published theorem, and we do not assume either is in error. The published proof first approximates separator value functions by polynomials to obtain positive bag polynomials and then applies a sparse Jackson operator; the tensor Jackson kernel and its preordering positivity already appear there.

Dense box preordering hierarchies have error $O(r^{-2})$ by Laurent and Slot [22], using the same kind of kernel, which de Klerk, Hess and Laurent [7] used earlier for upper-bound hierarchies. Dense full preorderings on products of standard domains are treated by Magron [24]. For actual measures on the torus, equal low-order Fourier moments give equal Jackson convolutions, as observed by Catala, Hockmann, Kunis and Wageringel [5]. Applying the cosine map gives the corresponding statement for Chebyshev moments. Gluing consistent local measures under running intersection is classical in sparse optimization [19]. Gamertsfelder and Mourrain [8] transfer effective positivity rates to generalized moment relaxations under an assumption that a dual with finitely supported, here polynomial, separator multipliers is attained; Section 5 gives a quadratic example in which no exact polynomial separator decomposition exists.

What Theorem 3.4 adds, to our knowledge, is a direct finite-order primal construction: every feasible sparse preordering family is turned into exactly consistent local probability densities, with the explicit constant $3A/D_m$ and without approximating separator value functions or assuming dual attainment. Independently approximating each local functional by a measure, as in fixed-degree truncated moment approximation [36], does not by itself produce equal separator marginals.

4 The ordinary sparse quadratic module

The ordinary box quadratic module uses $v + 1$ PSD blocks for a bag of size v , against up to 2^v for the full preordering. This section proves that the sparse ordinary-module hierarchy still has error $C_f O_{w,d_\infty}(\log^3 r/r^2)$, where C_f is the sum of the local nonconstant Chebyshev norms (Theorem 4.6 and Corollary 4.7). The implicit constant and the order threshold depend only on the width and the largest coordinate degree; they do not depend on the number of bags or the shape of the tree. The size of the SDP grows with the number of bags, and the absolute error through C_f can grow with it as well.

Why the construction of Section 3 does not apply. Positivity of a product of univariate kernels is certified in Lemma 3.2 by multiplying interval certificates, which creates products of distinct box generators. Such products are not in the ordinary module, and their values under an ordinary-module functional can be negative. For example, on two variables let $L(1) = 1$, let L vanish on every monomial with an odd exponent, and let $L(x^2) = L(y^2) = L(x^4) = L(y^4) = \frac{3}{4}$ and $L(x^2y^2) = \frac{3}{8}$. The moment matrix of order two is PSD: the block on $(1, x^2, y^2)$ has Schur complement $\frac{3}{16} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$, and the remaining diagonal entries are $\frac{3}{4}, \frac{3}{4}, \frac{3}{8}$. The localizing matrices of $1 - x^2$ and $1 - y^2$ on $(1, x, y)$ are $\text{diag}(\frac{1}{4}, 0, \frac{3}{8})$ and $\text{diag}(\frac{1}{4}, \frac{3}{8}, 0)$. So L is M_2 -positive, yet $L((1 - x^2)(1 - y^2)) = -\frac{1}{8}$.

Products of polynomials that are *globally* sums of squares, by contrast, are sums of squares, so a kernel that is SOS in its input variable for every fixed output gives nonnegative densities under ordinary-module positivity. Gribling, de Klerk and Vera [9] built such a kernel for the dense hierarchy: a squared Fejér kernel times a geometric-series multiplier, whose mass is $1 - \varrho(x)$ with a residual ϱ that is exponentially small in the length of the series. The mass cannot be exactly one (Remark 4.3), and in a sparse hierarchy unequal mass defects in different bags break the exact equality of separator marginals on which gluing relies.

The repair. We add the residual back: the kernel $\bar{\Psi} = \Psi + \varrho$ has mass exactly one, so the resulting *signed* densities have exactly equal separator marginals, as in [Section 3](#). Expanding the tensor product of $\bar{\Psi}$, the terms with at most one residual factor are nonnegative under the ordinary module, since a single interval certificate times SOS factors stays in the module. Terms with two or more residual factors may be negative. A weighted moment Cauchy–Schwarz inequality bounds a term with j residual factors by $\delta^j \Lambda^{v-j}$, where δ is the sup norm of the residual on the interval and Λ bounds the Chebyshev norm of one kernel factor. This gives a lower bound $-\Delta_w$ on every density, with the same Δ_w in every bag. Adding Δ_w times the reference density and renormalizing keeps the separator marginals equal, because the reference marginals are all equal to one. The correction costs Δ_w times the oscillation of each local objective, which is negligible for a suitable series length.

4.1 The squared Fejér kernel and its normalization

Fix integers $s \geq 2$ and $N \geq 2$ with N even. Let $\chi_j = (1 - j/s)_+$ for $j \geq 0$ and $\bar{\chi}_j = 1 - \chi_j = \min\{j/s, 1\}$. Define

$$\Phi_s(x, u) = 1 + 2 \sum_{j=1}^{s-1} \chi_j T_j(x) T_j(u), \quad M_s(x) = \int \Phi_s(x, u)^2 d\mu(u) = 1 + 2 \sum_{j=1}^{s-1} \chi_j^2 T_j(x)^2, \quad (4.1)$$

$$C_s = 1 + 2 \sum_{j=1}^{s-1} \chi_j^2 = \frac{2s^2 + 1}{3s}, \quad z_s(x) = 1 - \frac{M_s(x)}{C_s}, \quad p_{s,N}(x) = \frac{1}{C_s} \sum_{j=0}^N z_s(x)^j, \quad (4.2)$$

and

$$\begin{aligned} \Psi(x, u) &= p_{s,N}(x) \Phi_s(x, u)^2, & \omega(x) &= \int \Psi(x, u) d\mu(u), \\ \varrho &= 1 - \omega, & \bar{\Psi}(x, u) &= \Psi(x, u) + \varrho(x). \end{aligned} \quad (4.3)$$

Put

$$D = 2(s-1)(N+1), \quad \delta = 2^{-(N+1)}, \quad \Lambda = \frac{s^2(N+1)}{C_s}. \quad (4.4)$$

For a univariate polynomial q let $\mathcal{A}q(x) = \int q(u) \Psi(x, u) d\mu(u)$ and $\bar{\mathcal{A}}q(x) = \int q(u) \bar{\Psi}(x, u) d\mu(u)$.

Lemma 4.1 (Squared Fejér kernel). *The following hold.*

- (a) $\Phi_s \geq 0$ on $[-1, 1]^2$, and $C_s/2 \leq M_s \leq C_s$ on $[-1, 1]$; hence $0 \leq z_s \leq \frac{1}{2}$ on $[-1, 1]$.
- (b) $p_{s,N}$ is a sum of squares of polynomials of degree at most $(s-1)N$. For each fixed u , $\Psi(\cdot, u)$ is a sum of squares of polynomials of degree at most $D/2$; its degree in x is at most D and its degree in u is at most $2(s-1)$.
- (c) $\omega = p_{s,N} M_s = 1 - z_s^{N+1}$ identically and is globally SOS, so $\varrho = z_s^{N+1}$ has degree at most D and $0 \leq \varrho \leq \delta$ on $[-1, 1]$.
- (d) $\int \bar{\Psi}(x, u) d\mu(u) = 1$ identically in x , $\bar{\mathcal{A}}1 = 1$, and $\bar{\mathcal{A}}T_k = \mathcal{A}T_k$ for $k \geq 1$.
- (e) $\|\Psi(\cdot, u)\|_{\text{ch}} \leq \Lambda$ for every $u \in [-1, 1]$, and $\Lambda \geq 1$.
- (f) For $0 \leq k \leq s$,

$$\|\bar{\mathcal{A}}T_k - T_k\|_{\text{ch}} \leq \eta_k := \frac{N+1}{C_s} \left(\frac{k^2}{s^2} + \frac{3k^2}{2s} \right) + \sqrt{2(D+1)} \delta [k \geq 1], \quad (4.5)$$

where $[k \geq 1]$ is one if $k \geq 1$ and zero if $k = 0$.

The construction is due to Gribling, de Klerk and Vera [\[9\]](#), who use it for the dense ordinary module. We rederive its properties because the sparse argument needs them in a precise form: N

must be even for the SOS identity (4.7) below; SOS is claimed for $\Psi(\cdot, u)$, not for images $\mathcal{A}T_k$ of signed inputs, which can be negative (for $s = 2$, $\mathcal{A}T_1(x) = x p_{2,N}(x)$); and the source degree D is computed directly from the definitions.

The proof of (f) uses an exact formula for the defect of the unnormalized squared kernel.

Lemma 4.2 (Defect identity). *Let $\mathcal{K}q(x) = \int q(u)\Phi_s(x, u)^2 d\mu(u)$. For every $k \geq 0$,*

$$\mathcal{K}T_k - M_s T_k = -\bar{\chi}_k^2 T_k - \sum_{l=1}^{s-1} (\bar{\chi}_{l+k} - \bar{\chi}_l)^2 T_l T_{l+k} - \frac{1}{2} \sum_{j=1}^{k-1} (\bar{\chi}_{k-j} - \bar{\chi}_j)^2 T_j T_{k-j}. \quad (4.6)$$

Consequently, for $0 \leq k \leq s$, $\|\mathcal{K}T_k - M_s T_k\|_{\text{ch}} \leq k^2/s^2 + 3k^2/(2s)$.

Proof. For $m \geq 0$, orthogonality gives $\int T_m(u)\Phi_s(x, u) d\mu(u) = \chi_m T_m(x)$, with $\chi_0 = 1$ and $\chi_m = 0$ for $m \geq s$. By (2.2),

$$T_k(u)\Phi_s(x, u) = T_k(u) + \sum_{j \geq 1} \chi_j T_j(x) (T_{j+k}(u) + T_{|j-k|}(u)).$$

Multiplying by $\Phi_s(x, u)$ and integrating in u replaces each $T_m(u)$ by $\chi_m T_m(x)$:

$$\mathcal{K}T_k = \chi_k T_k + \sum_{j \geq 1} \chi_j T_j (\chi_{j+k} T_{j+k} + \chi_{|j-k|} T_{|j-k|}).$$

Also $M_s T_k = T_k + \sum_{j \geq 1} \chi_j^2 T_j (2T_j T_k) = T_k + \sum_{j \geq 1} \chi_j^2 T_j (T_{j+k} + T_{|j-k|})$. Subtracting,

$$\mathcal{K}T_k - M_s T_k = (\chi_k - 1)T_k + \sum_{j \geq 1} \chi_j (\chi_{j+k} - \chi_j) T_j T_{j+k} + \sum_{j \geq 1} \chi_j (\chi_{|j-k|} - \chi_j) T_j T_{|j-k|}.$$

For $k = 0$ every term vanishes, in agreement with (4.6). Let $k \geq 1$ and split the last sum. Its terms with $j = l + k$, $l \geq 1$, combine with the terms $j = l$ of the middle sum to $[\chi_l(\chi_{l+k} - \chi_l) + \chi_{l+k}(\chi_l - \chi_{l+k})]T_l T_{l+k} = -(\chi_l - \chi_{l+k})^2 T_l T_{l+k}$; these vanish for $l \geq s$. Its term $j = k$ is $\chi_k(1 - \chi_k)T_k$, which combines with $(\chi_k - 1)T_k$ to $-(1 - \chi_k)^2 T_k$. Its terms $1 \leq j \leq k - 1$ are $\chi_j(\chi_{k-j} - \chi_j)T_j T_{k-j}$; symmetrizing under $j \mapsto k - j$ gives $-\frac{1}{2}(\chi_j - \chi_{k-j})^2 T_j T_{k-j}$. Since $\chi_a - \chi_b = \bar{\chi}_b - \bar{\chi}_a$, this proves (4.6).

For the bound, every product $T_i T_j$ has Chebyshev norm one, and $|\bar{\chi}_a - \bar{\chi}_b| \leq |a - b|/s$. Hence the first term has norm at most k^2/s^2 , and the last at most $\frac{1}{2}(k-1)k^2/s^2 \leq k^2/(2s)$ because $k \leq s$. For the middle sum, $0 \leq \bar{\chi}_{l+k} - \bar{\chi}_l \leq k/s$ and $\sum_{l \geq 1} (\bar{\chi}_{l+k} - \bar{\chi}_l) = \sum_{i=1}^k (1 - \bar{\chi}_i) \leq k$ by telescoping, so its norm is at most $(k/s) \cdot k = k^2/s$. $\square$

Proof of Lemma 4.1. (a) With $x = \cos \theta$ and $u = \cos \phi$, $\Phi_s(x, u) = \frac{1}{2}(F_s(\theta - \phi) + F_s(\theta + \phi))$, where $F_s(t) = \sum_{|j| < s} (1 - |j|/s) e^{ijt} = \frac{1}{s} |\sum_{j=0}^{s-1} e^{ijt}|^2 \geq 0$ is the Fejér kernel. The formula for M_s follows from orthogonality, and $C_s = 1 + (2/s^2) \sum_{i=1}^{s-1} i^2 = (2s^2 + 1)/(3s)$. The composed kernel $R_s(a, c) = \int \Phi_s(a, u)\Phi_s(u, c) d\mu(u) = 1 + 2 \sum_j \chi_j^2 T_j(a)T_j(c)$ is nonnegative on $[-1, 1]^2$. Since $2T_j(x)^2 = 1 + T_{2j}(x) = 1 + T_j(T_2(x))$ and $T_j(1) = 1$,

$$M_s(x) = \frac{1}{2}(C_s + R_s(T_2(x), 1)).$$

For $x \in [-1, 1]$ also $T_2(x) \in [-1, 1]$, so $0 \leq R_s(T_2(x), 1) \leq C_s$. This gives $C_s/2 \leq M_s \leq C_s$ and $0 \leq z_s \leq \frac{1}{2}$.

(b) For even N ,

$$\sum_{j=0}^N z^j = \frac{1}{2} \left[1 + \sum_{j=0}^{N/2-1} (z^j(1+z))^2 + (z^{N/2})^2 \right], \quad (4.7)$$

as one checks by collecting the even and odd powers of z . Substituting $z = z_s(x)$, of degree at most $2(s-1)$, shows that $p_{s,N}$ is a sum of squares of polynomials of degree at most $(s-1)N$. Multiplying each of these by $\Phi_s(\cdot, u)$, of degree at most $s-1$, gives the claim for $\Psi(\cdot, u)$.

(c) $M_s = C_s(1 - z_s)$, so $p_{s,N}M_s = (1 - z_s) \sum_{j=0}^N z_s^j = 1 - z_s^{N+1}$. Both $p_{s,N}$ and M_s are globally SOS, hence so is their product ω . By (a), $0 \leq z_s^{N+1} \leq 2^{-(N+1)}$ on $[-1, 1]$.

(d) $\int \bar{\Psi}(x, u) d\mu(u) = \omega(x) + \varrho(x) = 1$. Moreover $\bar{\mathcal{A}}q = \mathcal{A}q + \varrho \int q d\mu$, and $\int T_k d\mu = 0$ for $k \geq 1$.

(e) For $|u| \leq 1$, $\|\Phi_s(\cdot, u)\|_{\text{ch}} \leq 1 + 2 \sum_{j=1}^{s-1} (1 - j/s) = s$. Expanding M_s with $2T_j^2 = 1 + T_{2j}$ gives $z_s = \frac{C_s-1}{2C_s} - \frac{1}{C_s} \sum_{j=1}^{s-1} \chi_j^2 T_{2j}$, so $\|z_s\|_{\text{ch}} = (C_s - 1)/C_s \leq 1$ and $\|p_{s,N}\|_{\text{ch}} \leq (N+1)/C_s$ by Lemma 2.3. Hence $\|\Psi(\cdot, u)\|_{\text{ch}} \leq (N+1)s^2/C_s = \Lambda$. Finally $\Lambda = 3s^3(N+1)/(2s^2+1) \geq 1$ for $s \geq 2$.

(f) For $k = 0$, (d) gives $\bar{\mathcal{A}}T_0 - T_0 = 0$. For $1 \leq k \leq s$, (d) and $\mathcal{A}T_k = p_{s,N}\mathcal{K}T_k$ give

$$\bar{\mathcal{A}}T_k - T_k = p_{s,N}(\mathcal{K}T_k - M_s T_k) + (p_{s,N}M_s - 1)T_k = p_{s,N}(\mathcal{K}T_k - M_s T_k) - \varrho T_k.$$

By Lemmas 2.3 and 4.2, the first term has norm at most $(N+1)C_s^{-1}(k^2/s^2 + 3k^2/(2s))$. For a univariate polynomial $q = \sum_{j=0}^D c_j T_j$, the Cauchy–Schwarz inequality and $\int q^2 d\mu = c_0^2 + \frac{1}{2} \sum_{j \geq 1} c_j^2$ give $\|q\|_{\text{ch}} \leq \sqrt{2D+1} (\int q^2 d\mu)^{1/2} \leq \sqrt{2(D+1)} \|q\|_{\infty}$. Applied to ϱ , with (c), the second term has norm at most $\sqrt{2(D+1)} \delta$. The two terms may separately have degree above D ; only their sum, which is $\bar{\mathcal{A}}T_k - T_k$ and has degree at most D because $\bar{\Psi}$ has x -degree at most D and $k \leq s \leq D$, is ever evaluated by a truncated functional. $\square$

Remark 4.3 (Exact normalization of a globally SOS kernel forces source independence). Let $K(x, u)$ be a polynomial such that $K(\cdot, u)$ is nonnegative on all of $\mathbb{R}$ for every $u \in [-1, 1]$ and $\int K(x, u) d\mu(u) = 1$ for all x . Then K does not depend on x . Indeed, write $K = \sum_{l=0}^m a_l(u) x^l$ with $a_m \not\equiv 0$ and suppose $m \geq 1$. Nonnegativity on $\mathbb{R}$ forces m even and $a_m(u) \geq 0$ for $u \in [-1, 1]$, while normalization forces $\int a_m d\mu = 0$. Since μ has full support and a_m is continuous, $a_m = 0$ on $[-1, 1]$, hence identically, a contradiction. Thus a polynomial kernel that depends nontrivially on its input and is globally SOS there cannot have mass identically one. The constant kernel $K = 1$ is normalized and SOS, but does not approximate nonconstant functions. This obstruction concerns globally nonnegative source kernels; it does not rule out exactly normalized kernels certified by an interval quadratic module. The observation is elementary and not claimed as new; it explains why this SOS construction needs a correction.

4.2 Signed densities with exact separator consistency

From now on let $s \geq \max\{2, d_{\infty}\}$, $N \geq 2$ even, and $r \geq wD$. Then every local polynomial has total degree $\deg f_b \leq v_b d_{\infty} \leq v_b s \leq v_b D \leq r$. For a bag B and $u \in [-1, 1]^B$ put

$$\bar{\Psi}_B(x, u) = \prod_{i \in B} \bar{\Psi}(x_i, u_i), \quad \bar{\mathcal{A}}_B = \bigotimes_{i \in B} \bar{\mathcal{A}},$$

so that $\bar{\mathcal{A}}_B q(x) = \int q(u) \bar{\Psi}_B(x, u) d\mu^B(u)$. As a polynomial in x , $\bar{\Psi}_B(\cdot, u)$ has total degree at most $|B|D$.

Lemma 4.4 (Signed densities). *Let (L_b) be a feasible ordinary-module family of order $r \geq wD$, and define*

$$\bar{h}_b(u) = L_b(\bar{\Psi}_{B_b}(\cdot, u)), \quad \bar{h}_S^{(b)}(u_S) = L_b(\bar{\Psi}_S(\cdot, u_S)) \quad (S \subseteq B_b),$$

with $\bar{h}_{\emptyset}^{(b)} = 1$. Then:

(a) $\int \bar{h}_b d\mu^{B_b \setminus S} = \bar{h}_S^{(b)}$ for every $S \subseteq B_b$; in particular $\int \bar{h}_b d\mu^{B_b} = 1$;

- (b) $\bar{h}_{S_e}^{(b)} = \bar{h}_{S_e}^{(c)}$ for every edge $e = bc$;
(c) $|\int f_b \bar{h}_b d\mu^{B_b} - L_b(f_b)| \leq C_b \Gamma_{v_b}$, where $\Gamma_v = (1 + \eta)^v - 1$ and

$$\eta = \frac{N+1}{C_s} \left(\frac{d_\infty^2}{s^2} + \frac{3d_\infty^2}{2s} \right) + \sqrt{2(D+1)} \delta. \quad (4.8)$$

Proof. (a) and (b) follow exactly as in Lemma 3.2(b),(c), using Lemma 4.1(d) and the fact that $\bar{\Psi}_{S_e}(\cdot, u)$ has degree at most $|S_e|D \leq r \leq 2r$ in x .

(c) Write $f_b = c_{b,0} + g_b$ with $g_b = \sum_{\alpha \neq 0} c_{b,\alpha} T_\alpha$. By linearity, $\int f_b \bar{h}_b d\mu^{B_b} = L_b(\bar{\mathcal{A}}_{B_b} f_b)$, and $\bar{\mathcal{A}}_{B_b} 1 = 1$. Since $L_b(1) = 1$,

$$\int f_b \bar{h}_b d\mu^{B_b} - L_b(f_b) = L_b(\bar{\mathcal{A}}_{B_b} g_b - g_b).$$

Fix α with $\alpha_i \leq d_\infty$ and put $a_i = \bar{\mathcal{A}}_{B_b} T_\alpha(x_i)$, $t_i = T_\alpha(x_i)$. Then $\bar{\mathcal{A}}_{B_b} T_\alpha - T_\alpha = \prod_i a_i - \prod_i t_i = \sum_i (\prod_{j < i} a_j)(a_i - t_i)(\prod_{j > i} t_j)$. The Chebyshev norm is multiplicative for polynomials in distinct variables, $\|t_j\|_{\text{ch}} = 1$, and $\|a_j\|_{\text{ch}} \leq 1 + \eta_{\alpha_j}$ by (4.5). Hence $\|\bar{\mathcal{A}}_{B_b} T_\alpha - T_\alpha\|_{\text{ch}} \leq \prod_i (1 + \eta_{\alpha_i}) - 1 \leq \Gamma_{v_b}$, since $\eta_k \leq \eta$ for $k \leq d_\infty$. Summing, $\|\bar{\mathcal{A}}_{B_b} g_b - g_b\|_{\text{ch}} \leq C_b \Gamma_{v_b}$. This polynomial has total degree at most $v_b D \leq r$, so its tensor Chebyshev expansion involves only T_β with $|\beta| \leq r$, and Lemma 2.8 gives $|L_b(\bar{\mathcal{A}}_{B_b} g_b - g_b)| \leq C_b \Gamma_{v_b}$. $\square$

4.3 Products of residuals

The density $\bar{h}_b$ may be negative. Expanding the product defining $\bar{\Psi}_{B_b}$ by the set $J \subseteq B_b$ of coordinates where the residual is chosen,

$$\bar{h}_b(u) = \sum_{J \subseteq B_b} L_b(G_J(\cdot, u) H_J), \quad G_J(x, u) = \prod_{i \in B_b \setminus J} \Psi(x_i, u_i), \quad H_J(x) = \prod_{i \in J} \varrho(x_i). \quad (4.9)$$

Lemma 4.5 (Residual products). *Let (L_b) be a feasible ordinary-module family of order $r \geq wD$, fix a bag b with $v = v_b$, a point $u \in [-1, 1]^{B_b}$, and $J \subseteq B_b$ with $j = |J|$. Write $G = G_J(\cdot, u)$ and $H = H_J$.*

- (a) *If $j \leq 1$, then $L_b(GH) \geq 0$.*
(b) *$0 \leq L_b(G) \leq \Lambda^{v-j}$.*
(c) *$0 \leq L_b(GH^2) \leq \delta^{2j} L_b(G)$.*
(d) *$|L_b(GH)| \leq \delta^j L_b(G) \leq \delta^j \Lambda^{v-j}$.*

Proof. By Lemma 4.1(b), $G = \sum_l q_l^2$ with $\deg q_l \leq (v-j)D/2$. All degrees below are collected in Table 2; each is at most r or $2r$ as required because $(v+j)D \leq 2vD \leq 2wD \leq 2r$.

(a) For $j = 0$, $GH = G \in \Sigma_{2r}$ since $vD \leq 2r$. For $J = \{i\}$, the polynomial ϱ is nonnegative on $[-1, 1]$ and has degree at most D , which is even. Lemma 2.11 gives $\varrho = \sigma_0 + (1 - x_i^2)\sigma_1$ with $\sigma_0 \in \Sigma_D$, $\sigma_1 \in \Sigma_{D-2}$. Then $GH = G\sigma_0 + (1 - x_i^2)G\sigma_1$ with $G\sigma_0 \in \Sigma_{vD}$ and $G\sigma_1 \in \Sigma_{vD-2}$. Since $vD \leq 2r$, this lies in $M_r(B_b)$.

(b) $L_b(G) \geq 0$ by its SOS representation at the start of the proof. The polynomial G has degree at most $(v-j)D \leq r$, and by Lemma 4.1(e) and multiplicativity of the Chebyshev norm over distinct variables, $\|G\|_{\text{ch}} \leq \Lambda^{v-j}$. Lemma 2.8 gives $L_b(G) \leq \Lambda^{v-j}$.

(c) Order $J = \{i_1, \dots, i_j\}$ and write $\varrho_\ell = \varrho(x_{i_\ell})$. Then

$$\delta^{2j} - H^2 = \sum_{\ell=1}^j \delta^{2(j-\ell)} (\delta^2 - \varrho_\ell^2) \prod_{\ell' < \ell} \varrho_{\ell'}^2.$$

Table 2: Degree ledger for [Lemmas 4.4](#) and [4.5](#) in a bag of size v , with j residual factors. All bounds use $r \geq wD$.

Polynomial in x	Total degree at most	Use
$\Psi(\cdot, u), \bar{\Psi}(\cdot, u), \omega, \varrho$	D	kernel input
$\bar{\Psi}_B(\cdot, u), \bar{\mathcal{A}}_B f_b, f_b$	$vD \leq r$	evaluation via Lemma 2.8
$\bar{\Psi}_S(\cdot, u_S)$ for a separator	$ S D \leq r$	exact separator consistency
G	$(v-j)D \leq r$	SOS and coefficient bound
GH	$vD \leq r$	term of (4.9)
single-residual certificate of GH	$vD \leq 2r$	ordinary-module positivity
GH^2 ; each term of $G(\delta^{2j} - H^2)$	$(v+j)D \leq 2r$	residual bound
square roots $q_l H, q_l(a + cH)$	$(v+j)D/2 \leq r$	weighted Cauchy–Schwarz

Each $\delta^2 - \varrho_\ell^2$ is nonnegative on $[-1, 1]$ and has degree at most $2D$, so [Lemma 2.11](#) gives $\delta^2 - \varrho_\ell^2 = \sigma_{0,\ell} + (1 - x_{i_\ell}^2)\sigma_{1,\ell}$ with $\sigma_{0,\ell} \in \Sigma_{2D}$ and $\sigma_{1,\ell} \in \Sigma_{2D-2}$. Multiplying the ℓ -th term by the sum of squares $G \prod_{\ell' < \ell} \varrho_{\ell'}^2$, of degree at most $(v-j)D + 2(\ell-1)D$, gives a sum of squares of degree at most $(v-j+2\ell)D \leq (v+j)D \leq 2r$ plus $(1 - x_{i_\ell}^2)$ times a sum of squares of degree at most $(v+j)D - 2 \leq 2r - 2$. Only one box generator occurs in each term. Hence $G(\delta^{2j} - H^2) \in M_r(B_b)$ and $L_b(GH^2) \leq \delta^{2j} L_b(G)$. Also $GH^2 = \sum_l (q_l H)^2$ with $\deg(q_l H) \leq (v-j)D/2 + jD = (v+j)D/2 \leq r$, so $L_b(GH^2) \geq 0$.

(d) For real a, c , $G(a + cH)^2 = \sum_l (q_l(a + cH))^2$ is a sum of squares of polynomials of degree at most $(v+j)D/2 \leq r$, so $L_b(G(a + cH)^2) \geq 0$. [Lemma 2.7](#) with weight G and $V = \text{span}\{1, H\}$ gives $L_b(GH)^2 \leq L_b(G)L_b(GH^2) \leq \delta^{2j} L_b(G)^2$ by (c). This includes the case $L_b(G) = 0$; no division is used. Part (b) completes the proof. $\square$

4.4 The finite-order theorem

For $v \geq 0$ let

$$\Delta_v = \sum_{j=2}^v \binom{v}{j} \delta^j \Lambda^{v-j}, \quad (4.10)$$

with $\Delta_0 = \Delta_1 = 0$ (empty sums), and let $W = \sum_b \text{osc}(f_b) \leq 2C_f$.

Theorem 4.6 (Rounding for the sparse ordinary module). *Let $s \geq \max\{2, d_\infty\}$ and $N \geq 2$ be integers with N even, and define $C_s, D, \delta, \Lambda, \eta, \Gamma_v, \Delta_w$ by [\(4.2\)](#), [\(4.4\)](#), [\(4.8\)](#) and [\(4.10\)](#). Let $r \geq wD$. For every feasible ordinary-module family (L_b) of order r there is a Borel probability measure ν on $[-1, 1]^n$ whose marginal on $[-1, 1]^{B_b}$ has the polynomial density*

$$h_b^+ = \frac{\bar{h}_b + \Delta_w}{1 + \Delta_w} \quad (4.11)$$

with respect to μ^{B_b} , for every bag, and

$$\left| \int f d\nu - \sum_b L_b(f_b) \right| \leq \sum_b C_b \Gamma_{v_b} + \Delta_w W \leq C_f (\Gamma_w + 2\Delta_w). \quad (4.12)$$

Consequently $0 \leq f^* - \rho_r^{\text{mod}} \leq \sum_b C_b \Gamma_{v_b} + \Delta_w W$.

Proof. Lower bound on the signed densities. Fix b and $u \in [-1, 1]^{B_b}$, and use (4.9). By Lemma 4.5(a) the terms with $|J| \leq 1$ are nonnegative, and by part (d) the terms with $|J| = j \geq 2$ are at least $-\delta^j \Lambda^{v_b-j}$. Hence $\bar{h}_b(u) \geq -\Delta_{v_b}$. Since $\Lambda \geq 1$, the sequence Δ_v is nondecreasing in v ; explicitly, for $v \geq 1$, $\Delta_{v+1} = (\Lambda + \delta)\Delta_v + v\Lambda^{v-1}\delta^2 \geq \Delta_v$, and $\Delta_0 = \Delta_1 = 0$. So $\bar{h}_b \geq -\Delta_w$ on $[-1, 1]^{B_b}$.

Corrected laws. By the lower bound and Lemma 4.4(a), h_b^+ is nonnegative with integral one. Its marginal on $S \subseteq B_b$ is $(\bar{h}_S^{(b)} + \Delta_w)/(1 + \Delta_w)$, because the constant Δ_w integrates to Δ_w over the removed coordinates. The same scalar Δ_w is used in every bag, so Lemma 4.4(b) gives equal marginals on every separator, including empty ones and bags of different sizes. For an empty bag, $\bar{h}_b = 1$ and $h_b^+ = 1$, so its constant objective is preserved exactly. Lemma 2.12 (or Remark 3.5) gives a probability measure ν on $[-1, 1]^n$ with marginals $h_b^+ d\mu^{B_b}$. A bagwise choice of different shifts would in general destroy the equality of separator marginals.

Objective. Write $I_b = \int f_b d\mu^{B_b}$, $\bar{I}_b = \int f_b \bar{h}_b d\mu^{B_b}$ and $I_b^+ = \int f_b h_b^+ d\mu^{B_b} = (\bar{I}_b + \Delta_w I_b)/(1 + \Delta_w)$. Then

$$I_b^+ - \bar{I}_b = \Delta_w(I_b - I_b^+).$$

Both I_b and I_b^+ are expectations of f_b under probability measures on the bag box, so $|I_b - I_b^+| \leq \text{osc}(f_b)$. No probability inequality is applied to the signed density $\bar{h}_b$. With Lemma 4.4(c),

$$\left| \int f d\nu - \sum_b L_b(f_b) \right| \leq \sum_b (|I_b^+ - \bar{I}_b| + |\bar{I}_b - L_b(f_b)|) \leq \sum_b (\Delta_w \text{osc}(f_b) + C_b \Gamma_{v_b}).$$

The second inequality in (4.12) uses $\Gamma_{v_b} \leq \Gamma_w$ and $W \leq 2C_f$. The gap bound follows as in Theorem 3.4: $f^* \leq \int f d\nu$ for every feasible family, and Lemma 2.6 gives $\rho_r^{\text{mod}} \leq f^*$. $\square$

Additive constants in the f_b cancel in every step, and no reciprocal of a possibly small mass appears. The bound has no term that grows with the number of edges or with the diameter of the tree: the positivity correction couples bags only through the common scalar Δ_w .

4.5 The rate and the certificate

The two error terms in Theorem 4.6 pull the parameters in different directions. The smoothing term $\Gamma_w \approx w\eta$ is of order N/s^2 , while the correction Δ_w is of order $\delta^2 \Lambda^{w-2}$ with Λ of order sN . A series length N of order $\log s$, with a width-dependent constant, makes the correction negligible; the degree constraint $wD \leq r$ then allows s of order $r/(w \log r)$.

Corollary 4.7 (Rate for the sparse ordinary module). *Let $c_w = \max\{3, (w+1)/2\}$ and $\ell_r = \log_2(r+2)$. Suppose*

$$r \geq 2w(c_w + 3) \ell_r \max\{2, d_\infty\} \quad \text{and} \quad 16w^3(3d_\infty^2 + 1)(c_w + 3)^3 \ell_r^3 \leq r^2. \quad (4.13)$$

Then

$$0 \leq f^* - \rho_r^{\text{mod}} \leq C_f \left[\frac{16e w^3(3d_\infty^2 + 1)(c_w + 3)^3 \ell_r^3}{r^2} + \binom{w}{2} \frac{2^{w+3} w^3 (c_w + 3)^{w+1} \ell_r^{w+1}}{r^3} \right]. \quad (4.14)$$

In particular $f^ - \rho_r^{\text{mod}} \leq C_f O_{w, d_\infty}(\log^3 r / r^2)$ for all sufficiently large r , where the implicit constant and the threshold depend only on w and d_∞ . If $C_f = 0$, every f_b is constant and the gap is zero.*

Proof. Let $s = \lfloor r/(2w(c_w + 3)\ell_r) \rfloor$ and $N = 2\lceil (c_w/2) \log_2 s \rceil$. The first condition in (4.13) gives $s \geq \max\{2, d_\infty\}$. Then N is even, $N \geq 2$, and $c_w \log_2 s \leq N < c_w \log_2 s + 2$. We check the hypotheses of Theorem 4.6 and bound its right-hand side.

Degree. $N + 1 \leq c_w \log_2 s + 3 \leq (c_w + 3)\ell_r$, because $\log_2 s \leq \ell_r$ and $\ell_r \geq 1$. Hence $wD \leq 2ws(N + 1) \leq 2w(c_w + 3)\ell_r s \leq r$.

Smoothing term. Since $N + 1 \geq c_w \log_2 s + 1$, we have $\delta \leq \frac{1}{2}s^{-c_w} \leq \frac{1}{2}s^{-3}$. Using $C_s \geq 2s/3$ and $s \geq 2$,

$$\frac{N + 1}{C_s} \left(\frac{d_\infty^2}{s^2} + \frac{3d_\infty^2}{2s} \right) \leq (N + 1)d_\infty^2 \left(\frac{3}{2s^3} + \frac{9}{4s^2} \right) \leq \frac{3d_\infty^2(N + 1)}{s^2}.$$

Also $D + 1 \leq 2s(N + 1)$, so $\sqrt{2(D + 1)}\delta \leq 2\sqrt{s(N + 1)} \cdot \frac{1}{2s^3} = \sqrt{N + 1}s^{-5/2} \leq (N + 1)/s^2$. Thus $\eta \leq (3d_\infty^2 + 1)(N + 1)/s^2$. Since $r/(2w(c_w + 3)\ell_r) \geq 1$, the floor satisfies $s \geq r/(4w(c_w + 3)\ell_r)$, so

$$\frac{N + 1}{s^2} \leq (c_w + 3)\ell_r \cdot \frac{16w^2(c_w + 3)^2\ell_r^2}{r^2} = \frac{16w^2(c_w + 3)^3\ell_r^3}{r^2}.$$

By the second condition in (4.13), $w\eta \leq 1$, and then $\Gamma_w = (1 + \eta)^w - 1 \leq e^{w\eta} - 1 \leq w\eta e^{w\eta} \leq e w\eta$. This gives the first term of (4.14).

Correction term. For $w = 1$, $\Delta_w = 0$. For $w \geq 2$, Taylor's formula for $F(t) = (\Lambda + t)^w$ gives $\Delta_w = F(\delta) - F(0) - \delta F'(0) \leq \binom{w}{2}\delta^2(\Lambda + \delta)^{w-2}$. Moreover $\Lambda \leq 3s(N + 1)/2$, so $\Lambda + \delta \leq 2s(N + 1)$, and $\delta^2 \leq \frac{1}{4}s^{-2c_w}$ with $2c_w \geq \max\{6, w + 1\}$. Hence

$$\Delta_w \leq \binom{w}{2} \frac{2^{w-2}(N + 1)^{w-2}s^{w-2}}{4s^{2c_w}} \leq \binom{w}{2} 2^{w-4} \frac{(N + 1)^{w-2}}{s^3},$$

because $w - 2 - 2c_w \leq -3$. With $N + 1 \leq (c_w + 3)\ell_r$ and $s^{-3} \leq 64w^3(c_w + 3)^3\ell_r^3/r^3$, the term $2\Delta_w C_f$ is at most the second term of (4.14). Inserting both bounds into $C_f(\Gamma_w + 2\Delta_w)$ proves (4.14). The second term is $O_w(\ell_r^{w+1}/r^3)$, which is $o(\ell_r^3/r^2)$ for fixed w . $\square$

The corollary states the dependence on the problem only through C_f , w and d_∞ . For bounded width and bounded coordinate degree, the order needed to reach a prescribed error per unit of C_f is independent of the number of bags and of the tree. For $w \leq 5$ the series length is $N = 2\lceil \frac{3}{2} \log_2 s \rceil$; larger widths need a longer series so that Δ_w stays negligible. The constants and the threshold depend strongly on w , and the corollary makes no statement for growing width.

Remark 4.8 (Size of the constants). The finite bound $C_f(\Gamma_w + 2\Delta_w)$ in (4.12) is conservative. For $d_\infty = 2$, the first three rows below use the prescription for N in the proof of Corollary 4.7 and the largest admissible s with $wD \leq r$ under that prescription. The final row uses a different admissible pair (s, N) . The displayed values of $\Gamma_w + 2\Delta_w$ are rounded evaluations of the worst-case formula, not observed relaxation gaps.

w	r	s	N	$\Gamma_w + 2\Delta_w$
2	10,000	109	22	0.0354
5	10,000	53	18	0.600
10	100,000	122	40	1.87×10^8
10	100,000	82	60	1.21

The last two rows show that the prescription in Corollary 4.7 is chosen for the asymptotic statement and is far from optimal at finite order: for $w = 10$ the admissible choice $s = 82$, $N = 60$ reduces the bound by eight orders of magnitude. The orders at which the bound becomes small are far beyond the orders used in practice; the theorem is a statement about the rate, not a practical accuracy guarantee.

Corollary 4.9 (Sparse ordinary-module certificate). *Under the hypotheses of [Theorem 4.6](#), with $E = \sum_b C_b \Gamma_{v_b} + \Delta_w W$,*

$$f - f^* + E \in \sum_{b=1}^t M_r(B_b),$$

with real coefficients and every summand of total degree at most $2r$. Under the hypotheses of [Corollary 4.7](#), the same holds with E the right-hand side of [\(4.14\)](#).

Proof. By [Lemma 2.10](#), $f - \rho_r^{\text{mod}} \in \sum_b M_r(B_b)$, and $\rho_r^{\text{mod}} - f^* + E \geq 0$ by [Theorem 4.6](#); add this constant to the square part of one bag. For the second statement, the proof of [Corollary 4.7](#) shows that E is at most the right-hand side of [\(4.14\)](#) for the parameters chosen there. $\square$

4.6 Relation to prior work and scope

Remark 4.10 (What is prior in [Theorem 4.6](#)). The squared Fejér kernel, its geometric-series SOS normalization, the coefficient-norm method, and the dense rate $O(\log^3 r / r^2)$ for the ordinary box module are due to Gribbling, de Klerk and Vera [\[9\]](#); their theorem concerns the dense hierarchy. The earlier dense ordinary-module rate was $O(1/r)$ [\[1\]](#). Korda, Magron and Ríos-Zertuche [\[17\]](#), [Theorem 8](#), treat sparse ordinary modules on general semialgebraic domains under running intersection, normalized local Archimedean certificates and local Łojasiewicz inequalities; for the box their rate exponent is a small negative power of r that decreases with w . Their theorem covers broader domains than ours. Gamertsfelder and Mourrain [\[8\]](#) give a general transfer of positivity rates to generalized moment problems, which include separator equalities, under the assumption that a dual with finitely supported multipliers is attained; [Theorem 4.6](#) needs no such assumption, and its finite SDP duality ([Lemma 2.10](#)) is a different statement. Magron’s lecture slides [\[25\]](#) mention a sparse Putinar rate without stating its exponent.

To our knowledge the following are new: the exactly normalized signed kernel together with the residual-product bound of [Lemma 4.5](#), which uses only ordinary-module positivity, and the resulting transfer of the dense rate to the sparse ordinary module for arbitrary feasible truncated functionals, with an error normalized by C_f and no dependence on the number of bags or the tree. The common affine correction in the last step is elementary.

The theorem concerns minimization over the box. The corrected laws are supported on the box but need not satisfy additional constraints even if the relaxation contains their localizing matrices; constrained problems are treated in [Section 8](#) under additional hypotheses. We do not know whether the logarithmic factor in [Corollary 4.7](#) is necessary: the example of [Section 5](#) gives a lower bound of order r^{-2} for the ordinary module, which leaves a gap of a factor $\log^3 r$.

5 Separator information is the obstruction

The upper bounds of [Sections 3](#) and [4](#) hold however the local functionals are chosen. This section shows that, for the full preordering, the inverse-square rate cannot be improved in general, even for a quadratic objective in three variables, and that the obstruction does not come from local positivity. We exhibit a fixed problem with two bags of size two whose sparse gap is of exact order r^{-2} ([Theorem 5.4](#)). The lower bound is witnessed by *actual* local probability measures whose separator moments agree through degree $2r$. It therefore applies to every relaxation whose feasible set contains these integration functionals and whose only coupling between bags is agreement of these separator moments. The dense preordering on the same problem is exact at order two. For the first two orders we compute the sparse values exactly ([Proposition 5.5](#)).

Minimizing each bag over its private coordinate leaves the two separator functions $-\varphi(y)$ and $\varphi(y)$, where $\varphi(y) = \max\{y, 0\}^2$. Two separator laws with equal moments through degree n cannot tell φ apart from its best polynomial approximation of degree n , and the best approximation error of φ decays exactly like n^{-2} . The example shows that finite separator information can limit the rate even with exact local measures; it does not determine the rate for every sparse problem.

5.1 Moment matching and uniform approximation

For a compact interval I , a function $\psi \in C(I)$ and $n \geq 0$, let $E_n(\psi; I) = \min\{\|\psi - q\|_{\infty, I} : q \in \mathbb{R}[y], \deg q \leq n\}$, and write $E_n(\psi) = E_n(\psi; [-1, 1])$.

Lemma 5.1 (Moment matching). *Let I be a compact interval, $\psi \in C(I)$, and $n \geq 0$. Then*

$$\max\left\{\int \psi d\mu_1 - \int \psi d\mu_2\right\} = 2E_n(\psi; I),$$

where the maximum is over pairs of Borel probability measures μ_1, μ_2 on I with $\int y^k d\mu_1 = \int y^k d\mu_2$ for $k = 0, \dots, n$, and it is attained.

Proof. For any such pair and any polynomial q of degree at most n , $\int \psi d\mu_1 - \int \psi d\mu_2 = \int (\psi - q) d\mu_1 - \int (\psi - q) d\mu_2 \leq 2\|\psi - q\|_{\infty}$. Minimizing over q gives the upper bound.

If $E_n(\psi; I) = 0$, take $\mu_1 = \mu_2$. Otherwise, the polynomials of degree at most n form a finite-dimensional, hence closed, subspace P_n of $C(I)$ at distance $E_n(\psi; I)$ from ψ . By the Hahn–Banach theorem there is a linear functional ℓ on $C(I)$ of norm one that vanishes on P_n and satisfies $\ell(\psi) = E_n(\psi; I)$. By the Riesz representation theorem [34], $\ell(g) = \int g d\lambda$ for a finite signed Borel measure λ of total variation one. Since $\ell(1) = 0$, the Jordan parts $\lambda_{\pm}$ have equal mass $\frac{1}{2}$. The probability measures $\mu_1 = 2\lambda_+$ and $\mu_2 = 2\lambda_-$ satisfy $\int y^k d\mu_1 - \int y^k d\mu_2 = 2\ell(y^k) = 0$ for $k \leq n$, and $\int \psi d\mu_1 - \int \psi d\mu_2 = 2\ell(\psi) = 2E_n(\psi; I)$. $\square$

This identity is known; it is stated explicitly, for example, by Han, Jiao and Weissman [12], Lemma 25, and we include it for completeness. The lower bounds below do not need it: they use the explicit witness of Lemma 5.3.

5.2 The example

Consider

$$f^* = \min\{f(x, y, z) : x, z \in [0, 1], y \in [-1, 1]\}, \quad f = \underbrace{x^2 - 2xy}_{f_1(x, y)} + \underbrace{y^2 + z^2 + 2yz}_{f_2(y, z)}, \quad (5.1)$$

with bags $B_1 = \{x, y\}$ and $B_2 = \{y, z\}$ and separator $\{y\}$. The affine change $x = (1 + \hat{x})/2$, $z = (1 + \hat{z})/2$ maps the problem to $[-1, 1]^3$, keeps both bags and all total degrees, and maps the generators $1 - \hat{x}^2$, $1 - \hat{z}^2$ to $4g_x$, $4g_z$, where

$$g_x = x(1 - x), \quad g_y = 1 - y^2, \quad g_z = z(1 - z).$$

Thus the sparse hierarchies of Definition 2.5 for the transformed problem are, in the original coordinates, the hierarchies whose cones are generated by g_x, g_y on B_1 and by g_y, g_z on B_2 , with every certificate summand of total degree at most $2r$. We write ρ_r^{mod} and ρ_r^{pre} for their moment values on this instance.

Since $f = (y - x + z)^2 + 2xz$, we have $f \geq 0$ on the box and $f^* = f(0, 0, 0) = 0$. Let $\varphi(y) = \max\{y, 0\}^2$. For every $y \in [-1, 1]$,

$$\min_{x \in [0, 1]} f_1(x, y) = -\varphi(y), \quad \min_{z \in [0, 1]} f_2(y, z) = \varphi(y), \quad (5.2)$$

attained at $x = \max\{y, 0\}$ and $z = \max\{-y, 0\}$. Indeed $f_1 = (x - y)^2 - y^2$ and $f_2 = (y + z)^2$, and these choices lie in $[0, 1]$.

For $n \geq 0$, let v_n be the infimum of $\int f_1 d\pi_1 + \int f_2 d\pi_2$ over Borel probability measures π_1 on $[0, 1] \times [-1, 1]$ and π_2 on $[-1, 1] \times [0, 1]$ whose y -marginals have equal moments of degrees $0, \dots, n$. This is the *local-measure relaxation*: each bag is an actual measure, and only the separator moments are shared.

Proposition 5.2 (Local-measure value). *For every $n \geq 0$, $v_n = -2E_n(\varphi)$, and the infimum is attained. Moreover, $\rho_r^{\text{mod}} \leq \rho_r^{\text{pre}} \leq v_{2r}$ for every $r \geq 1$.*

Proof. Let μ_1, μ_2 be the y -marginals of a feasible pair. By (5.2), $\int f_1 d\pi_1 + \int f_2 d\pi_2 \geq -(\int \varphi d\mu_1 - \int \varphi d\mu_2) \geq -2E_n(\varphi)$ by Lemma 5.1. Conversely, let (μ_1, μ_2) attain the maximum in Lemma 5.1, let π_1 be the law of $(\max\{Y, 0\}, Y)$ for $Y \sim \mu_1$, and let π_2 be the law of $(Y, \max\{-Y, 0\})$ for $Y \sim \mu_2$. By (5.2) this pair has objective $-2E_n(\varphi)$.

For the second claim, integration against a feasible pair with $n = 2r$ is a feasible preordering family of order r : each local functional is integration against a measure on its rectangle, hence nonnegative on every square times a product of generators, and separator moments agree through degree $2r$. Lemma 2.6 gives $\rho_r^{\text{mod}} \leq \rho_r^{\text{pre}}$. $\square$

5.3 An explicit approximation lower bound

Lemma 5.3 (Lower bound for φ). *For every $n \geq 0$ there are Borel probability measures μ_+, μ_- on $[-1, 1]$ with equal moments of degrees $0, \dots, n$ and*

$$\int \varphi d\mu_+ - \int \varphi d\mu_- \geq \frac{2}{27\pi(n+2)^2}. \quad (5.3)$$

Consequently $E_n(\varphi) \geq 1/(27\pi(n+2)^2)$.

Proof. Let N be the smallest even integer with $N \geq n+1$, so $2 \leq N \leq n+2$. For $N < k < 3N$ let $w_k = 1 - |k - 2N|/N$, and for odd k let $\epsilon_k = -\sin(k\pi/2) \in \{\pm 1\}$. Define the polynomial

$$q_N(y) = \sum_{\substack{N < k < 3N \\ k \text{ odd}}} \epsilon_k w_k T_k(y)$$

and the signed measure $d\lambda = q_N d\mu$ on $[-1, 1]$.

Annihilation. Every T_k in q_N has $k \geq N+1 > n$, so $\int y^j d\lambda = 0$ for $j \leq n$ by orthogonality; in particular $\lambda([-1, 1]) = 0$.

Total variation. Let $F_N(t) = \sum_{|j| < N} (1 - |j|/N) e^{ijt} \geq 0$ be the Fejér kernel, which has integral one against $dt/(2\pi)$, and let $Q_N(\theta) = \sin(2N(\theta - \frac{\pi}{2})) F_N(\theta - \frac{\pi}{2})$. Using $2\sin(2Nt) \cos(jt) = \sin((2N+j)t) + \sin((2N-j)t)$,

$$Q_N(\theta) = \sum_{k=N+1}^{3N-1} w_k \sin(k(\theta - \frac{\pi}{2})) = \sum_k w_k [\cos(\frac{k\pi}{2}) \sin(k\theta) - \sin(\frac{k\pi}{2}) \cos(k\theta)].$$

The even part $\frac{1}{2}(Q_N(\theta) + Q_N(-\theta))$ is $\sum_{k \text{ odd}} \epsilon_k w_k \cos(k\theta) = q_N(\cos \theta)$. Since $\cos \theta$ pushes $d\theta/(2\pi)$ on the circle forward to μ ,

$$\int |q_N| d\mu = \int_{-\pi}^{\pi} |q_N(\cos \theta)| \frac{d\theta}{2\pi} \leq \int_{-\pi}^{\pi} |Q_N(\theta)| \frac{d\theta}{2\pi} \leq \int_{-\pi}^{\pi} F_N\left(\theta - \frac{\pi}{2}\right) \frac{d\theta}{2\pi} = 1.$$

Pairing with φ . For odd k ,

$$\int \varphi T_k d\mu = \frac{1}{\pi} \int_0^{\pi/2} \cos^2 \theta \cos(k\theta) d\theta = \frac{\sin(k\pi/2)}{2\pi} \left(\frac{1}{k} - \frac{k}{k^2 - 4} \right) = -\frac{2 \sin(k\pi/2)}{\pi k(k^2 - 4)},$$

using $\cos^2 \theta = \frac{1}{2}(1 + \cos 2\theta)$ and $\sin((k \pm 2)\pi/2) = -\sin(k\pi/2)$. Since $\sin^2(k\pi/2) = 1$ for odd k ,

$$\int \varphi d\lambda = \frac{2}{\pi} \sum_{\substack{N < k < 3N \\ k \text{ odd}}} \frac{w_k}{k(k^2 - 4)}.$$

Every k in the sum satisfies $3 \leq k < 3N$, so $0 < k(k^2 - 4) < 27N^3$. The odd k are $k = 2N \pm j$ with j odd and $1 \leq j \leq N - 1$; since N is even there are $N/2$ such j , with sum $N^2/4$, so $\sum_{k \text{ odd}} w_k = 2 \sum_j (1 - j/N) = N/2$. Hence $\int \varphi d\lambda \geq (2/\pi)(N/2)/(27N^3) = 1/(27\pi N^2)$.

Measures. Let $\lambda = \lambda_+ - \lambda_-$ be the Jordan decomposition. Since $\lambda([-1, 1]) = 0$ and $|\lambda|([-1, 1]) \leq 1$, both parts have the same mass $t \in (0, \frac{1}{2}]$; $t > 0$ because $\int \varphi d\lambda > 0$. Put $\mu_{\pm} = \lambda_{\pm}/t$. These are probability measures with equal moments through degree n , and $\int \varphi d\mu_+ - \int \varphi d\mu_- = t^{-1} \int \varphi d\lambda \geq 2/(27\pi N^2) \geq 2/(27\pi(n+2)^2)$. The bound on $E_n(\varphi)$ follows from the elementary half of [Lemma 5.1](#). $\square$

5.4 The sharp rate

Theorem 5.4 (A fixed quadratic with sparse gap of exact order r^{-2}). *For problem (5.1), with $f^* = 0$:*

- (a) (Lower bound by local measures.) *For every $r \geq 1$ there are probability measures π_1 on $[0, 1] \times [-1, 1]$ and π_2 on $[-1, 1] \times [0, 1]$ whose y -marginals have equal moments through degree $2r$ and*

$$\int f_1 d\pi_1 + \int f_2 d\pi_2 \leq -\frac{1}{54\pi(r+1)^2}.$$

Consequently $\rho_r^{\text{mod}} \leq \rho_r^{\text{pre}} \leq v_{2r} \leq -1/(54\pi(r+1)^2)$. The same bound holds for every relaxation whose feasible set contains the integration functionals of such pairs.

- (b) (Upper bound, preordering.) *For every $r \geq 2$,*

$$\rho_r^{\text{pre}} \geq -\frac{30}{2(\lfloor r/2 \rfloor + 1)^2 + 1} \geq -\frac{60}{r^2}.$$

- (c) (Ordinary module.) *For all sufficiently large r , $\rho_r^{\text{mod}} \geq -c \log^3 r / r^2$ for an absolute constant c . Thus $-\rho_r^{\text{mod}}$ lies between $\Omega(r^{-2})$ and $O(\log^3 r / r^2)$.*
- (d) (Dense exactness.) *With the generators g_x, g_y, g_z on the single bag $\{x, y, z\}$, f belongs to the dense truncated preordering of order two:*

$$f = (y - x + z)^2 + 2((xz)^2 + z^2 g_x + x^2 g_z + g_x g_z). \quad (5.4)$$

- (e) (No exact polynomial separator.) *There are no polynomials $q_1(x, y)$ and $q_2(y, z)$, nonnegative on $[0, 1] \times [-1, 1]$ and $[-1, 1] \times [0, 1]$ respectively, with $f = q_1 + q_2$.*

In particular, $-\rho_r^{\text{pre}} = \Theta(r^{-2})$, and the same order holds for the local-measure relaxation: $-v_{2r} = \Theta(r^{-2})$.

Proof. (a) Apply [Lemma 5.3](#) with $n = 2r$, lift μ_+ and μ_- as in the proof of [Proposition 5.2](#), and use (5.2): the objective is $-(\int \varphi d\mu_+ - \int \varphi d\mu_-) \leq -2/(27\pi(2r+2)^2) = -1/(54\pi(r+1)^2)$. The chain of inequalities is [Proposition 5.2](#).

(b) In the transformed coordinates of [Section 5.2](#),

$$f_1 = \frac{3}{8} + \frac{1}{2}\hat{x} + \frac{1}{8}T_2(\hat{x}) - \hat{x}y - y, \quad f_2 = \frac{7}{8} + \frac{1}{2}T_2(y) + \frac{1}{8}T_2(\hat{z}) + \frac{1}{2}\hat{z} + \hat{z}y + y,$$

so $A(f_1) = \frac{1}{2} + \frac{4}{8} + 2 + 1 = 4$ and $A(f_2) = \frac{4}{2} + \frac{4}{8} + \frac{1}{2} + 2 + 1 = 6$. Here $w = 2$ and $d = 2$, so [Theorem 3.4](#) applies for $r \geq 2$ with $m = \lfloor r/2 \rfloor + 1$ and $A = 10$. Since $m > r/2$, $2m^2 + 1 > r^2/2$.

(c) The expansions in (b) give $C_f = \frac{21}{8} + \frac{25}{8} = \frac{23}{4}$, and $w = 2$, $d_\infty = 2$. Apply [Corollary 4.7](#); with (a) this gives the two-sided statement.

(d) Expanding verifies (5.4). The terms $(y - x + z)^2$ and $(xz)^2$ are squares of polynomials of degree at most two, z^2g_x and x^2g_z are squares of degree two times one generator, and g_xg_z is a product of two generators with the constant square one; every term has total degree at most four.

(e) Suppose $f = q_1 + q_2$ as stated. Then $q_1 - f_1 = f_2 - q_2$ lies in $\mathbb{R}[x, y] \cap \mathbb{R}[y, z] = \mathbb{R}[y]$, so $q_1 = f_1 + p(y)$ and $q_2 = f_2 - p(y)$ for a polynomial p . By (5.2), nonnegativity of q_1 and q_2 on their rectangles is equivalent to $p \geq \varphi$ and $p \leq \varphi$ on $[-1, 1]$. Then $p = \varphi$ on $[-1, 1]$; p vanishes on $[-1, 0]$, hence identically, but $\varphi(1) = 1$, a contradiction. $\square$

Part (e) explains part (a). A sparse certificate of $f - f^*$ would be such a decomposition, so none exists at any order, and a certificate of $f + \epsilon$ requires a polynomial p of degree at most $2r$ with $\varphi \leq p \leq \varphi + \epsilon$ on $[-1, 1]$; by [Lemma 5.3](#), $\epsilon \geq 2E_{2r}(\varphi)$, with this lower bound of order r^{-2} . This is a property of the hierarchy, not of the problem: the problem is solved by (5.2), and the dense certificate (5.4) uses the product xz , which crosses the two bags. The ordinary-module upper bound in (c) carries the logarithmic factor of [Corollary 4.7](#); we do not claim that the ordinary-module gap is of exact order r^{-2} .

With the linear generators $x, 1 - x, z, 1 - z$ in place of g_x, g_z , the identity $f = (y - x + z)^2 + 2xz$ is already a dense certificate of degree two. The choice of generators does not affect part (a), whose witnesses are actual measures.

5.5 The first two orders

For $r = 1$ and $r = 2$ the sparse values can be computed exactly. Write $\sqrt{3}$ explicitly and set

$$a = 3\sqrt{3} - 5, \quad b = \sqrt{3} - 1, \quad c_\star = \frac{3 + 2\sqrt{3}}{18}, \quad E_\star = \frac{7\sqrt{3}}{9} - \frac{4}{3}, \quad p_\star(y) = c_\star(y + 1)(y + a)^2, \quad (5.5)$$

so that $0 < a < b < 1$ and $E_\star > 0$ (numerically $a \approx 0.196$, $b \approx 0.732$, $E_\star \approx 0.0138$).

Proposition 5.5 (Exact values at orders one and two). *For problem (5.1):*

(a) $\rho_1^{\text{mod}} = \rho_1^{\text{pre}} = -\frac{1}{4}$, while $v_2 = -(3 - 2\sqrt{2}) \approx -0.1716$.

(b) $\rho_2^{\text{mod}} = \rho_2^{\text{pre}} = v_4 = -2E_\star = \frac{8}{3} - \frac{14\sqrt{3}}{9} \approx -0.027635$, and $E_3(\varphi) = E_4(\varphi) = E_\star$.

At order one the finite SDP has a larger gap than the local-measure relaxation: $\rho_1^{\text{pre}} < v_2$. At order two the entire gap is caused by the separator truncation, even for the ordinary module.

Proof. (a) At order one the product $g_x g_y$ has degree four and is absent, so the two cones coincide and $\rho_1^{\text{mod}} = \rho_1^{\text{pre}}$. The identity

$$x^2 - 2xy + \frac{1}{2}y^2 + \frac{1}{2}y + \frac{1}{8} = \frac{1}{2}(2x - y - \frac{1}{2})^2 + g_x \quad (5.6)$$

exhibits the left side in the order-one module of B_1 . Substituting $(x, y) \mapsto (z, -y)$ gives $z^2 + 2yz + \frac{1}{2}y^2 - \frac{1}{2}y + \frac{1}{8}$ in the order-one module of B_2 , and the two left sides add to $f + \frac{1}{4}$. Hence $\lambda_1^{\text{mod}} \geq -\frac{1}{4}$, and $\rho_1^{\text{mod}} \geq -\frac{1}{4}$ by [Lemma 2.6](#).

For the reverse inequality, let L_1 be integration against the atomic measure $\frac{3}{4}\delta_{(0, -1/2)} + \frac{1}{4}\delta_{(1, 3/2)}$ in the variables (x, y) , restricted to polynomials of degree at most two. Its moment matrix is PSD because L_1 is integration against a positive measure, and

$$L_1(y) = 0, \quad L_1(y^2) = \frac{3}{4}, \quad L_1(g_x) = 0, \quad L_1(g_y) = \frac{1}{4}, \quad L_1(f_1) = -\frac{1}{2}.$$

The generator tests of order one are $L_1(g_x) \geq 0$ and $L_1(g_y) \geq 0$, so L_1 is feasible for B_1 . The atom $(1, \frac{3}{2})$ lies outside the rectangle. Moreover, $L_1(xy) = 3/8 > L_1(x) = 1/4$, whereas $xy \leq x$ on the rectangle. Thus no probability measure supported on the bag represents these truncated moments. Let L_2 be integration against the image measure under $(X, Y) \mapsto (y, z) = (-Y, X)$. It is feasible for B_2 by symmetry, its separator moments $L_2(y) = 0$, $L_2(y^2) = \frac{3}{4}$ agree with those of L_1 , and $L_2(f_2) = \frac{3}{4} \cdot \frac{1}{4} + \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{4}$, the value of $(y + z)^2 = (X - Y)^2$ at the two atoms. The total objective is $-\frac{1}{4}$.

For v_2 , let $c = \sqrt{2} - 1$ and $S_1(y) = \frac{1}{2}y^2 + cy$. The error $e = \varphi - S_1$ is odd, equals $\frac{1}{2}y^2 - cy$ on $[0, 1]$, and has extreme values $-\frac{1}{2}c^2$ at $y = c$ and $\frac{1}{2} - c = \frac{1}{2}c^2$ at $y = 1$. Hence $\|e\|_\infty = \frac{1}{2}c^2$, attained with alternating signs at $-1, -c, c, 1$. If a polynomial q of degree at most two had $\|\varphi - q\|_\infty < \frac{1}{2}c^2$, then $q - S_1 = e - (\varphi - q)$ would have the sign of e at these four points, hence at least three zeros, so $q = S_1$, a contradiction. Thus $E_2(\varphi) = \frac{1}{2}c^2$ and $v_2 = -c^2 = -(3 - 2\sqrt{2})$ by [Proposition 5.2](#).

(b) *Approximation.* Direct expansion gives the identities

$$p_\star(y) - y^2 = c_\star(y + 7 - 4\sqrt{3})(y - b)^2, \quad p_\star(y) + p_\star(-y) = y^2 + 2E_\star. \quad (5.7)$$

Since $7 - 4\sqrt{3} > 0$, the first gives $p_\star \geq y^2 = \varphi$ on $[0, 1]$, and $p_\star \geq 0 = \varphi$ on $[-1, 0]$ by (5.5). With $\varphi(y) + \varphi(-y) = y^2$, the second gives $p_\star(y) - \varphi(y) = 2E_\star - (p_\star(-y) - \varphi(-y)) \leq 2E_\star$. So $0 \leq p_\star - \varphi \leq 2E_\star$ on $[-1, 1]$, and $p_\star - \varphi$ takes the values $0, 2E_\star, 0, 2E_\star, 0, 2E_\star$ at $-1, -b, -a, a, b, 1$ (use the factorizations and the second identity). The cubic $p_\star - E_\star$ therefore approximates φ with error $E_\star$, alternating in sign at six points. A polynomial of degree at most four with smaller error would differ from $p_\star - E_\star$ by a nonzero polynomial with five sign changes, which is impossible. Hence $E_4(\varphi) = E_3(\varphi) = E_\star$.

Certificate. Let

$$V = \begin{pmatrix} x(x-b) \\ x(y-b) \\ (y+1)(y+a) - 3bx \end{pmatrix}, \quad U = \begin{pmatrix} x-b \\ y-b \end{pmatrix}, \quad Z = (b+a)x - b(y+a),$$

$$G_0 = \begin{pmatrix} 2 - \sqrt{3} & \frac{3\sqrt{3}-7}{4} & \frac{\sqrt{3}-1}{12} \\ \frac{3\sqrt{3}-7}{4} & \frac{7}{6} & -\frac{\sqrt{3}+1}{12} \\ \frac{\sqrt{3}-1}{12} & -\frac{\sqrt{3}+1}{12} & \frac{1}{12} + \frac{\sqrt{3}}{18} \end{pmatrix}, \quad G_1 = \begin{pmatrix} 2 - \sqrt{3} & \frac{3\sqrt{3}-7}{4} \\ \frac{3\sqrt{3}-7}{4} & 1 \end{pmatrix}, \quad k_\star = \frac{1}{6} + \frac{7\sqrt{3}}{72}.$$

(The third entry of V is $(y+1)(y+a) - \frac{(b+1)(b+a)}{b}x$, simplified using $(b+1)(b+a)/b = 3b$.) Direct expansion gives the polynomial identity

$$x^2 - 2xy + p_\star(y) = V^\top G_0 V + g_x U^\top G_1 U + g_y k_\star Z^2. \quad (5.8)$$

The leading principal minors of G_0 are $2 - \sqrt{3}$, $(35\sqrt{3} - 58)/24$ and $(38 - 15\sqrt{3})/864$; those of G_1 are $2 - \sqrt{3}$ and $(13\sqrt{3} - 22)/8$. All are positive, since $\sqrt{3} < 2$, $35^2 \cdot 3 = 3675 > 58^2 = 3364$, $15\sqrt{3} < 30 < 38$, and $13^2 \cdot 3 = 507 > 22^2 = 484$; also $k_\star > 0$. Hence G_0, G_1 are positive definite, their quadratic forms are sums of squares, and every term of (5.8) has total degree at most four with at most one generator. So the left side of (5.8) is in the order-two ordinary module of B_1 . Substituting $(x, y) \mapsto (z, -y)$, which fixes g_y , puts $z^2 + 2yz + p_\star(-y)$ in the order-two module of B_2 . By (5.7) the two left sides add to $f + 2E_\star$, so $\lambda_2^{\text{mod}} \geq -2E_\star$ and $\rho_2^{\text{mod}} \geq -2E_\star$.

Witness. Let μ_1 put masses

$$w_{-1} = \frac{8 - 4\sqrt{3}}{9}, \quad w_{-a} = \frac{5}{18} + \frac{\sqrt{3}}{6}, \quad w_b = \frac{5\sqrt{3}}{18} - \frac{1}{6}$$

on the points $-1, -a, b$. Direct substitution of (5.5) gives

$$w_{-1} + w_{-a} + w_b = 1, \quad -w_{-1} - aw_{-a} + bw_b = 0, \quad -w_{-1} - a^3w_{-a} + b^3w_b = 0.$$

The masses are positive because $\sqrt{3} < 2$ and $\sqrt{3} > 1$. Thus the first and third moments of μ_1 vanish. Let μ_2 be the reflection of μ_1 under $y \mapsto -y$; then μ_1 and μ_2 have equal moments through degree four. Lifting as in Proposition 5.2 gives local measures on the two rectangles with objective $-\int \varphi d\mu_1 + \int \varphi d\mu_2 = -w_b b^2 + w_{-1} + w_{-a} a^2 = -2E_\star$. Hence $v_4 \leq -2E_\star$, and

$$-2E_\star \leq \rho_2^{\text{mod}} \leq \rho_2^{\text{pre}} \leq v_4 \leq -2E_\star$$

by Proposition 5.2. This also gives $v_4 = -2E_4(\varphi)$ directly. $\square$

The proof of Proposition 5.5 uses neither numerical SDP solutions nor strong SDP duality: certificates bound the values from below and explicit functionals bound them from above. The values for $r = 1, 2$ agree, for both cones, with $-2E_{2r}(\varphi)$ at $r = 2$ and exceed it in magnitude at $r = 1$.

Remark 5.6 (An optimal majorant need not be certifiable at the same order). The quadratic $\frac{1}{2}(y + c)^2 = S_1 + \frac{1}{2}c^2$, with $c = \sqrt{2} - 1$, is a majorant of φ of minimal uniform width among polynomials of degree at most two, so $q(x, y) = x^2 - 2xy + \frac{1}{2}(y + c)^2$ is nonnegative on $[0, 1] \times [-1, 1]$. It does not belong to the order-one module of B_1 . Suppose $q = \sigma_0 + \gamma_x g_x + \gamma_y g_y$ with σ_0 a sum of squares of affine polynomials and $\gamma_x, \gamma_y \geq 0$. At $(0, -c)$ we have $q = 0$, $g_x = 0$ and $g_y > 0$, so $\gamma_y = 0$ and every affine factor of σ_0 vanishes there; write them as $\alpha_i x + \beta_i(y + c)$. Comparing the coefficients of y^2 , xy , x and x^2 gives $\sum \beta_i^2 = \frac{1}{2}$, $\sum \alpha_i \beta_i = -1$, $\gamma_x = 2c$ and $\sum \alpha_i^2 = 1 + 2c$. The Cauchy–Schwarz inequality then requires $1 \leq \frac{1}{2}(1 + 2c) = \sqrt{2} - \frac{1}{2} < 1$, a contradiction. Thus polynomial nonnegativity of a local term does not imply membership in the local cone of the same degree, and Proposition 5.2 cannot be transferred to the SOS hierarchy at a fixed order without a degree-specific certificate such as (5.8).

Whether $\rho_r^{\text{mod}} = \rho_r^{\text{pre}} = -2E_{2r}(\varphi)$ for every $r \geq 2$ is open; floating-point SDP solutions for $3 \leq r \leq 6$ are numerically close to $-2E_{2r}(\varphi)$, but they are not certified and we do not use them.

5.6 Relation to prior work

Remark 5.7 (What is prior in Theorem 5.4). Failure of finite exactness of sparse hierarchies, explained by separator functions that are not polynomials, is known. Nie, Qu, Tang and Zhang [27], Example 6.7, give a quartic two-bag problem whose dense hierarchy is exact at order two and whose sparse hierarchy is never exact; its separator function is rational. Grimm, Netzer and

Schweighofer [10] obtain sparse representations under strict positivity by approximating separator fiber minima by polynomials, which the strict margin permits. The identity of Lemma 5.1 is known [12]. Baldi and Slot [1] prove a degree lower bound for the dense ordinary module on a polynomial that lies in the dense preordering at degree four; that obstruction is of a different kind, coming from the missing generator products rather than from restricted bags, and neither lower bound implies the other.

What Theorem 5.4 adds, to our knowledge, is a quantitative version: a fixed quadratic objective with a piecewise quadratic separator function, a matching upper bound for the sparse preordering, and explicit local-measure witnesses showing that the rate r^{-2} is optimal for that hierarchy even when every local functional is a measure. Proposition 5.5 is a small-order benchmark for this instance, not a general theorem.

6 Large private convex quadratic blocks

This section studies the recourse model of Definition 2.13 with the private-degree-two hierarchy of Definition 2.14. Each bag carries, besides its shared coordinates, a private vector that may be much larger than the bag, enters the objective as a convex quadratic, and is constrained by linear inequalities. The hierarchy smooths only the shared coordinates; private moments are kept at degree two at every order.

We prove four results. If the private polytopes do not depend on the shared coordinates, the gap is $O(r^{-2})$ (Theorem 6.10). The redundant private quadratic bounds (R3) and private convexity cannot be dropped from this statement (Propositions 6.11 and 6.12). If the polytopes move affinely with the shared coordinates, have a fixed constraint matrix, and are nonempty at every shared point, the gap is $O(r^{-1})$ (Theorem 6.15), and this order is attained by a three-variable example (Theorem 6.17). Finally, for every feasible model the moment value of the hierarchy equals its certificate value, and certificates exist at every level strictly below it (Proposition 6.9).

The mechanism is the one of Section 3, applied to the shared coordinates only. A common kernel turns each local functional into a nonnegative density on its shared bag box, together with a positive semidefinite matrix of conditional private moments. The densities have exactly matching separator marginals and glue to a global shared law. The private decision of each bag is then a measurable function of its own shared coordinates: the conditional private mean in the fixed case, and a Euclidean projection of that mean in the affine case. No private moment is ever compared across bags, so no matrix-valued gluing is needed.

6.1 Constants and kernel facts

Throughout Sections 6 and 7, $H_b(x)$ is the augmented coefficient matrix of (2.9), a symmetric matrix of order $p_b + 1$ with polynomial entries, and

$$H_b(x) = \sum_{\alpha} H_{b\alpha} T_{\alpha}(x), \quad H_{b\alpha} \in \mathbb{R}^{(p_b+1) \times (p_b+1)} \text{ symmetric}, \quad (6.1)$$

is its tensor Chebyshev expansion in the shared variables of B_b . For a matrix M write $\|M\|_1 = \sum_{i,j} |M_{ij}|$ (both off-diagonal entries are counted). Put

$$d = \max\{|\alpha| : H_{b\alpha} \neq 0 \text{ for some } b\}, \quad A_b = \sum_{\alpha} \|H_{b\alpha}\|_1 \sum_{i \in B_b} \alpha_i^2, \quad A = \sum_b A_b, \quad (6.2)$$

with $d = 0$ for the zero objective. In Sections 6 and 7 the symbol A always denotes this matrix budget. In terms of the objective, A_b adds the weighted absolute Chebyshev coefficients of c_b , of each

entry of q_b once, and of each ordered entry of Q_b . When $p_b = 0$ it equals $A(f_b)$ of (2.3). Coefficients that do not depend on x contribute nothing.

We use the kernel of Lemma 3.1. For an integer $m \geq 2$ let $b_j = (m - |j|)_+$ for $j \in \mathbb{Z}$, $a_k = \sum_{j \in \mathbb{Z}} b_j b_{j-k}$, and $\gamma_k = a_k/a_0$ for $0 \leq k \leq 2m - 2$, $\gamma_k = 0$ for $k > 2m - 2$. Then

$$J_m(t) = \frac{1}{a_0} \left| \sum_{j=0}^{m-1} e^{ijt} \right|^4 = 1 + 2 \sum_{k \geq 1} \gamma_k \cos(kt) \geq 0, \quad K_m(x, u) = 1 + 2 \sum_{k \geq 1} \gamma_k T_k(x) T_k(u), \quad (6.3)$$

and Lemma 3.1 gives $K_m \geq 0$ on $[-1, 1]^2$, $\int K_m(x, u) d\mu(u) = 1$, $\int T_k(u) K_m(x, u) d\mu(u) = \gamma_k T_k(x)$ for all $k \geq 0$, and $0 \leq 1 - \gamma_k \leq 3k^2/D_m$, where $D_m = 2m^2 + 1$. For a bag put

$$K_{m,b}(x, u) = \prod_{i \in B_b} K_m(x_i, u_i), \quad \Gamma_\alpha = \prod_{i \in B_b} \gamma_{\alpha_i},$$

so that $\int T_\alpha(u) K_{m,b}(x, u) d\mu^{B_b}(u) = \Gamma_\alpha T_\alpha(x)$. Empty products equal one, and $\mu^\emptyset$ is the unit mass on the single point of $[-1, 1]^\emptyset$; an empty bag therefore causes no exception below. Define

$$V_m = \frac{3(4m - 3)}{2mD_m} = \frac{6}{D_m} \cdot \frac{4m - 3}{4m} < \frac{6}{D_m}. \quad (6.4)$$

Lemma 6.1 (Kernel facts). *Let $m \geq 2$.*

- (a) $a_0 = mD_m/3$, $1 - \gamma_1 = 3/D_m$, and $1 - \gamma_2 = (4m - 3)/a_0$.
- (b) For every $x \in [-1, 1]$,

$$\int K_m(x, u) (x - u) d\mu(u) = (1 - \gamma_1) x, \quad \int K_m(x, u) (x - u)^2 d\mu(u) = V_m + \frac{3 - 2m}{a_0} x^2 \leq V_m. \quad (6.5)$$

- (c) For a bag B_b and a fixed $u \in [-1, 1]^{B_b}$ there are polynomials $p_{I,l}$ in x_{B_b} , finitely many for each $I \subseteq B_b$, with

$$K_{m,b}(x, u) = \sum_{I \subseteq B_b} w_I(x) \sum_l p_{I,l}(x)^2, \quad \deg p_{I,l} \leq v_b(m - 1) - |I|. \quad (6.6)$$

Proof. (a) We have $a_0 = \sum_{|j| < m} (m - |j|)^2 = m^2 + 2 \sum_{i=1}^{m-1} i^2 = (2m^3 + m)/3 = mD_m/3$. Since $\sum_j b_j^2 = \sum_j b_j^2 = a_0$,

$$a_0 - a_k = \frac{1}{2} \sum_{j \in \mathbb{Z}} (b_j - b_{j-k})^2.$$

For $k = 1$ there are exactly $2m$ nonzero differences, each of absolute value one, so $a_0 - a_1 = m$ and $1 - \gamma_1 = m/a_0 = 3/D_m$. For $k = 2$ the nonzero differences $b_j - b_{j-2}$ are 1 at $j = 1 - m$, 2 for $2 - m \leq j \leq 0$, -2 for $2 \leq j \leq m$, and -1 at $j = m + 1$. Their squares sum to $2 + 8(m - 1)$, so $a_0 - a_2 = 4m - 3$.

(b) Integrate against $K_m(x, u) d\mu(u)$, using $u = T_1(u)$, $u^2 = \frac{1}{2}(1 + T_2(u))$ and the eigenvalue relation: $\int K_m(x, u) u d\mu(u) = \gamma_1 x$ and $\int K_m(x, u) u^2 d\mu(u) = \frac{1}{2} + \frac{1}{2} \gamma_2 (2x^2 - 1)$. Hence the first integral is $(1 - \gamma_1)x$ and the second is

$$x^2 - 2\gamma_1 x^2 + \frac{1}{2} + \frac{1}{2} \gamma_2 (2x^2 - 1) = \frac{1}{2} (1 - \gamma_2) + (2(1 - \gamma_1) - (1 - \gamma_2)) x^2.$$

By (a), $\frac{1}{2}(1 - \gamma_2) = (4m - 3)/(2a_0) = V_m$ and $2(1 - \gamma_1) - (1 - \gamma_2) = (2m - 4m + 3)/a_0 = (3 - 2m)/a_0 < 0$.

(c) For fixed u_i , the polynomial $x_i \mapsto K_m(x_i, u_i)$ is nonnegative on $[-1, 1]$ and has degree at most $2m - 2$. By Lemma 2.11 it equals $\sigma_{0,i} + (1 - x_i^2)\sigma_{1,i}$ with $\sigma_{0,i} \in \Sigma_{2m-2}$ and $\sigma_{1,i} \in \Sigma_{2m-4}$. Multiply these representations over $i \in B_b$ and expand. The term indexed by $I \subseteq B_b$ is w_I times a product of sums of squares, which is a sum of squares of polynomials of degree at most $(v_b - |I|)(m - 1) + |I|(m - 2) = v_b(m - 1) - |I|$. $\square$

The representation in (c) need not depend continuously on u ; it is used at each fixed u separately.

Remark 6.2 (Private and shared dimensions). The two kinds of variables enter the hierarchy differently. The order r needed for a given accuracy is set by the kernel, which acts only on shared coordinates, through $m = \lfloor r/w \rfloor + 1$ or $m = \lceil r/w \rceil$ below. For fixed r , by (2.10), the shared bag size v_b appears in the exponent of the block sizes, about r^{v_b} , while the private dimension enters only through the factor $p_b + 1$ of the largest block, the number $J_b + p_b$ of scalar localizers, and at most $\binom{p_b+2}{2}$ private monomials. Accuracy constants are a different matter: $\mathbf{A}$, and in the affine case the Hoffman and gradient constants of Theorem 6.15, can grow with p_b and with the data. These statements concern the sizes of finite semidefinite programs. We make no claim about the time needed to solve them, about their numerical conditioning, or about a practical advantage over other formulations; see also Remark 6.20.

6.2 Conditional moment matrices

Fix a feasible family (L_b) of order r for Definition 2.14. Restricted to polynomials in x_{B_b} alone, (R1) with $s = 0$ shows that L_b is $P_r(B_b)$ -positive, so Lemmas 2.7 and 2.8 apply to it. We write $\mathbf{z} = (z_0, z_1, \dots, z_{p_b})^\top$ with $z_0 = 1$ and $z_k = y_{b,k}$.

Lemma 6.3 (Mixed moments). *For every multi-index α with $|\alpha| \leq r$ and all $0 \leq i, j \leq p_b$,*

$$|L_b(T_\alpha z_i z_j)| \leq 1.$$

Proof. By Lemma 2.8, $0 \leq L_b(T_\alpha^2) \leq 1$. For $k \geq 1$, (R1) gives $L_b(T_\alpha^2 y_{b,k}^2) \geq 0$, and (R3) with $I = \emptyset$ and $s = T_\alpha$ gives $L_b(T_\alpha^2(1 - y_{b,k}^2)) \geq 0$. Hence $0 \leq L_b(T_\alpha^2 z_i^2) \leq 1$ for every i , and in particular $0 \leq L_b(z_j^2) \leq 1$. Apply Lemma 2.7 with $G = 1$ to the space of polynomials $s_0(x) + s(x)^\top y_b$ with $\deg s_k \leq r$, on which (R1) with $I = \emptyset$ is a positive semidefinite form; both $T_\alpha z_i$ and z_j belong to it. Thus $L_b(T_\alpha z_i z_j)^2 \leq L_b(T_\alpha^2 z_i^2) L_b(z_j^2) \leq 1$. $\square$

The private quadratic bounds (R3) are used here: affine localizers alone do not bound second private moments. With no shared variables, the functional $L(1) = 1$, $L(y) = 0$, $L(y^2) = R$ on $\mathbb{R}[y]_{\leq 2}$ satisfies (R1)–(R2) for $P = [-1, 1]$ and every $R \geq 0$.

For a kernel degree $m \geq 2$ and a bag b , and for every output point $u \in [-1, 1]^{B_b}$, define the conditional moments

$$h_b(u) = L_b(K_{m,b}(\cdot, u)), \quad \psi_b(u) = L_b(K_{m,b}(\cdot, u) y_b), \quad Y_b(u) = L_b(K_{m,b}(\cdot, u) y_b y_b^\top), \quad (6.7)$$

where L_b acts on (x, y_b) entrywise and u is a parameter, and set

$$M_b(u) = L_b(K_{m,b}(\cdot, u) \mathbf{z} \mathbf{z}^\top) = \begin{pmatrix} h_b(u) & \psi_b(u)^\top \\ \psi_b(u) & Y_b(u) \end{pmatrix}.$$

These are defined when $2v_b(m-1) \leq 2r$, and they are polynomials in u . When $2v_b(m-1) \leq 2r-2$ we also use

$$U_b(u) = L_b(K_{m,b}(\cdot, u) x_{B_b}) \in \mathbb{R}^{v_b}, \quad \widehat{M}_b(u) = L_b(K_{m,b}(\cdot, u) \zeta \zeta^\top), \quad \zeta = (1, x_{B_b}^\top, y_b^\top)^\top.$$

Lemma 6.4 (Conditional matrices). *Let $m \geq 2$ and $\tau \in \{0, 1\}$ satisfy $v_b(m-1) \leq r - \tau$. Then for every $u \in [-1, 1]^{B_b}$:*

(a) $M_b(u) \succeq 0$ and $0 \leq Y_b(u)_{kk} \leq h_b(u)$ for every k . If $h_b(u) = 0$ then $M_b(u) = 0$.

- (b) For every fiber row j with $\epsilon_{bj} \leq \tau$, that is, every row if $\tau = 1$ and every row with E_b -row zero if $\tau = 0$,

$$a_{bj}h_b(u) + (E_b U_b(u))_j - (A_b \psi_b(u))_j \geq 0,$$

where the middle term is absent when $\tau = 0$ (it is then multiplied by a zero row).

- (c) If $\tau = 1$, then $\widehat{M}_b(u) \succeq 0$ and $0 \leq L_b(K_{m,b}(\cdot, u)x_i^2) \leq h_b(u)$ for $i \in B_b$. If moreover $h_b(u) = 0$, then $\widehat{M}_b(u) = 0$; in particular $U_b(u) = 0$.
- (d) If $h_b(u) > 0$, then $\bar{y}_b(u) = \psi_b(u)/h_b(u) \in [-1, 1]^{p_b}$, and $A_b \bar{y}_b(u) \leq a_b$ holds in every row with zero E_b -row. If $\tau = 1$, then also $\bar{x}_b(u) = U_b(u)/h_b(u) \in [-1, 1]^{B_b}$ and

$$\bar{y}_b(u) \in P_b(\bar{x}_b(u)). \quad (6.8)$$

Proof. Write $K = K_{m,b}(\cdot, u)$ and use (6.6); each square polynomial $p_{I,l}$ has degree at most $r - \tau - |I|$.

(a) For a constant vector $c \in \mathbb{R}^{p_b+1}$, $c^\top M_b(u)c = \sum_{I,l} L_b(w_I(p_{I,l} c^\top \mathbf{z})^2)$. Each $p_{I,l} c^\top \mathbf{z}$ has the form $s_0 + s^\top y_b$ with all s_k of degree at most $r - |I|$, so (R1) gives $c^\top M_b(u)c \geq 0$. Similarly $h_b(u) - Y_b(u)_{kk} = \sum_{I,l} L_b(w_I p_{I,l}^2 (1 - y_{b,k}^2)) \geq 0$ by (R3), and $Y_b(u)_{kk} \geq 0$ because $M_b(u) \succeq 0$. If $h_b(u) = 0$, every diagonal entry of $M_b(u)$ vanishes, and so does every entry, because $M_{ij}^2 \leq M_{ii}M_{jj}$ for a positive semidefinite matrix.

(b) The left side equals $L_b(K \ell_{bj}) = \sum_{I,l} L_b(w_I p_{I,l}^2 \ell_{bj})$. Since $\deg p_{I,l} \leq r - \tau - |I| \leq r - |I| - \epsilon_{bj}$, each term is nonnegative by (R2).

(c) For $c = (c_0, c_x, c_y)$, $c^\top \widehat{M}_b(u)c = \sum_{I,l} L_b(w_I(p_{I,l}(c_0 + c_x^\top x) + p_{I,l} c_y^\top y_b)^2)$. The coefficient polynomials have degree at most $r - 1 - |I| + 1 = r - |I|$, so (R1) applies. Next, $h_b(u) - L_b(K x_i^2) = \sum_{I,l} L_b(w_I(1 - x_i^2)p_{I,l}^2)$. If $i \notin I$, the term is $w_{I \cup \{i\}} p_{I,l}^2$ with $\deg p_{I,l} \leq r - |I \cup \{i\}|$. If $i \in I$, it is $w_{I \setminus \{i\}}((1 - x_i^2)p_{I,l})^2$, whose square polynomial has degree at most $r - 1 - |I| + 2 = r - |I \setminus \{i\}|$. Both are allowed in (R1), so $L_b(K x_i^2) \leq h_b(u)$. Nonnegativity of the diagonal of $\widehat{M}_b(u)$ and the minor argument of (a) finish the proof, the private diagonal being bounded by (a).

(d) By the 2×2 minors of $M_b(u)$ and (a), $\psi_b(u)_k^2 \leq h_b(u)Y_b(u)_{kk} \leq h_b(u)^2$, so $|\bar{y}_{b,k}| \leq 1$. The same argument with (c) gives $|\bar{x}_{b,i}| \leq 1$ when $\tau = 1$. Dividing the inequalities of (b) by $h_b(u) > 0$ gives the remaining claims; when $\tau = 1$ every row of $A_b \bar{y}_b \leq a_b + E_b \bar{x}_b$ holds, and together with $\bar{y}_b \in [-1, 1]^{p_b}$ this is (6.8). $\square$

Part (d) guarantees feasibility at the conditional mean $\bar{x}_b(u)$ of the *input* shared variables. In the affine case the private mean need not be feasible at the output u . Bounding the cost of repairing it there gives the general $O(r^{-1})$ term in Theorem 6.15.

Lemma 6.5 (Matrix cost). *Let $m \geq 2$, $v_b(m-1) \leq r$, and $d \leq r$. Then*

$$\int \langle H_b(u), M_b(u) \rangle d\mu^{B_b}(u) \leq L_b(f_b) + \frac{3A_b}{D_m},$$

where $\langle H, M \rangle = \sum_{i,j} H_{ij} M_{ij}$.

Proof. By (6.1), $\langle H_b(u), M_b(u) \rangle = \sum_\alpha \sum_{i,j} (H_{b\alpha})_{ij} T_\alpha(u) M_b(u)_{ij}$. Expand $K_{m,b}(x, u)$ as a finite sum of products of polynomials in x and in u . By linearity of L_b ,

$$\int T_\alpha(u) M_b(u)_{ij} d\mu^{B_b}(u) = L_b\left(z_i z_j \int T_\alpha(u) K_{m,b}(x, u) d\mu^{B_b}(u)\right) = \Gamma_\alpha L_b(T_\alpha z_i z_j).$$

Also $L_b(f_b) = \sum_\alpha \sum_{i,j} (H_{b\alpha})_{ij} L_b(T_\alpha z_i z_j)$. Every α with $H_{b\alpha} \neq 0$ has $|\alpha| \leq d \leq r$, so Lemma 6.3 bounds the difference by $\sum_\alpha \|H_{b\alpha}\|_1 (1 - \Gamma_\alpha)$. Since all γ_k lie in $[0, 1]$, induction on the number of factors gives $0 \leq 1 - \Gamma_\alpha \leq \sum_i (1 - \gamma_{\alpha_i}) \leq 3 \sum_i \alpha_i^2 / D_m$. Multipliers with $\alpha_i > 2m - 2$ are zero; this case is included. $\square$

Lemma 6.6 (Consistent shared laws). *Let $m \geq 2$ with $v_b(m-1) \leq r$ for every b . Then each h_b is nonnegative on $[-1, 1]^{B_b}$, $\int h_b d\mu^{B_b} = 1$, and for every edge $e = bc$ the laws $h_b d\mu^{B_b}$ and $h_c d\mu^{B_c}$ have the same marginal on $[-1, 1]^{S_e}$. Let $\phi_b : [-1, 1]^{B_b} \rightarrow \mathbb{R}^{p_b}$ be Borel maps with $\phi_b(u) \in P_b(u)$ for every u . Then there is a probability law ν on the feasible set of the recourse model with*

$$\int f d\nu = \sum_b \int f_b(u, \phi_b(u)) h_b(u) d\mu^{B_b}(u),$$

and consequently f^* is at most the right side.

Proof. Nonnegativity is $M_b(u)_{00} \geq 0$ in Lemma 6.4(a). Integrating $K_m(x_i, u_i)$ in u_i gives one, so $\int h_b d\mu^{B_b} = L_b(1) = 1$, and the marginal of $h_b d\mu^{B_b}$ on S_e has density $L_b(\prod_{i \in S_e} K_m(x_i, u_i))$ with respect to μ^{S_e} . The argument of L_b is a polynomial in x_{S_e} of degree at most $2v_b(m-1) \leq 2r$, so (R4) makes this density the same for b and c . By Lemma 2.12 there is a law ν^{sh} on $[-1, 1]^n$ with marginals $h_b d\mu^{B_b}$. Let ν be its image under $x \mapsto (x, (\phi_b(x_{B_b}))_b)$. It is concentrated on the feasible set, and its objective is the displayed sum because each f_b depends only on (x_{B_b}, y_b) . The objective is at least f^* at every feasible point. $\square$

6.3 Finiteness, attainment, and certificates

For the box hierarchies, product arcsine moments are strictly feasible and Lemma 2.10 gives zero duality gap and dual attainment. That argument does not transfer to the recourse hierarchy: a fiber may be lower-dimensional, for instance when an equality is written as two inequalities, and then no feasible family has positive definite localizing matrices. We use instead an order-unit argument, which needs no strict feasibility and gives slightly less: certificates at every level below the moment value, without a claim at the moment value itself.

Let $V_{b,r} = \mathbb{R}[x_{B_b}]_{\leq 2r} \otimes \mathbb{R}[y_b]_{\leq 2}$, let $C_{b,r} \subseteq V_{b,r}$ be the convex cone of finite sums of the polynomials tested in (R1)–(R3), with the degree allowances of Definition 2.14, and let

$$W_r = \sum_b V_{b,r}, \quad C_r = \sum_b C_{b,r} \subseteq W_r$$

be the sums inside the global polynomial ring in $(x, y_1, \dots, y_t)$. The *certificate value* of the private-degree-two hierarchy is

$$\lambda_r^{\text{rec}} = \sup\{\lambda \in \mathbb{R} : f - \lambda \in C_r\}, \quad f = \sum_b f_b, \quad (6.9)$$

that is, the supremum of λ for which $f - \lambda = \sum_b \sigma_b$ as polynomials, with $\sigma_b \in C_{b,r}$.

Lemma 6.7 (Decompositions of zero with private variables). *If $p_b \in V_{b,r}$ and $\sum_b p_b = 0$, then there are polynomials $\phi_e \in \mathbb{R}[x_{S_e}]_{\leq 2r}$ such that each p_b is the signed sum of the ϕ_e over the edges incident to b , as in Lemma 2.2. In particular, every feasible family (L_b) defines a linear functional F on W_r by $F(\sum_b p_b) = \sum_b L_b(p_b)$.*

Proof. A monomial containing a variable of y_b occurs only in p_b , because private vectors of different bags are distinct variables. Its coefficient in $\sum_b p_b = 0$ is therefore its coefficient in p_b , which must vanish. Hence each p_b is a polynomial in x_{B_b} of degree at most $2r$, and Lemma 2.2 applies. If $\sum_b p_b = \sum_b p'_b$, the differences $p_b - p'_b$ are signed sums of separator polynomials of degree at most $2r$, on which adjacent functionals agree by (R4); so F is well defined. $\square$

Lemma 6.8 (Order unit). *For every bag b , every multi-index α with $|\alpha| \leq 2r$, and all $0 \leq i, j \leq p_b$, both $1 + x^\alpha z_i z_j$ and $1 - x^\alpha z_i z_j$ belong to $C_{b,r}$.*

Proof. Write $\alpha = \beta + \beta'$ with $|\beta|, |\beta'| \leq r$, and put $g = x^\beta z_i$ and $g' = x^{\beta'} z_j$. Order the coordinates of B_b . For $|\beta| \leq r$, telescoping and $1 - x_h^{2c} = (1 - x_h^2) \sum_{l=0}^{c-1} x_h^{2l}$ give

$$1 - x^{2\beta} = \sum_{h \in B_b} \sum_{l=0}^{\beta_h-1} (1 - x_h^2) \left(x_h^l \prod_{h' < h} x_{h'}^{\beta_{h'}} \right)^2. \quad (6.10)$$

Each square polynomial has degree at most $|\beta| - 1 \leq r - 1$, so every term is allowed in (R1) with $I = \{h\}$. If $i = 0$, this shows $1 - g^2 \in C_{b,r}$. If $i \geq 1$, write $1 - g^2 = (1 - x^{2\beta}) + x^{2\beta}(1 - y_{b,i}^2)$; the last term is allowed in (R3) with $I = \emptyset$ and $s = x^\beta$. The same holds for g' . Finally

$$1 \pm gg' = \frac{1}{2}[(1 - g^2) + (1 - g'^2) + (g \pm g')^2],$$

and $(g \pm g')^2$ is allowed in (R1) with $I = \emptyset$, because $g \pm g'$ is affine in y_b with shared coefficients of degree at most r . $\square$

Proposition 6.9 (Moment and certificate values). *Let $r \geq 1$ with $f_b \in V_{b,r}$ for every bag, and suppose the recourse model has at least one feasible point (this holds under nonempty fixed polytopes or complete recourse). Then:*

- (i) *the set of feasible families of order r is compact, every local moment of a feasible family has absolute value at most one, and ρ_r^{rec} is finite and attained;*
 - (ii) $\lambda_r^{\text{rec}} = \rho_r^{\text{rec}}$;
 - (iii) $f - \lambda \in C_r$ for every $\lambda < \rho_r^{\text{rec}}$;
 - (iv) *if $f^* - \rho_r^{\text{rec}} \leq \mathcal{E}$, then $f - (f^* - \mathcal{E} - \varepsilon) \in C_r$ for every $\varepsilon > 0$.*
- We make no claim that $f - \rho_r^{\text{rec}} \in C_r$.*

Proof. Global functionals. By Lemma 6.7, a feasible family defines a linear functional F on W_r with $F(1) = 1$, $F \geq 0$ on C_r , and $F(f) = \sum_b L_b(f_b)$. Conversely, if F is a linear functional on W_r with $F(1) = 1$ and $F \geq 0$ on C_r , its restrictions $L_b = F|_{V_{b,r}}$ satisfy (R0)–(R3), and they satisfy (R4) because a separator polynomial lies in both $V_{b,r}$ and $V_{c,r}$. Hence

$$\rho_r^{\text{rec}} = \inf\{F(f) : F \in C_r^*, F(1) = 1\},$$

where C_r^* is the dual cone in the dual space of W_r .

Interior point. Choose a basis of W_r consisting of monomials. Each basis monomial lies in some $V_{b,r}$ and has the form $x^\alpha z_i z_j$ of Lemma 6.8, so $1 \pm e \in C_r$ for every basis element e . If $g = \sum_e c_e e$ with $\sum_e |c_e| \leq \varepsilon$, then

$$\varepsilon + g = \left(\varepsilon - \sum_e |c_e| \right) \cdot 1 + \sum_e |c_e| (1 + \text{sign}(c_e) e) \in C_r,$$

using $1 \in C_r$. Hence C_r contains a coefficient ball around each $\varepsilon \cdot 1$, $\varepsilon > 0$; in particular 1 is an interior point of C_r in W_r . We do not use, and do not claim, that C_r is closed.

(i) If $F \in C_r^*$, then $|F(e)| \leq F(1)$ for every basis monomial e . Thus every normalized $F \in C_r^*$ has all coordinates in $[-1, 1]$, and the same bound holds for the local moments of every feasible family. The set of such F is closed, being cut out by the closed half-spaces $F(c) \geq 0$ and the hyperplane $F(1) = 1$. It is nonempty, because evaluation at a feasible point (x, y) is nonnegative on every generator: $w_I \geq 0$ on the box, and $\ell_{bj} \geq 0$ and $1 - y_{b,k}^2 \geq 0$ at feasible points. A linear function attains its minimum on a nonempty compact set.

(iii) Let $\lambda < \rho_r^{\text{rec}}$ and suppose $p = f - \lambda \notin C_r$. Since C_r is convex with nonempty interior, the separation theorem for convex sets in finite dimension [3, Theorem 2.4.2] gives a nonzero linear

functional F on W_r with $F(c) \geq F(p)$ for all $c \in C_r$. Because C_r is a cone containing 0, this gives $F \in C_r^*$ and $F(p) \leq 0$. By (i), $F \neq 0$ forces $F(1) > 0$. Then $G = F/F(1)$ is normalized and $G(f) \leq \lambda < \rho_r^{\text{rec}}$, contradicting the formula for ρ_r^{rec} .

(ii) If $f - \lambda \in C_r$, then every normalized $F \in C_r^*$ has $F(f) \geq \lambda$, so $\lambda_r^{\text{rec}} \leq \rho_r^{\text{rec}}$; with (iii), equality holds.

(iv) Apply (iii) with $\lambda = f^* - \mathcal{E} - \varepsilon < \rho_r^{\text{rec}}$. $\square$

Each element of C_r is a finite sum of the tested polynomials, so (iii) asserts a finite certificate with real coefficients: positive semidefinite Gram matrices for (R1) and sum-of-squares multipliers for the scalar localizers. The proposition gives no bound on the size of these coefficients, no rational certificate, and no method to compute one. The order-unit argument uses only (R1) and (R3). Without (R3) it fails, and [Proposition 6.11](#) shows that ρ_r^{rec} can then be $-\infty$. Every gap bound proved below for ρ_r^{rec} is also a bound on $f^* - \lambda_r^{\text{rec}}$, by (ii).

6.4 Fixed private polytopes

Theorem 6.10 (Fixed private domain). *Consider the recourse model with a fixed private domain ($E_b = 0$ for all b), nonempty polytopes P_b , and private convexity. Let $r \geq \max(w, d)$ and $m = \lfloor r/w \rfloor + 1$. Every feasible family (L_b) of order r admits a probability law ν on the feasible set with*

$$\int f d\nu \leq \sum_b L_b(f_b) + \frac{3A}{D_m}. \quad (6.11)$$

Consequently

$$0 \leq f^* - \rho_r^{\text{rec}} = f^* - \lambda_r^{\text{rec}} \leq \frac{3A}{2m^2 + 1} \leq \frac{3w^2 A}{2r^2}.$$

In particular, if no entry of any H_b depends on x , the hierarchy is exact at every order $r \geq w$.

Proof. Here $v_b(m-1) \leq w \lfloor r/w \rfloor \leq r$, so [Lemma 6.4](#) applies with $\tau = 0$; since $E_b = 0$, every row has $\epsilon_{bj} = 0$. Fix a point $y_b^\circ \in P_b$ and define $\phi_b(u) = \bar{y}_b(u)$ if $h_b(u) > 0$ and $\phi_b(u) = y_b^\circ$ otherwise. The set $\{h_b > 0\}$ is open and $\bar{y}_b$ is continuous there, so ϕ_b is Borel. By [Lemma 6.4\(d\)](#), $\phi_b(u) \in P_b$ for every u .

We claim that for every u ,

$$h_b(u) f_b(u, \phi_b(u)) \leq \langle H_b(u), M_b(u) \rangle. \quad (6.12)$$

If $h_b(u) = 0$, both sides vanish by [Lemma 6.4\(a\)](#). If $h = h_b(u) > 0$, write $\psi = \psi_b(u)$, $Y = Y_b(u)$, and $\bar{y} = \psi/h$. Then

$$\langle H_b(u), M_b(u) \rangle - h f_b(u, \bar{y}) = \text{tr}(Q_b(u)(Y - h^{-1}\psi\psi^\top)) \geq 0,$$

because $Y - h^{-1}\psi\psi^\top$ is the Schur complement of h in $M_b(u) \succeq 0$ and $Q_b(u) \succeq 0$.

Integrate (6.12), apply [Lemma 6.5](#) (its hypotheses hold because $d \leq r$), and sum over bags. [Lemma 6.6](#) gives a law ν on the feasible set with $\int f d\nu = \sum_b \int f_b(u, \phi_b(u)) h_b(u) d\mu^{B_b} \leq \sum_b L_b(f_b) + 3A/D_m$. Taking the infimum over feasible families gives the gap bound, and [Proposition 6.9](#) gives the equality with λ_r^{rec} . Finally $m > r/w$ gives $2m^2 + 1 > 2r^2/w^2$. $\square$

The comparison (6.11) is one-sided. The Jensen step (6.12) can strictly decrease the private cost, so the rounded objective can lie well below $\sum_b L_b(f_b)$; no bound on $|\int f d\nu - \sum_b L_b(f_b)|$ is claimed. The conditional-mean inequality (6.12) is the quadratic case of the pseudomoment Jensen

inequality for SOS-convex polynomials [20]. Kahl and Henrion’s partial lifting linearizes only a limited nonlinear subset; it does not guarantee asymptotic convergence in general [15]. Guo and Wang hold an SOS-convex decision-variable module at fixed degree while increasing the uncertainty-variable order in a robust polynomial matrix inequality hierarchy [11]. Neither result supplies the sparse private-degree-two recourse rate proved here. What the theorem adds is the quantitative rate for this sparse hierarchy, with fixed private polytopes that need not be full-dimensional. Because the private matrices are eliminated inside each bag and only scalar shared densities are glued, the failure of generic sparse matrix hierarchies under running intersection [26] does not arise here.

Both hypotheses beyond the fixed domain are needed, in a strong sense.

Proposition 6.11 (The private quadratic bounds are needed). *Remove (R3) from Definition 2.14. Take one bag with shared coordinates $x = (x_1, x_2, x_3)$, one private variable y with the two fiber rows $1 - y \geq 0$ and $1 + y \geq 0$ (so $P = [-1, 1]$ and $E = 0$), and*

$$f(x, y) = q(x) y^2, \quad q(x) = x_1^4 x_2^2 + x_1^2 x_2^4 + x_3^6 - 3x_1^2 x_2^2 x_3^2.$$

This instance satisfies private convexity and has $f^ = 0$. For every order $r \geq 3$, the relaxation without (R3) has infimum $-\infty$.*

The polynomial q is the Motzkin polynomial. The proof, in Appendix A, shows that q lies in no truncated preordering $P_r(\{1, 2, 3\})$, separates it from that closed cone by a normalized functional $\mathcal{L}$, and uses the feasible functionals $L_R(a(x) + b(x)y + c(x)y^2) = a(0) + R\mathcal{L}(c)$, whose objective $R\mathcal{L}(q)$ tends to $-\infty$.

Proposition 6.12 (Private convexity is needed). *Take one bag with one shared coordinate x_1 , which does not appear in the data, a private vector $y \in \mathbb{R}^2$ with fixed polytope $P = \{y \in [-1, 1]^2 : y_1 \geq 0, y_2 \geq 0, y_1 + y_2 \leq 1\}$, and the objective $f = -(y_1 + y_2)^2$, which violates private convexity. Then $f^* = -1$, while $\rho_r^{\text{rec}} \leq -4$ for every $r \geq 1$. Since $A = 0$ for this instance, Theorem 6.10 fails without private convexity.*

Proof. On P we have $0 \leq y_1 + y_2 \leq 1$, so $f^* = -1$. Let Λ be the linear functional on $\mathbb{R}[y]_{\leq 2}$ with $\Lambda(1) = 1$, $\Lambda(y_1) = \Lambda(y_2) = \frac{1}{2}$, and $\Lambda(y_1^2) = \Lambda(y_2^2) = \Lambda(y_1 y_2) = 1$. Its moment matrix

$$\begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 & 1 \\ \frac{1}{2} & 1 & 1 \end{pmatrix}$$

is positive semidefinite: the Schur complement of the leading entry is $\frac{3}{4} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$. Define $L(p) = \Lambda(p(0, \cdot))$ on $\mathbb{R}[x_1]_{\leq 2r} \otimes \mathbb{R}[y]_{\leq 2}$, evaluating the shared variable at zero. For a polynomial tested in (R1), $p(0, y) = w_I(0)(s_0(0) + s(0)^\top y)^2$, and Λ of a square of an affine form is nonnegative. For (R2), $p(0, y) = w_I(0)s(0)^2 \ell_j(y)$ with $\Lambda(y_1) = \Lambda(y_2) = \frac{1}{2}$ and $\Lambda(1 - y_1 - y_2) = 0$. For (R3), $\Lambda(1 - y_k^2) = 0$. There are no separator conditions. Hence L is feasible at every order, and $L(f) = -\Lambda(y_1^2 + 2y_1 y_2 + y_2^2) = -4$. $\square$

The shared coordinate in Proposition 6.12 only places the instance in the setting $w \geq 1$ of Definition 2.1; it plays no role. The private moment matrix above has no representing measure on P , since $\Lambda(y_1^2) = 1 > \Lambda(y_1)$ is impossible for a law on $[0, 1]$.

6.5 Affine recourse

When the fibers move with x , the conditional private mean is guaranteed to be feasible at the conditional input mean $\bar{x}_b(u)$ of Lemma 6.4(d), but need not be feasible at the output u . We repair it by Euclidean projection onto the output fiber and bound the cost using a Hoffman error bound. Write the fiber with its box rows as

$$P_b(x) = \{y : \hat{A}_b y \leq \hat{a}_b + \hat{E}_b x\}, \quad \hat{A}_b = \begin{pmatrix} A_b \\ I \\ -I \end{pmatrix}, \quad \hat{a}_b = \begin{pmatrix} a_b \\ \mathbf{1} \\ \mathbf{1} \end{pmatrix}, \quad \hat{E}_b = \begin{pmatrix} E_b \\ 0 \\ 0 \end{pmatrix}. \quad (6.13)$$

By Hoffman's theorem [14], in the form that is uniform over consistent right-hand sides [29], there is a constant $\kappa_b < \infty$, depending only on $\hat{A}_b$, such that for every vector c for which $\{z : \hat{A}_b z \leq c\}$ is nonempty and every $y \in \mathbb{R}^{p_b}$,

$$\text{dist}(y, \{z : \hat{A}_b z \leq c\}) \leq \kappa_b \|(\hat{A}_b y - c)_+\|_2. \quad (6.14)$$

Distances and norms of vectors are Euclidean, $\|E_b\|_2$ is the spectral norm, and $(\cdot)_+$ acts component-wise. Put

$$\Lambda_b = \max_{x \in [-1, 1]^{B_b}, y \in [-1, 1]^{p_b}} \|q_b(x) + 2Q_b(x)y\|_2, \quad \Gamma_b = \Lambda_b \kappa_b \|E_b\|_2, \quad \mathcal{R} = \sum_b \Gamma_b \sqrt{v_b}. \quad (6.15)$$

Since $\nabla_y f_b = q_b + 2Q_b y$, Λ_b is a Lipschitz constant of $f_b(x, \cdot)$ on the private box, uniformly in x . A crude bound is $\Lambda_b \leq \sum_\alpha \|q_{b\alpha}\|_2 + 2\sqrt{p_b} \sum_\alpha \|Q_{b\alpha}\|_2$, from the Chebyshev expansions of q_b and Q_b .

Lemma 6.13 (Continuous fibers and projections). *Assume complete recourse. For $x, x' \in [-1, 1]^{B_b}$ the Hausdorff distance satisfies $d_H(P_b(x), P_b(x')) \leq \kappa_b \|E_b\|_2 \|x - x'\|_2$, and the map $(x, z) \mapsto \text{proj}_{P_b(x)}(z)$ is continuous on $[-1, 1]^{B_b} \times \mathbb{R}^{p_b}$. Moreover, if $\bar{x} \in [-1, 1]^{B_b}$, $\bar{y} \in P_b(\bar{x})$ and $u \in [-1, 1]^{B_b}$, then*

$$\|\text{proj}_{P_b(u)}(\bar{y}) - \bar{y}\|_2 \leq \kappa_b \|E_b\|_2 \|\bar{x} - u\|_2. \quad (6.16)$$

Proof. For (6.16): $P_b(u)$ is nonempty, compact, and convex, so the projection exists and is unique. The box rows of $\hat{A}_b \bar{y} - \hat{a}_b - \hat{E}_b u$ are nonpositive, and in the rows of A_b , $A_b \bar{y} - a_b - E_b u \leq E_b(\bar{x} - u)$ because $A_b \bar{y} \leq a_b + E_b \bar{x}$. Hence $\|(\hat{A}_b \bar{y} - \hat{a}_b - \hat{E}_b u)_+\|_2 \leq \|E_b(\bar{x} - u)\|_2$, and (6.14) gives (6.16). Applying (6.16) to every point of $P_b(x')$ with $u = x$, and symmetrically, gives the Hausdorff bound.

For continuity, let $(x_n, z_n) \rightarrow (x, z)$ and $\pi_n = \text{proj}_{P_b(x_n)}(z_n)$. The π_n lie in the private box. Let $\bar{\pi}$ be the limit of a subsequence. Passing to the limit in $\hat{A}_b \pi_n \leq \hat{a}_b + \hat{E}_b x_n$ shows $\bar{\pi} \in P_b(x)$. For any $\zeta \in P_b(x)$, the Hausdorff bound gives $\zeta_n \in P_b(x_n)$ with $\zeta_n \rightarrow \zeta$, and $\|\pi_n - z_n\| \leq \|\zeta_n - z_n\|$ gives $\|\bar{\pi} - z\| \leq \|\zeta - z\|$ in the limit. So $\bar{\pi} = \text{proj}_{P_b(x)}(z)$. Since every subsequence has a further subsequence with this limit, π_n converges to it. $\square$

Lemma 6.14 (Integrated displacement). *Let $m \geq 2$ with $v_b(m-1) \leq r-1$. For every u with $h_b(u) > 0$,*

$$h_b(u) \|\bar{x}_b(u) - u\|_2^2 \leq \sum_{i \in B_b} L_b(K_{m,b}(\cdot, u) (x_i - u_i)^2), \quad (6.17)$$

and, defining the left side to be zero where $h_b(u) = 0$,

$$\int h_b(u) \|\bar{x}_b(u) - u\|_2^2 d\mu^{B_b}(u) \leq v_b V_m, \quad \int h_b(u) \|\bar{x}_b(u) - u\|_2 d\mu^{B_b}(u) \leq \sqrt{v_b V_m}.$$

Proof. Write $K = K_{m,b}(\cdot, u)$, $h = h_b(u)$, and $U_i = L_b(Kx_i)$. Then $L_b(K(x_i - u_i)^2) - h(\bar{x}_{b,i} - u_i)^2 = L_b(Kx_i^2) - U_i^2/h \geq 0$, by the 2×2 minor of $\widehat{M}_b(u) \succeq 0$ formed by the entries for 1 and x_i (Lemma 6.4(c)). Summing over i gives (6.17), which also holds where $h = 0$. The right side is a polynomial in (x, u) inside L_b . Integrating over u removes the factors $K_m(x_j, u_j)$, $j \neq i$, and Lemma 6.1(b) gives

$$\int L_b(K(x_i - u_i)^2) d\mu^{B_b}(u) = V_m + \frac{3 - 2m}{a_0} L_b(x_i^2) \leq V_m,$$

since $L_b(x_i^2) \geq 0$. Sum over i . The last bound follows from the Cauchy–Schwarz inequality and $\int h_b d\mu^{B_b} = 1$. $\square$

The constant V_m is the best possible in this lemma: for the evaluation functional at $x = 0$, equality holds in (6.17) and in the integral. This sharpness concerns only the displacement estimate.

Theorem 6.15 (Affine recourse). *Consider the recourse model with complete recourse and private convexity. Let $r \geq \max(w + 1, d)$ and $m = \lceil r/w \rceil$. Every feasible family (L_b) of order r admits a probability law ν on the feasible set with*

$$\int f d\nu \leq \sum_b L_b(f_b) + \frac{3A}{D_m} + \sqrt{V_m} \mathcal{R}. \quad (6.18)$$

Consequently

$$0 \leq f^* - \rho_r^{\text{rec}} = f^* - \lambda_r^{\text{rec}} \leq \frac{3A}{2m^2 + 1} + \sqrt{V_m} \mathcal{R} \leq \frac{3w^2 A}{2r^2} + \frac{\sqrt{3} w \mathcal{R}}{r}.$$

Proof. Since $m = \lfloor (r - 1)/w \rfloor + 1 \geq 2$, we have $v_b(m - 1) \leq r - 1$, so Lemma 6.4 applies with $\tau = 1$ and Lemma 6.14 applies. Define $\phi_b(u) = \text{proj}_{P_b(u)}(\bar{y}_b(u))$ if $h_b(u) > 0$ and $\phi_b(u) = \text{proj}_{P_b(u)}(0)$ otherwise. By Lemma 6.13, ϕ_b is continuous on the open set $\{h_b > 0\}$ and on its closed complement, so it is Borel, and $\phi_b(u) \in P_b(u)$ for every u ; complete recourse is used here at every output point.

Fix u with $h_b(u) > 0$. By (6.8) and (6.16), $\|\phi_b(u) - \bar{y}_b(u)\|_2 \leq \kappa_b \|E_b\|_2 \|\bar{x}_b(u) - u\|_2$. Both points lie in the private box, hence so does the segment between them, and

$$f_b(u, \phi_b(u)) \leq f_b(u, \bar{y}_b(u)) + \Gamma_b \|\bar{x}_b(u) - u\|_2.$$

Multiply by $h_b(u)$ and use (6.12), whose proof uses only $M_b(u) \succeq 0$ and $Q_b(u) \succeq 0$ and not feasibility of $\bar{y}_b(u)$ at u . Together with the case $h_b(u) = 0$, where both sides vanish, this gives for every u

$$h_b(u) f_b(u, \phi_b(u)) \leq \langle H_b(u), M_b(u) \rangle + \Gamma_b h_b(u) \|\bar{x}_b(u) - u\|_2.$$

Integrate, and use Lemmas 6.5 and 6.14: $\int h_b f_b(u, \phi_b(u)) d\mu^{B_b} \leq L_b(f_b) + 3A_b/D_m + \Gamma_b \sqrt{v_b V_m}$. Summing over bags and applying Lemma 6.6 gives (6.18). The infimum over feasible families and Proposition 6.9 give the gap bound. Finally $m \geq r/w$, so $D_m > 2r^2/w^2$ and $\sqrt{V_m} < \sqrt{6/D_m} < \sqrt{3}w/r$. $\square$

If every E_b is zero, then $\mathcal{R} = 0$ and Theorem 6.10 already gives the second-order bound without the one-degree reserve. The objective smoothing still costs only $O(r^{-2})$; the first-order term comes entirely from moving the private mean from the fiber at $\bar{x}_b(u)$ to the fiber at u . Repairing a relaxation’s private first moments by a fixed-matrix error bound has precedents in moment relaxations for distributionally robust optimization [39], and Hoffman’s error bound is classical [14],

with computational treatments of its constants [29]. The contribution here is the combination with exactly consistent kernel densities in this sparse hierarchy and the resulting explicit rate.

The two structural hypotheses are used essentially. Complete recourse is needed because the kernel can place mass at every shared point, and a point with an empty fiber admits no repair. The fixed matrix A_b is needed twice: to pass from (R2) to source feasibility by taking conditional means, and for the uniform Hoffman bound. With a shared-dependent matrix neither step is available. For example $P(x) = \{y \in [-1, 1] : xy = 0\}$ has nonempty fibers, but $P(0) = [-1, 1]$ and $P(x) = \{0\}$ for $x \neq 0$, so the fibers are not Hausdorff continuous at 0. Discontinuous recourse of this type is a known phenomenon [40]. The constants κ_b can be large for nearly inconsistent constraint systems, and the theorem gives no method for computing a sharp Hoffman constant.

6.6 The inverse-order rate is sharp

We now show that the order $1/r$ in Theorem 6.15 cannot be improved under its hypotheses. The example has one shared coordinate x , a private $y_1 \in [-1, 1]$ in the first bag, a private $y_2 \in [0, 1]$ in the second bag, and

$$\min f = -xy_1 + y_2 \quad \text{subject to} \quad x, y_1 \in [-1, 1], \quad y_2 \in [0, 1], \quad y_2 \geq x, \quad y_2 \geq -x. \quad (6.19)$$

For fixed x , $\min_{y_1}(-xy_1) = -|x|$ and the smallest feasible y_2 is $|x|$, so $f^* = 0$. Both private problems are linear programs.

We consider two different hierarchies for this problem.

Private-degree-two hierarchy. Substitute $y_2 = (1 + \zeta)/2$ with $\zeta \in [-1, 1]$. The bags are $B_1 = B_2 = \{1\}$ with private vectors y_1 and ζ , the first fiber is $[-1, 1]$ (no rows), and the second fiber has the rows $1 - 2x + \zeta \geq 0$ and $1 + 2x + \zeta \geq 0$, that is, $A_2 = (-1, -1)^\top$, $a_2 = (1, 1)^\top$, and $E_2 = (-2, 2)^\top$. The objectives are $f_1 = -xy_1$ and $f_2 = (1 + \zeta)/2$. This is a recourse model with complete recourse ($\zeta = 1$ is always feasible) and private convexity, and Definition 2.14 gives values ρ_r^{rec} .

Total-degree hierarchy. Use the bags $\{y_1, x\}$ and $\{x, y_2\}$ and the generator lists $G_1 = (1 - y_1^2, 1 - x^2)$ and $G_2 = (1 - x^2, y_2, 1 - y_2, y_2 - x, y_2 + x)$. Let $T_{b,r}$ be the full preordering of G_b truncated at total degree $2r$: finite sums of $\sigma_J \prod_{g \in J} g$ over subsets $J \subseteq G_b$, with σ_J a sum of squares and each summand of total degree at most $2r$. Let ρ_r^{tot} be the infimum of $-L_1(xy_1) + L_2(y_2)$ over linear functionals L_b on polynomials of total degree at most $2r$ in the variables of bag b with $L_b(1) = 1$, $L_b \geq 0$ on $T_{b,r}$, and $L_1(x^j) = L_2(x^j)$ for $0 \leq j \leq 2r$. Let $\lambda_r^{\text{tot}} = \sup\{\lambda : f - \lambda \in T_{1,r} + T_{2,r}\}$. As in Lemma 2.6, Lemma 2.2 for the junction tree with these two bags gives $\lambda_r^{\text{tot}} \leq \rho_r^{\text{tot}} \leq f^* = 0$.

The two hierarchies are not compared by inclusion: the second has products of private constraints and higher private degrees, while the first has mixed moments of joint degree up to $2r + 2$.

Lemma 6.16 (A moment-matching witness for $|x|$). *For every integer $n \geq 0$ there are probability measures α, β on $[-1, 1]$ with $\int x^j d\alpha = \int x^j d\beta$ for $0 \leq j \leq n$ and*

$$\int |x| d\alpha - \int |x| d\beta \geq \frac{2}{9\pi(n+2)}.$$

Proof. Let N be the even integer with $n + 1 \leq N \leq n + 2$ and let $F_N(s) = \sum_{|j| < N} (1 - |j|/N) e^{ijs}$ be the Fejér kernel, which is nonnegative with $\int F_N ds / (2\pi) = 1$. Put

$$Q(\theta) = -\cos(2N(\theta - \frac{\pi}{2})) F_N(\theta - \frac{\pi}{2}) = -\sum_{|j| < N} \left(1 - \frac{|j|}{N}\right) \cos((2N + j)(\theta - \frac{\pi}{2})),$$

where the second form follows from $\cos(2Ns)e^{ijs} + \cos(2Ns)e^{-ijs}$ pairing. Then $\int |Q| d\theta/(2\pi) \leq 1$, and all frequencies $2N + j$ lie in $[N + 1, 3N - 1]$. Since $p(\cos \theta)$ is a trigonometric polynomial of frequencies at most $n < N + 1$ for $\deg p \leq n$, $\int p(\cos \theta) Q(\theta) d\theta/(2\pi) = 0$.

The function $|\cos \theta|$ is even with period π ; its cosine coefficients c_k vanish for odd k , and for even $k = 2l \geq 2$ a direct integration gives $c_k = -4(-1)^l/(\pi(k^2 - 1))$. Since $\cos(k(\theta - \frac{\pi}{2})) = \cos(k\theta)\cos(k\pi/2) + \sin(k\theta)\sin(k\pi/2)$ and $|\cos \theta|$ is even,

$$\int |\cos \theta| \cos(k(\theta - \frac{\pi}{2})) \frac{d\theta}{2\pi} = \frac{c_k \cos(k\pi/2)}{2} = -\frac{2}{\pi(k^2 - 1)} \quad (k \geq 2 \text{ even}),$$

and it is zero for odd k . The frequency $2N + j$ is even exactly when j is even, and then $(2N + j)^2 - 1 < 9N^2$. Hence

$$\int |\cos \theta| Q(\theta) \frac{d\theta}{2\pi} = \sum_{\substack{|j| < N \\ j \text{ even}}} \left(1 - \frac{|j|}{N}\right) \frac{2}{\pi((2N + j)^2 - 1)} \geq \frac{2}{9\pi N^2} \cdot \frac{N}{2} = \frac{1}{9\pi N},$$

because, with $N = 2M$, the even-index weights sum to $1 + 2 \sum_{i=1}^{M-1} (1 - i/M) = M$.

Let ς be the image of the signed measure $Q(\theta) d\theta/(2\pi)$ under $\theta \mapsto \cos \theta$. It has total variation at most one, total mass zero (frequency zero is absent), annihilates all polynomials of degree at most n , and satisfies $\int |x| d\varsigma \geq 1/(9\pi N) > 0$. Write its Jordan decomposition $\varsigma = \varsigma_+ - \varsigma_-$. Both parts have the same mass t , with $0 < t \leq \frac{1}{2}$. The probability measures $\alpha = \varsigma_+/t$ and $\beta = \varsigma_-/t$ have equal moments through degree n , and $\int |x| d(\alpha - \beta) = t^{-1} \int |x| d\varsigma \geq 2/(9\pi N) \geq 2/(9\pi(n + 2))$. $\square$

The lemma is a quantitative form of the classical fact that $|x|$ is approximated by polynomials of degree n only to accuracy of order $1/n$; sharper constants are classical. The identity relating the best moment matching difference to best uniform approximation is recorded in [Lemma 5.1](#); we need only the explicit lower bound.

Theorem 6.17 (Sharpness of the inverse-order rate). *For problem (6.19) and every integer $r \geq 2$:*

(a) (private-degree-two hierarchy)

$$\frac{1}{9\pi(r + 1)} \leq -\rho_r^{\text{rec}} \leq \frac{3}{2r^2 + 1} + \sqrt{2V_r} < \frac{3}{2r^2 + 1} + \frac{2\sqrt{3}}{\sqrt{2r^2 + 1}};$$

(b) (total-degree hierarchy)

$$\frac{1}{9\pi(r + 1)} \leq -\rho_r^{\text{tot}} \leq -\lambda_r^{\text{tot}} \leq 2\sqrt{V_r} < \frac{2\sqrt{6}}{\sqrt{2r^2 + 1}} \leq \frac{2\sqrt{3}}{r}.$$

In both hierarchies the gap has exact order $1/r$, and neither is exact at any finite order.

Proof. Lower bounds. Apply [Lemma 6.16](#) with $n = 2r$. Give the first bag the law of $(x, \text{sign}(x))$ with $x \sim \alpha$ and $\text{sign}(0) = 0$, and the second bag the law of $(x, |x|)$ with $x \sim \beta$ (in the private-degree-two encoding, $\zeta = 2|x| - 1$). These laws are supported on the local feasible sets. Integration against them defines functionals that are nonnegative on every polynomial tested in either hierarchy, since each such polynomial is nonnegative on the local feasible set, and the separator moments agree through degree $2r$. Their objective is $-\int |x| d\alpha + \int |x| d\beta \leq -2/(9\pi(2r + 2))$. Hence both moment values are at most $-1/(9\pi(r + 1))$.

Upper bound in (a). We apply [Theorem 6.15](#) with $w = 1$, $d = 1$, and $m = r$. Here $H_1(x) = xH_{1,(1)}$ with $H_{1,(1)} = \begin{pmatrix} 0 & -1/2 \\ -1/2 & 0 \end{pmatrix}$, so $A_1 = 1$, and H_2 is constant, so $A = 1$. The first bag has no rows, so

$\Gamma_1 = 0$. For the second, $\Lambda_2 = \frac{1}{2}$ and $\|E_2\|_2 = 2\sqrt{2}$. All rows of $\hat{A}_2 = (-1, -1, 1, -1)^\top$ have absolute value one, so a nonempty solution set is an interval cut out by half-lines, and the distance from a point to that interval is the largest distance to one of the half-lines, which equals the largest positive residual. Hence $\kappa_2 = 1$ is admissible, $\Gamma_2 = \sqrt{2}$, and $\mathcal{R} = \sqrt{2}$. The bound follows from $V_r < 6/(2r^2 + 1)$.

Upper bound in (b). Let $m = r$, let $s_m(x) = \int K_m(x, t) \text{sign}(t) d\mu(t)$ and $\delta_m = 2\sqrt{V_m}$. Since $K_m \geq 0$ integrates to one, $|s_m| \leq 1$. By (6.3), $s_m = \sum_{k \geq 1} 2\gamma_k T_k \int T_k \text{sign} d\mu$, and the integrals vanish for even k ; so s_m is odd of degree at most $2m - 3$. For all $x, t \in [-1, 1]$, $0 \leq |x| - x \text{sign}(t) \leq 2|t - x|$; the middle expression is 0 if the signs agree, $|x|$ if $t = 0$, and $2|x| \leq 2|t - x|$ if the signs are opposite. Integrating against $K_m(x, t) d\mu(t)$ and using the Cauchy–Schwarz inequality and (6.5),

$$0 \leq |x| - x s_m(x) \leq 2 \left(\int K_m(x, t) (x - t)^2 d\mu(t) \right)^{1/2} \leq \delta_m.$$

Put $p_m = x s_m + \delta_m$, so $p_m \geq |x|$ on $[-1, 1]$ and $\deg p_m \leq 2m - 2$. Then

$$f + \delta_m = \underbrace{\left(\frac{1 + y_1}{2} (p_m - x) + \frac{1 - y_1}{2} (p_m + x) \right)}_{= p_m - x y_1} + \underbrace{\left(\frac{1 + s_m}{2} (y_2 - x) + \frac{1 - s_m}{2} (y_2 + x) \right)}_{= y_2 - x s_m}.$$

The polynomials $p_m \pm x$ and $(1 \pm s_m)/2$ are nonnegative on $[-1, 1]$ and have degree at most $2m - 2$. By Lemma 2.11 each equals $\sigma_0 + (1 - x^2)\sigma_1$ with $\sigma_0 \in \Sigma_{2m-2}$ and $\sigma_1 \in \Sigma_{2m-4}$. Also $(1 \pm y_1)/2 = (1 \pm y_1)^2/4 + (1 - y_1^2)/4$. Expanding, the first bracket is a sum of terms σ , $(1 - x^2)\sigma$, $(1 - y_1^2)\sigma$, and $(1 - y_1^2)(1 - x^2)\sigma$ with sums of squares σ , each of total degree at most $2m$; it lies in $T_{1,m}$. The second bracket is a sum of terms $\sigma(y_2 \mp x)$ and $(1 - x^2)\sigma(y_2 \mp x)$ of total degree at most $2m - 1$; it lies in $T_{2,m}$. Thus $\lambda_r^{\text{tot}} \geq -\delta_r$. The final inequalities use $V_r < 6/(2r^2 + 1)$ and $6/(2r^2 + 1) \leq 3/r^2$. $\square$

The lower bound is a statement about actual local measures: it holds for every relaxation whose local conditions are valid for measures on the local feasible sets and whose separator information is polynomial of degree at most $2r$, however strong its local positivity conditions are. The upper bounds are proved separately for each hierarchy; the certificate in (b) uses the products $(1 - x^2)(y_2 \mp x)$ of the full preordering. Qualitative failure of finite sparse exactness with a dense certificate is known from a two-bag example whose separator function is rational [27]; here the separator function is $|x|$, and the rate is identified. Bernstein’s theorem on best uniform approximation of $|x|$ gives the exact asymptotics of the local-measure gap; the leading constants of the finite hierarchy gaps are not determined here.

Remark 6.18 (Merging or splitting removes the gap for this instance). The obstruction concerns the fixed two-bag decomposition, not the difficulty of (6.19). The identity

$$f = \frac{(1 + y_1)^2}{4} (y_2 - x) + \frac{(1 - y_1)^2}{4} (y_2 + x) + \frac{1 - y_1^2}{2} y_2$$

is a degree-three certificate in the dense preordering of the merged bag $\{y_1, x, y_2\}$; its last term multiplies generators of different bags. Alternatively, splitting the problem at $x = 0$ and adding the generator x (respectively $-x$) to the first bag gives the exact certificates $f = (y_2 - x) + x(1 - y_1)$ and $f = (y_2 + x) + (-x)(1 + y_1)$, where $1 \mp y_1 = \frac{1}{2}(1 \mp y_1)^2 + \frac{1}{2}(1 - y_1^2)$; each has degree three and keeps the two bags. These are properties of this instance only.

6.7 Comparison with discretization

Corollary 6.19 (Comparison of a fixed grid with every moment family). *Let d_∞ be the largest individual shared degree in any entry of H_b , with zero for constant entries. Under the hypotheses and parameter choice of [Theorem 6.10](#), put $N = m + \lfloor d_\infty/2 \rfloor$. Under those of [Theorem 6.15](#), instead put $N = m + \lfloor \max(d_\infty, 2)/2 \rfloor$. In both cases let $\Xi_N = \{\cos((2j-1)\pi/(2N)) : 1 \leq j \leq N\}$ and $F_b(u) = \min_{y \in P_b(u)} f_b(u, y)$. Every feasible family satisfies*

$$\min_{x \in \Xi_N^n} \sum_b F_b(x_{B_b}) \leq \sum_b L_b(f_b) + \frac{3A}{D_m} + \begin{cases} 0, & \text{fixed private domain,} \\ \sqrt{V_m} \mathcal{R}, & \text{affine recourse.} \end{cases}$$

After the $\sum_b N^{v_b}$ local QP values have been computed, junction-tree dynamic programming finds the grid minimum in $O(tN^w)$ arithmetic and comparison operations. This count excludes construction and certification of the QP tables.

Proof. By [Lemma 3.6](#), equal-weight quadrature on Ξ_N integrates univariate polynomials through degree $2N-1$ exactly. The masses $N^{-v_b} h_b(u)$ at tensor grid nodes are nonnegative, sum to one, and have exactly matching separator marginals: quadrature deletes each nonseparator kernel factor, and the remaining separator density is the one in [Lemma 6.6](#). Finite-state gluing therefore gives a global shared grid law with these bag marginals.

In the fixed case, the pointwise Jensen inequality gives $h_b(u)F_b(u) \leq \langle H_b(u), M_b(u) \rangle$, including zero-density nodes. The matrix expression is polynomial of individual degree at most $2m-2+d_\infty \leq 2N-1$. Exact quadrature and [Lemma 6.5](#) bound its grid expectation by $L_b(f_b) + 3A_b/D_m$.

In the affine case, use the repaired decision from [Theorem 6.15](#) at each grid node. The pointwise matrix and repair bounds give $h_b F_b \leq \langle H_b, M_b \rangle + \Gamma_b h_b \|\bar{x}_b - u\|_2$. The displacement polynomials $L_b(K_{m,b}(\cdot, u)(x_i - u_i)^2)$ have individual output degree at most $2m$, and the chosen N satisfies $2N-1 \geq 2m$. Hence quadrature reproduces the displacement integral in [Lemma 6.14](#). Finite-sum Cauchy–Schwarz bounds the grid expectation of $h_b \|\bar{x}_b - u\|_2$ by $\sqrt{v_b V_m}$. Summing the bag bounds gives the stated cost expectation, and some global grid assignment has no larger cost. The dynamic program visits at most N^{v_b} entries per bag and per parent–child update; the tree has $t-1$ edges, giving $O(tN^w)$ once the local tables are supplied. Grid nodes are algebraic and coefficients may be real; the count is an exact-arithmetic or oracle statement, not a bit-complexity bound. $\square$

Remark 6.20 (Grid baseline). The exponents of [Theorems 6.10](#) and [6.15](#) are also reached without any semidefinite program, by discretizing the shared coordinates and solving convex quadratic programs. Let $N \geq 1$, $\xi_j = \cos((2j-1)\pi/(2N))$ for $1 \leq j \leq N$, $\Xi = \{\xi_1, \dots, \xi_N\}$, and $F_b(x) = \min_{y \in P_b(x)} f_b(x, y)$. Then

$$\min_{x \in \Xi^n} \sum_b F_b(x_{B_b}) \leq f^* + \frac{\pi^2 A}{8N^2} + \begin{cases} 0, & \text{fixed private domain,} \\ \pi \mathcal{R}/N, & \text{complete affine recourse.} \end{cases} \quad (6.20)$$

Once the $\sum_b N^{v_b}$ values F_b on the grid are known, ordinary junction-tree dynamic programming minimizes the left side.

To prove (6.20), let (x^*, y^*) be optimal and write $x_i^* = \cos \theta_i^*$ with $\theta_i^* \in [0, \pi]$. The angles $(2j-1)\pi/(2N)$, $j \in \mathbb{Z}$, have spacing $\Delta = \pi/N$, and all their cosines lie in Ξ . Round each shared coordinate independently: pick the two neighboring angles of θ_i^* with the probabilities that make the mean angle equal to θ_i^* , and let X_i be the cosine of the chosen angle. Linear interpolation of $\theta \mapsto \cos(k\theta)$ has error at most $k^2 \Delta^2/8$, so $|\mathbb{E} T_k(X_i) - T_k(x_i^*)| \leq k^2 \Delta^2/8$. By independence and since all factors have absolute value at most one, $|\mathbb{E} T_\alpha(X) - T_\alpha(x^*)| \leq \Delta^2 \sum_i \alpha_i^2/8$. Since the

entries of $\mathbf{z}_b^*$ have absolute value at most one, $\mathbb{E} \sum_b f_b(X_{B_b}, y_b^*) \leq f^* + \pi^2 \mathbf{A} / (8N^2)$. With a fixed private domain $y_b^* \in P_b$, and $F_b(X_{B_b}) \leq f_b(X_{B_b}, y_b^*)$. Under complete recourse, replace y_b^* by its projection onto $P_b(X_{B_b})$. Since $|X_i - x_i^*| \leq \Delta$, (6.16) and the Lipschitz bound Λ_b increase the cost of bag b by at most $\Gamma_b \sqrt{v_b} \pi / N$. Some realization of X does no worse than the expectation.

With N proportional to r , (6.20) has the same exponents as the two theorems; the argument does not even use private convexity, which only makes the local problems convex. Chebyshev-grid bounds of this type are established [31]. The theorems of this section are of a different kind: they bound every feasible point of the private-degree-two hierarchy, and hence its moment value and certificate value, rather than the value of a particular discrete search. We make no comparison of the computational effort of the two approaches.

7 Regular projected recourse multipliers

The inverse-order term in Theorem 6.15 pays for a private feasibility repair using only its length. A regular optimal multiplier projection permits a finer estimate: the mean displacement cancels to first order under the normalized kernel. We prove an inverse-square bound under a weighted tensor Chebyshev condition, and an $O(r^{-(1+\beta)})$ bound under coordinatewise Hölder regularity, including the Lipschitz endpoint $\beta = 1$. Both statements concern the unchanged private-degree-two hierarchy of Definition 2.14. The multipliers are used in the proof; they are not variables or inputs of the semidefinite relaxation.

Throughout this section we assume complete affine recourse and private convexity, and retain d , $\mathbf{A}_b$, and $\mathbf{A}$ from (6.2). Set

$$r \geq \max(w + 1, d), \quad m = \left\lfloor \frac{r-1}{w} \right\rfloor + 1 = \left\lceil \frac{r}{w} \right\rceil \geq 2, \quad D_m = 2m^2 + 1, \quad (7.1)$$

and use V_m from (6.4). Thus $v_b(m-1) \leq r-1$ for every bag. This single degree reserve will permit affine localizers, source shared moments, and the multiplier commutator.

7.1 The KKT projection and its sign

For the stacked system (6.13), write $D_b = \hat{E}_b = [E_b; 0; 0]$ and

$$F_b(u) = \min_{y \in P_b(u)} f_b(u, y).$$

At each u suppose we choose an optimal private point $y_b^*(u)$ and nonnegative multipliers $\pi_b(u)$ satisfying

$$\begin{aligned} q_b(u) + 2Q_b(u)y_b^*(u) + \hat{A}_b^\top \pi_b(u) &= 0, \\ \pi_b(u)^\top (\hat{a}_b + D_b u - \hat{A}_b y_b^*(u)) &= 0. \end{aligned} \quad (7.2)$$

Such pairs exist pointwise even for lower-dimensional or degenerate fibers. To see this directly, let I be the active rows at an optimal point and let g be its private objective gradient. If $-g$ were outside the cone generated by the active row normals, separation from that closed finitely generated cone would give a direction v with $\hat{A}_{b,I} v \leq 0$ and $g^\top v < 0$. For sufficiently small positive t , the inactive rows remain satisfied at $y_b^* + tv$, and the active rows are satisfied as well. Differentiability would then give a smaller objective, a contradiction. Hence $-g = \hat{A}_{b,I}^\top \pi_I$ with $\pi_I \geq 0$; extending by zero supplies (7.2). Existence of an optimizer follows from nonempty compact fibers.

Our substantive assumptions concern only the projected multipliers

$$a_b^{\text{dual}}(u) = D_b^\top \pi_b(u) \in \mathbb{R}^{B_b}. \quad (7.3)$$

The regular projection need not arise from a bounded, continuous, or measurable selection of the full vector π_b . No integral below uses that full vector.

The sign in (7.3) follows our convention $\hat{A}_b y \leq \hat{a}_b + D_b u$. Quadratic expansion, stationarity, and complementarity give, for every private vector y ,

$$f_b(u, y) - F_b(u) - \pi_b(u)^\top (\hat{a}_b + D_b u - \hat{A}_b y) = (y - y_b^*(u))^\top Q_b(u) (y - y_b^*(u)) \geq 0. \quad (7.4)$$

In particular this is a lower bound on $f_b(u, y)$ even if y is not feasible at u .

For a feasible moment family, use $h_b, \psi_b, Y_b, M_b, U_b$ from Section 6.2, with m as in (7.1). When $h_b(u) > 0$, the means $\bar{x}_b = U_b/h_b$ and $\bar{y}_b = \psi_b/h_b$ obey $\bar{y}_b \in P_b(\bar{x}_b)$. Apply (7.4) at $\bar{y}_b$ and use

$$\pi_b(u)^\top (\hat{a}_b + D_b u - \hat{A}_b \bar{y}_b) = \pi_b(u)^\top (\hat{a}_b + D_b \bar{x}_b - \hat{A}_b \bar{y}_b) + a_b^{\text{dual}}(u)^\top (u - \bar{x}_b).$$

The first term is nonnegative. Matrix Jensen (6.12) therefore gives

$$h_b(u) F_b(u) \leq \langle H_b(u), M_b(u) \rangle + a_b^{\text{dual}}(u)^\top (U_b(u) - u h_b(u)). \quad (7.5)$$

The correction has a plus sign and uses $U_b - u h_b$. At zero density $M_b = U_b = 0$ by Lemma 6.4, so the same inequality holds without division by zero.

7.2 A moment bound at the required frequencies

We first give the positivity statement that will control the correction for truncated functionals. The usual bound requiring $|\alpha| \leq r$ does not reach all its frequencies.

Lemma 7.1 (Half-degree Chebyshev moments). *Let $L : \mathbb{R}[x_B]_{\leq 2r} \rightarrow \mathbb{R}$ be $P_r(B)$ -positive. If $\sum_{i \in B} \lceil \alpha_i/2 \rceil \leq r$, then*

$$1 \pm T_\alpha \in P_r(B), \quad |L(T_\alpha)| \leq L(1).$$

Proof. For each k , $1 \pm T_k$ is nonnegative on the interval, and Lemma 2.11 gives a certificate of degree at most $2\lceil k/2 \rceil$. The even and odd cases can also be read directly from

$$\begin{aligned} 1 + T_{2j} &= 2T_j^2, & 1 - T_{2j} &= 2(1 - x^2)U_{j-1}^2 \quad (j \geq 1), \\ 1 + T_{2j+1} &= (1+x)(U_j - U_{j-1})^2, & 1 - T_{2j+1} &= (1-x)(U_j + U_{j-1})^2 \quad (j \geq 0), \end{aligned}$$

with $U_{-1} = 0$. In the odd case substitute $1 \pm x = ((1 \pm x)^2 + (1 - x^2))/2$; the resulting degree is at most $2j + 2$. For $k = 0$ the certificates are constants.

Put $z_i = T_{\alpha_i}(x_i)$ and let $k = |B| > 0$. For $\varepsilon \in \{-1, 1\}$ the parity identity is

$$1 + \varepsilon \prod_{i \in B} z_i = 2^{1-k} \sum_{\substack{\eta \in \{-1, 1\}^B \\ \prod_i \eta_i = \varepsilon}} \prod_{i \in B} (1 + \eta_i z_i). \quad (7.6)$$

Expansion proves the identity: averaging over the prescribed parity retains only the empty product and the product over all coordinates. Multiply the univariate certificates in each term. Products of their sums of squares are sums of squares, and all generators concern distinct coordinates, so the resulting polynomials belong to the full box preordering. Their degree is at most $2 \sum_i \lceil \alpha_i/2 \rceil \leq 2r$. Applying L to both signs proves the bound. For an empty bag $T_\alpha = 1$ and the assertion is immediate. $\square$

For a bounded Borel function a on the shared box of bag b , define

$$(\mathcal{K}_b a)(x) = \int K_{m,b}(x, u) a(u) d\mu^{B_b}(u).$$

The kernel has a finite polynomial expansion, so $\mathcal{K}_b a$ is a polynomial. Integrating the correction in (7.5) and interchanging integration with the finite-dimensional functional gives

$$\begin{aligned} \int a_b^{\text{dual}}(u)^\top (U_b(u) - u h_b(u)) d\mu^{B_b}(u) &= L_b(R_b), \\ R_b(x) &= \sum_{i \in B_b} \left\{ x_i \mathcal{K}_b(a_{bi}^{\text{dual}})(x) - \mathcal{K}_b(u_i a_{bi}^{\text{dual}})(x) \right\}. \end{aligned} \quad (7.7)$$

Every tensor Chebyshev frequency in R_b has individual degrees at most $2m - 2$, except possibly one coordinate with degree at most $2m - 1$. Thus

$$|\alpha| \leq 2v_b(m - 1) + 1 \leq 2r - 1, \quad \sum_i \left\lceil \frac{\alpha_i}{2} \right\rceil \leq v_b(m - 1) + 1 \leq r. \quad (7.8)$$

Consequently R_b lies in the actual shared moment domain and every one of its basis moments is bounded by Lemma 7.1. Membership in the degree- $2r$ domain alone would not justify replacing $L_b(R_b)$ by a pointwise bound for a multivariate polynomial.

7.3 The two regularity bounds

Theorem 7.2 (Weighted tensor Chebyshev regularity). *Suppose the pointwise KKT pairs can be chosen so that*

$$a_{bi}^{\text{dual}}(u) = \sum_{\alpha \in \mathbb{N}^{B_b}} a_{bi\alpha} T_\alpha(u), \quad \mathcal{M} = \sum_{b,i,\alpha} |a_{bi\alpha}| \max(1, 2\alpha_i) < \infty. \quad (7.9)$$

With the parameters (7.1), every feasible family (L_b) admits a law ν on the feasible set with

$$\int f d\nu \leq \sum_b L_b(f_b) + \frac{3(A + \mathcal{M})}{D_m}. \quad (7.10)$$

Hence

$$0 \leq f^* - \rho_r^{\text{rec}} = f^* - \lambda_r^{\text{rec}} \leq \frac{3(A + \mathcal{M})}{D_m} \leq \frac{3w^2(A + \mathcal{M})}{2r^2}.$$

The weight in (7.9) is directional: for component i it weights only frequency α_i . Because it is at least one, the assumption implies absolute uniform convergence on the box. Polynomial projections satisfy it; geometrically decaying tensor coefficients do as well. We do not infer it from arbitrary multivariate Lipschitz continuity.

Theorem 7.3 (Coordinatewise Hölder regularity). *Instead suppose that for some $0 < \beta \leq 1$ the projection satisfies*

$$a_{bi}^{\text{dual}}(u) = \phi_{bi}(u_i), \quad \|\phi_{bi}\|_\infty \leq M_{bi}, \quad |\phi_{bi}(v) - \phi_{bi}(v')| \leq H_{bi}|v - v'|^\beta. \quad (7.11)$$

Then every feasible family admits a law ν on the feasible set with

$$\int f d\nu \leq \sum_b L_b(f_b) + \frac{3A}{D_m} + \sum_{b,i} \left\{ \frac{3M_{bi}}{D_m} + H_{bi}V_m^{(1+\beta)/2} \right\}. \quad (7.12)$$

The same error bounds both $f^* - \rho_r^{\text{rec}}$ and $f^* - \lambda_r^{\text{rec}}$. In particular it is at most

$$\frac{3w^2}{2r^2} \left(A + \sum_{b,i} M_{bi} \right) + \left(\frac{\sqrt{3}w}{r} \right)^{1+\beta} \sum_{b,i} H_{bi}.$$

For fixed data the rate is $O(r^{-(1+\beta)})$, including $O(r^{-2})$ when $\beta = 1$. No sharpness is asserted for $0 < \beta < 1$.

Coordinatewise dependence is automatic when a bag has one shared coordinate. For larger bags it is an additional assumption on the projection. It places no restriction on coupled polynomial master costs c_b , which do not enter private stationarity. When every shared bag has size at most one, the model separates across distinct shared coordinates; many private blocks may still depend on the same coordinate. The scalar case alone is therefore not a theorem for general shared networks.

Correction estimate for Theorem 7.2. We extend γ_j by zero beyond $2m - 2$. For every integer $j \geq 0$,

$$|\gamma_{j+1} - \gamma_j| \leq (2j + 1)(1 - \gamma_1) = \frac{3(2j + 1)}{D_m}. \quad (7.13)$$

Indeed, (6.3) gives $\gamma_j = \int \cos(jt) J_m(t) dt / (2\pi)$ for every $j \geq 0$. The cosine difference formula and $|\sin(kt)| \leq k|\sin t|$ for integer $k \geq 1$ give

$$|\cos(jt) - \cos((j + 1)t)| \leq 2(2j + 1) \sin^2(t/2) = (2j + 1)(1 - \cos t).$$

The sine bound follows by telescoping $e^{2ikt} - 1$ into k terms of modulus $|e^{2it} - 1|$. Integrating against the nonnegative normalized circle kernel proves (7.13), including its support-boundary indices.

For a single tensor mode T_α in component i , all coordinates other than i contribute the factor $\prod_{j \neq i} \gamma_{\alpha_j} \leq 1$. Writing $k = \alpha_i \geq 1$, its one-coordinate contribution to the commutator is

$$\frac{1}{2}(\gamma_k - \gamma_{k+1})T_{k+1} + \frac{1}{2}(\gamma_k - \gamma_{k-1})T_{k-1}, \quad (7.14)$$

by $xT_k = (T_{k+1} + T_{k-1})/2$. Its absolute coefficient sum is at most $2k(1 - \gamma_1)$ by (7.13). For $k = 0$ the contribution is $(1 - \gamma_1)T_1$, so the bound is $\max(1, 2k)(1 - \gamma_1)$ in all cases.

A mode with $k = 2m - 1$ can survive through the term $\gamma_{k-1}T_{k-1}$ even though $\gamma_k = 0$; it is included in (7.14). Modes with $k > 2m - 1$ vanish, as does any mode with an out-of-range frequency in another coordinate. Thus the commutator has the finite support in (7.8). Absolute uniform convergence in (7.9) permits termwise integration; the coefficient bound also permits summing the absolute contributions. Lemma 7.1, applied to the normalized shared restriction of L_b , now gives

$$|L_b(R_b)| \leq \frac{3}{D_m} \sum_{i,\alpha} |a_{bi\alpha}| \max(1, 2\alpha_i). \quad (7.15)$$

□

Correction estimate for Theorem 7.3. Kernel normalization removes every coordinate other than i in (7.7). Hence $R_b(x) = \sum_{i \in B_b} R_{bi}(x_i)$, where the univariate polynomial

$$R_{bi}(x) = \int K_m(x, u) \phi_{bi}(u) (x - u) d\mu(u)$$

has degree at most $2m - 1$. Add and subtract $\phi_{bi}(x)$ in the integrand. By Lemma 6.1(b),

$$R_{bi}(x) = \phi_{bi}(x)(1 - \gamma_1)x + \int K_m(x, u) (\phi_{bi}(u) - \phi_{bi}(x)) (x - u) d\mu(u).$$

The first term has absolute value at most $3M_{bi}/D_m$. The second has absolute value at most $H_{bi} \int K_m(x, u)|x - u|^{1+\beta} d\mu(u)$. Since $K_m(x, u)d\mu(u)$ is a probability law and $t \mapsto t^{(1+\beta)/2}$ is concave for $0 < \beta \leq 1$,

$$\int K_m(x, u)|x - u|^{1+\beta} d\mu(u) \leq \left(\int K_m(x, u)(x - u)^2 d\mu(u) \right)^{(1+\beta)/2} \leq V_m^{(1+\beta)/2}.$$

This argument includes $\beta = 1$, when the concavity inequality is an equality. Put $\varepsilon_{bi} = 3M_{bi}/D_m + H_{bi}V_m^{(1+\beta)/2}$. The polynomial $\varepsilon_{bi} - R_{bi}$ is nonnegative on the interval and has degree at most $2m - 1$. By [Lemma 2.11](#) its certificate has degree at most $2m$. Since $m \leq r$, this certificate embeds in the shared scalar part of (R1), giving $L_b(R_{bi}) \leq \varepsilon_{bi}L_b(1) = \varepsilon_{bi}$. This step transfers the pointwise estimate to the truncated functional; it uses no representing measure for L_b . Summing over i proves the required correction bound. $\square$

Completion of [Theorems 7.2 and 7.3](#). We need Borel optimal private policies to realize the local values F_b . Their construction is independent of the KKT pair choices. For $\varepsilon > 0$, the strictly convex function $f_b(u, y) + \varepsilon\|y\|_2^2$ has a unique minimizer $y_{b,\varepsilon}(u)$ on $P_b(u)$. This minimizer is continuous in u . Indeed, along any convergent parameter sequence, private compactness gives a cluster point, and fiber Hausdorff continuity ([Lemma 6.13](#)) approximates every limit-fiber comparison point by feasible points in the sequence of fibers. Passing to the optimality inequalities shows that the cluster point is the unique regularized minimizer at the limit parameter. This proves continuity.

Fix u and any unregularized minimizer z . Regularized optimality gives

$$F_b(u) \leq f_b(u, y_{b,\varepsilon}(u)) \leq F_b(u) + \varepsilon\|z\|_2^2, \quad \|y_{b,\varepsilon}(u)\|_2^2 \leq \|z\|_2^2.$$

Every cluster point as $\varepsilon \downarrow 0$ is therefore an optimal point with minimal norm. The optimizer set is nonempty compact and convex, so its minimum-norm point is unique. Thus $y_{b,\varepsilon}(u)$ converges to that point. Taking, for example, $\varepsilon = 1/k$ proves that the limiting policy is Borel, being a pointwise limit of continuous policies. Its cost is $F_b(u)$.

Apply [Lemma 6.6](#) to these policies. The resulting feasible law has objective $\sum_b \int h_b F_b d\mu^{B_b}$. Integrate (7.5), use [Lemma 6.5](#) (which requires $d \leq r$), and then use the applicable correction estimate. This proves both rounding inequalities. Since $f^* \leq \int f d\nu$, taking the infimum over feasible families gives the moment gap. [Proposition 6.9](#) gives equality with the certificate supremum and certificates at every strict level. Finally $D_m > 2r^2/w^2$ and $V_m < 3w^2/r^2$ give the stated data-dependent rate bounds. $\square$

Neither theorem controls the size of the full multiplier vector. Their constants $\mathcal{M}$, M_{bi} , and H_{bi} can grow with private dimension and active-constraint conditioning. The largest PSD block is still $(p_b + 1)\binom{r+v_b}{v_b}$, as in (2.10); no KKT localizers have been added. A usable numerical accuracy guarantee requires bounds on the stated projection norms, and the theorems supply no general algorithm to compute them.

7.4 Sharpness and the boundary of the hypotheses

Example 7.4 (A sharp regular affine-recourse instance). Let $x \in [-1, 1]$ be shared, let $v, z \in [0, 1]$ be private to the two bags, and impose $z \geq x$. Minimize

$$f_1(x, v) + f_2(x, z) = v^2 - 2xv + z^2.$$

The first private minimum is $-x_+^2$ at $v = x_+$, and the second is x_+^2 at $z = x_+$, where $x_+ = \max(x, 0)$. Thus $f^* = 0$ and recourse is complete. In the first bag the projection is zero, since its constraints

are fixed. In the second, multiplier $2x_+$ on $-z \leq -x$ and zero multipliers on the box rows satisfy stationarity and complementarity, including at $x = 0$ and the endpoints. The projection is $a_2^{\text{dual}}(x) = -2x_+$, which has sup norm and Lipschitz constant two.

Use $v = (1 + \xi)/2$, $z = (1 + \zeta)/2$ to place private variables in the model box. The first bag has no additional rows and the second has $1 + \zeta - 2x \geq 0$. This is a positive scaling of the original row, so rescaling its multiplier leaves its projection unchanged. The only nonconstant shared matrix coefficient is

$$H_{1,(1)} = \begin{pmatrix} -1 & -1/2 \\ -1/2 & 0 \end{pmatrix},$$

giving $A = 2$. Consequently [Theorem 7.3](#), with $w = 1$, $d = 1$, $m = r$, $M = H = 2$, gives for every $r \geq 2$

$$-\rho_r^{\text{rec}} \leq \frac{12}{2r^2 + 1} + 2V_r < \frac{24}{2r^2 + 1} < \frac{12}{r^2}.$$

Conversely, choose the two measures of [Lemma 5.3](#) for $\varphi(x) = x_+^2$ and $n = 2r$. Lift the first by $v = x_+$ and the second by $z = x_+$, using the appropriate private affine scalings. Both lifts are locally feasible actual measures and their separator moments agree through degree $2r$. Their cost is $-\int \varphi d\mu_+ + \int \varphi d\mu_-$, whence

$$-\rho_r^{\text{rec}} \geq \frac{2}{27\pi(2r + 2)^2}.$$

Thus the regular affine-recourse rate is exactly $\Theta(r^{-2})$ for this fixed rectangular hierarchy. We transfer the separator measures, using the lift $z = x_+$ for this problem; the lift $z = (-x)_+$ from [Section 5](#) belongs to a different problem. No inclusion between the rectangular and total-degree cones is used.

Example 7.5 (Strict private convexity is insufficient). For $x \in [-1, 1]$, consider the scalar private problem

$$\min_{|x| \leq z \leq 1} (z^2 + z) = x^2 + |x|.$$

It has complete affine recourse, a constant positive-definite private Hessian, and the Lipschitz optimizer $z^*(x) = |x|$. For $0 < x < 1$, the only active non-box row is $-z \leq -x$, whose multiplier is $2x + 1$; the projection is $-(2x + 1)$. For $-1 < x < 0$, the active row is $-z \leq x$, whose multiplier is $1 - 2x$, giving projection $1 - 2x$. The box rows are inactive on both punctured neighborhoods of zero. Hence the projection has right limit -1 and left limit 1 . No choice of multipliers at zero can make it continuous. Strict private convexity and a Lipschitz primal policy therefore do not imply either regularity assumption of this section. This example tests the assumptions; [Theorem 6.17](#) separately proves an actual inverse-order sparse gap under the general affine-recourse hypotheses.

7.5 Comparison with parametric quadratic programming

Classical multiparametric QP results give piecewise-affine primal and multiplier policies under appropriate nondegeneracy assumptions [\[2, 35\]](#). They supply one useful route to our regularity assumptions. We record its elementary argument to make the closed-box boundary precise.

Proposition 7.6 (A sufficient scalar QP regime). *For each bag, suppose Q_b is a constant positive-definite matrix, q_b is affine, and the affine fibers remain nonempty on an open neighborhood of the closed shared box. Suppose that, at the unique private optimizer throughout this neighborhood, all active rows of $\hat{A}_b$, including active private box rows, are linearly independent (LICQ). Then the primal optimizer and the full KKT multiplier are continuous piecewise-affine functions on the closed*

box and are Lipschitz there. In particular, if each nonempty shared bag has size one, [Theorem 7.3](#) applies with $\beta = 1$.

Proof. Suppress the bag index. Positive definiteness gives a unique primal optimizer. For an independent active row set I , the KKT system is

$$\begin{pmatrix} 2Q & \hat{A}_I^\top \\ \hat{A}_I & 0 \end{pmatrix} \begin{pmatrix} y \\ \pi_I \end{pmatrix} = \begin{pmatrix} -q(u) \\ \hat{a}_I + D_I u \end{pmatrix}.$$

Its matrix is nonsingular: a homogeneous solution has $\hat{A}_I y = 0$, and multiplication of its first equation by $y^\top$ gives $2y^\top Q y = 0$, hence $y = 0$ and then $\pi_I = 0$ by row independence. The solution is affine in u . Extend π_I by zero; primal feasibility and $\pi_I \geq 0$ describe a closed polyhedral region on which this solution is optimal. There are only finitely many row sets, and every parameter under the hypotheses lies in one such region.

LICQ makes the multiplier at each optimizer unique, because its nonzero entries are supported on the active independent rows. Valid affine policies therefore agree wherever their regions overlap. Along any convergent parameter sequence, pass to a subsequence with one of the finitely many affine policies. Its limit is a valid KKT pair at the limit parameter, and uniqueness identifies it with that pair. Thus the policies are continuous, including at region boundaries and at the boundary of the closed box. Each line segment in the box is partitioned into finitely many intervals by these polyhedral regions. On every interval the policy has one affine slope. Summing along the intervals bounds the total change by the largest slope norm times the segment length, proving Lipschitz continuity. The projection by fixed $D^\top$ is Lipschitz as well. For scalar bags, each component is a function of its sole coordinate, so [\(7.11\)](#) follows. $\square$

The neighborhood assumption avoids applying an interior sensitivity statement to an unqualified boundary point. Strict complementarity and a single fixed active set are not required. LICQ is a sufficient route, rather than a hypothesis of [Theorems 7.2](#) and [7.3](#); the sharp example above need not satisfy it everywhere. For multivariate bags, the proposition gives a Lipschitz projection but does not establish weighted absolute tensor Chebyshev summability or coordinatewise dependence. Oblique active-region boundaries are therefore outside the automatic corollary.

On each critical region the affine KKT policy eliminates the private QP exactly, leaving $c_b(u)$ plus a quadratic function. Enumerating those regions is an established alternative to the hierarchy; no comparison of practical effort is proved here. Polynomial lower approximation of recourse values is also established in the joint+marginal approach [\[21\]](#) and in two-stage polynomial optimization [\[40\]](#). Sparse KKT reformulations explicitly introduce multiplier equations and auxiliary variables [\[33\]](#). Our theorems instead give finite-order bounds for the original hierarchy while keeping private degree two. The quantitative link uses the KKT inequality, the kernel commutator, and its certificate at the precise truncated degree. It is not a new QP sensitivity theorem or a claim that smoothness alone guarantees the stated sparse rate.

8 Polynomial constraints and finite-state variables

The box rounding laws need not satisfy additional polynomial inequalities. For constrained problems we first bound their expected squared violation, then use a global geometric bound to repair their support. This gives $O(r^{-\alpha})$ for the constrained preordering and $O((\log r/r)^\alpha)$ for the constrained ordinary module. Finite-state variables require a different extension: retain their labels exactly and smooth only the continuous coordinates. The same separator argument then gives both box rates without a further label-count factor.

8.1 Global feasible repair

Let the continuous bags and tree be as in [Definition 2.1](#), with $w \geq 1$. Assign finitely many polynomial inequalities to the bags and put

$$K = \{x \in [-1, 1]^n : g_{bj}(x_{B_b}) \geq 0 \text{ for every } b, j\} \neq \emptyset, \quad V(x) = \sum_{b,j} (-g_{bj}(x_{B_b}))_+^2. \quad (8.1)$$

Here $t_+ = \max\{t, 0\}$. In this subsection and the next two, $f^* = \min_K f$. Equalities can be supplied as opposite inequalities. A generator need be assigned only once; repeated assignments are counted repeatedly in our bounds. Define

$$d = \max\{1, \max_b \deg f_b, \max_{b,j} \deg g_{bj}\}, \quad G = \sum_{b,j} \|g_{bj}\|_{\text{ch}} \mathbf{A}(g_{bj}).$$

An empty constraint family contributes zero. Let L_f be a Euclidean Lipschitz constant of f on the entire box; one may take $L_f = \mathbf{A}$, since $|T'_k| \leq k^2$.

Assumption 8.1 (Global error bound). There are $H \geq 0$ and $0 < \alpha \leq 1$ such that

$$\text{dist}_2(x, K) \leq HV(x)^{\alpha/2} \quad (x \in [-1, 1]^n). \quad (8.2)$$

The distance in this assumption is to the common feasible intersection in all original coordinates. The exponent and constant depend on the supplied generators, as well as on the geometry of K .

Lemma 8.2 (Repair of a box law). *Under [Assumption 8.1](#), if a box-supported probability law σ satisfies $\int V d\sigma \leq U$, then a probability law ν supported on K satisfies*

$$\int f d\nu \leq \int f d\sigma + L_f H U^{\alpha/2}.$$

Some point of K has objective no greater than this upper bound.

Proof. For $\varepsilon > 0$ choose a finite ε -net of the nonempty compact set K . Map each box point to its nearest net point, breaking ties by a fixed ordering. This is a Borel map P_ε , and $\|x - P_\varepsilon x\|_2 \leq \text{dist}_2(x, K) + \varepsilon$. Lipschitz continuity, (8.2), and Jensen's inequality for the concave function $t^{\alpha/2}$ give

$$\int f(P_\varepsilon x) d\sigma(x) \leq \int f d\sigma + L_f H (\int V d\sigma)^{\alpha/2} + L_f \varepsilon.$$

The pushforward laws are supported on K . Compactness gives a weakly convergent subsequence as $\varepsilon \downarrow 0$; continuity of f gives the claim in the limit. The minimum on K is at most the resulting mean. $\square$

Proposition 8.3 (A common convex Slater point). *Suppose every g_{bj} is concave on the box, and one point x^0 in the box satisfies $g_{bj}(x^0) \geq \sigma_0 > 0$ for every supplied constraint. Then [Assumption 8.1](#) holds with $\alpha = 1$ and $H = 2\sqrt{n}/\sigma_0$. The objective need not be convex.*

Proof. For a box point x put $v = \max_{b,j} (-g_{bj}(x))_+$, with $v = 0$ for an empty family. Set $\theta = v/(\sigma_0 + v)$ and $z = (1 - \theta)x + \theta x^0$. Concavity yields $g_{bj}(z) \geq -(1 - \theta)v + \theta\sigma_0 = 0$, so $z \in K$. Therefore

$$\text{dist}_2(x, K) \leq \|z - x\|_2 \leq 2\sqrt{n}\theta \leq (2\sqrt{n}/\sigma_0)v \leq (2\sqrt{n}/\sigma_0)\sqrt{V(x)}.$$

$\square$

Example 8.4 (Local bounds do not give the global exponent). Take bags $\{x, y\}$ and $\{y, z\}$ in the three-dimensional box, with $g_1 = y - x^2$ and $g_2 = -y$. Each local set has a linear repair bound with constant one: increase y to $\max\{y, x^2\}$ for the first, or decrease it to $\min\{y, 0\}$ for the second. Both repairs stay in their bag boxes. Globally $K = \{x = y = 0\} \times [-1, 1]$. At $(t, 0, 0)$ its distance is $|t|$, whereas $V = t^4$. Thus a global bound with exponent α requires $\alpha \leq 1/2$. Both generators are concave and each has a local strictly feasible point, but they have no common Slater point. This example concerns repair geometry; it does not assert a hierarchy lower bound for these generators.

8.2 Constrained preordering

For $I \subseteq B_b$ write $q_I = \prod_{i \in I} (1 - x_i^2)$. The constrained preordering relaxation has normalized local functionals through degree $2r$, exact separator equalities through degree $2r$, and

$$L_b(q_I p^2) \geq 0, \quad L_b(g_{bj} q_I p^2) \geq 0 \quad (8.3)$$

whenever the displayed product has total degree at most $2r$. Its moment value is denoted $\rho_r^{c, \text{pre}}$. Products of two different additional generators are unnecessary. A full local preordering in all supplied generators satisfies these conditions; a module using only g_{bj} times squares need not satisfy the second condition for nonempty I .

Theorem 8.5 (Constrained preordering rounding). *Assume [Assumption 8.1](#) and $r \geq \max\{2d, d+w\}$. Put $m = \lfloor (r-d)/w \rfloor + 1$ and $D_m = 2m^2 + 1$. Every feasible family for (8.3) admits a probability law ν on K with*

$$\int f d\nu \leq \sum_b L_b(f_b) + \frac{3A}{D_m} + L_f H \left(\frac{12G}{D_m} \right)^{\alpha/2}. \quad (8.4)$$

The last two terms also bound $0 \leq f^* - \rho_r^{c, \text{pre}}$. For fixed data and width this is $O(r^{-\alpha})$.

Proof. Use K_{m, B_b} from [Lemma 3.2](#). Its preordering certificate has degree at most $2v_b(m-1) \leq 2(r-d)$. The same lemma gives nonnegative mass-one densities $h_b(u) = L_b(K_{m, B_b}(x, u))$ with exactly consistent separators, hence a box law σ with these marginals. As in [Theorem 3.4](#), $|\int f d\sigma - \sum_b L_b(f_b)| \leq 3A/D_m$.

Fix $g = g_{bj}$ and an output u . The conditional functional $F_u(p) = L_b(K_{m, B_b}(x, u)p(x))$ is positive on $(a + cg)^2$: multiplying the kernel certificate by this square costs at most $2d$ extra degrees. The second localizers in (8.3) also give $F_u(g) \geq 0$. Write $h = F_u(1) \geq 0$. If $g(u) < 0$, put $a = -g(u)$. Then $F_u(g - g(u)) \geq ah$ and conditional Cauchy–Schwarz gives $F_u(g - g(u))^2 \leq h F_u((g - g(u))^2)$. For $h > 0$ these imply $a^2 h \leq F_u((g - g(u))^2)$; for $h = 0$ the same inequality follows from positivity. If $g(u) \geq 0$ the left side is zero. Consequently, without normalizing by h ,

$$(-g(u))_+^2 h_b(u) \leq L_b(K_{m, B_b}(x, u)(g(x) - g(u))^2). \quad (8.5)$$

Let $\mathcal{J}_b$ be the kernel’s diagonal Chebyshev operator. Integrating (8.5) bounds the violation by $L_b(\mathcal{J}_b(g^2) - 2g\mathcal{J}_b g + g^2)$. This polynomial has degree at most $2 \deg g \leq 2d \leq r$, so [Lemma 2.8](#) applies to its coefficients. By [Lemmas 2.3](#) and [3.1](#),

$$\begin{aligned} \|\mathcal{J}_b(g^2) - 2g\mathcal{J}_b g + g^2\|_{\text{ch}} &\leq \frac{3}{D_m} [A(g^2) + 2\|g\|_{\text{ch}} A(g)] \\ &\leq \frac{12\|g\|_{\text{ch}} A(g)}{D_m}. \end{aligned}$$

Thus the same global law satisfies $\int V d\sigma \leq 12G/D_m$. Apply [Lemma 8.2](#). A point of K supplies a feasible Dirac family, so the moment value is at most f^* ; taking the infimum over families proves the gap bound. Finally $m = \Theta_w(r)$ for fixed d, w . $\square$

8.3 Constrained ordinary module

Here the local conditions are only

$$L_b(p^2) \geq 0, \quad L_b((1 - x_i^2)p^2) \geq 0, \quad L_b(g_{bj}p^2) \geq 0, \quad (8.6)$$

for products of degree at most $2r$, together with normalization and exact separator consistency. Denote the value by $\rho_r^{\text{c,mod}}$. Let d_∞ bound all coordinate degrees of objectives and constraints. Use $s, N, \Psi, \bar{\Psi}, \omega, \varrho, D, \delta, \Lambda, \eta, \Gamma_v$ and Δ_w from [Section 4](#), with this d_∞ , and set

$$\begin{aligned} a_0 &= 1 - \delta, \quad c_s = \frac{4}{sC_s} \leq \frac{6}{s^2}, \quad E = \sum_b C_b \Gamma_{v_b} + \Delta_w \sum_b \text{osc}(f_b), \\ U &= \frac{c_s \sum_{b,j} A(g_{bj})^2 + \sum_{b,j} \|g_{bj}\|_{\text{ch}}^2 (1 + \Delta_w - a_0^{v_b})}{1 + \Delta_w}. \end{aligned} \quad (8.7)$$

The stronger degree reserve below supports conditional positivity and constraint-difference certificates. The latter are necessary: a pointwise multivariate Lipschitz inequality alone cannot be evaluated under a truncated functional.

Lemma 8.6 (A constraint-difference certificate). *Write $g = \sum_\beta c_\beta T_\beta$ on a bag, and set $A_i(g) = \sum_\beta |c_\beta| \beta_i^2$, so $\sum_i A_i(g) = A(g)$. For every fixed u in the bag box,*

$$A(g) \sum_i A_i(g) (x_i - u_i)^2 - (g(x) - g(u))^2 \in M_{\deg g}(B). \quad (8.8)$$

Every summand of its certificate has degree at most $2 \deg g$; for constant g the polynomial is zero.

Proof. Order the bag coordinates and telescope each tensor Chebyshev term. For $\ell = (\beta, i)$ with $c_\beta \neq 0$ and $k = \beta_i > 0$, define

$$w_\ell = |c_\beta| k^2, \quad t_\ell = x_i - u_i, \quad R_\ell(x) = \text{sign}(c_\beta) \frac{T_k(x_i) - T_k(u_i)}{k^2(x_i - u_i)} \prod_{j < i} T_{\beta_j}(x_j) \prod_{j > i} T_{\beta_j}(u_j).$$

The quotient is its polynomial divided-difference extension, of degree $k - 1$. The bound $|T'_k| \leq k^2$ shows its normalized modulus is at most one on the interval. All factors of R_ℓ are bounded in modulus by one, and $\deg R_\ell \leq |\beta| - 1$. For each univariate polynomial factor a , the nonnegative polynomial $1 - a^2$ has an interval certificate through degree $2 \deg a$ by [Lemma 2.11](#). Constant factors contribute a nonnegative constant. The identity $1 - \prod_j a_j^2 = \sum_j (1 - a_j^2) \prod_{k < j} a_k^2$ then gives an ordinary-module certificate for $1 - R_\ell^2$ through degree $2(|\beta| - 1)$: all multipliers are squares and each term has at most one box generator. Telescoping the original tensor terms gives $g(x) - g(u) = \sum_\ell w_\ell t_\ell R_\ell(x)$ and $\sum_\ell w_\ell = A(g)$. With $A = A(g)$ the exact identity

$$\begin{aligned} A \sum_\ell w_\ell t_\ell^2 - \left(\sum_\ell w_\ell t_\ell R_\ell \right)^2 &= A \sum_\ell w_\ell t_\ell^2 (1 - R_\ell^2) \\ &\quad + \sum_{\ell < k} w_\ell w_k (t_\ell R_\ell - t_k R_k)^2 \end{aligned}$$

proves the claim. Multiplying each complement certificate by t_ℓ^2 raises its degree to at most $2|\beta| \leq 2 \deg g$; the remaining squares satisfy the same degree bound. $\square$

Lemma 8.7 (Squared displacement of the SOS kernel). *Let $j_s(x) = \int (x - u)^2 \Psi(x, u) d\mu(u)$. It is globally SOS, has degree at most $D + 2$, and $c_s - j_s \in \mathcal{M}_{(D+2)/2}(\{x\})$. For any ordinary-module-positive L_b of order $r \geq v_b D + 2$ and mass one,*

$$0 \leq L_b\left(j_s(x_i) \prod_{\ell \in B_b \setminus \{i\}} \omega(x_\ell)\right) \leq c_s.$$

Proof. Put $J_s(x) = \int (x - u)^2 \Phi_s(x, u)^2 d\mu(u)$, so $j_s = p_{s,N} J_s$. Integrating polynomial squares yields an SOS polynomial: its coefficient matrix is the integral of positive semidefinite rank-one matrices. Thus J_s and j_s are SOS, with degrees at most $2s$ and $D + 2$ respectively. To bound J_s , let $F_s(t) = \sum_{j \in \mathbb{Z}} b_j e^{ijt}$ with $b_j = (1 - |j|/s)_+$. For $x = \cos \theta$, $u = \cos \phi$, $\Phi_s(x, u) = [F_s(\theta - \phi) + F_s(\theta + \phi)]/2$. Convexity of the square and reflection of the ϕ integral give

$$\begin{aligned} J_s(\cos \theta) &\leq \int_{-\pi}^{\pi} [\cos \theta - \cos(\theta - t)]^2 F_s(t)^2 \frac{dt}{2\pi} \\ &\leq \int_{-\pi}^{\pi} 2(1 - \cos t) F_s(t)^2 \frac{dt}{2\pi} = \sum_{j \in \mathbb{Z}} (b_j - b_{j-1})^2 = \frac{2}{s}. \end{aligned}$$

The cosine difference bound follows from $|\cos \theta - \cos(\theta - t)| \leq 2|\sin(t/2)|$; the integral identity is Parseval's identity applied to $(1 - e^{it})F_s(t)$. Exactly $2s$ successive differences have modulus $1/s$. Since $0 \leq p_{s,N} \leq 2/C_s$ on the interval, $0 \leq j_s \leq c_s$. Its degree cap $D + 2$ is even, so [Lemma 2.11](#) certifies $c_s - j_s$ with that cap.

Each ω is globally SOS, and $1 - \omega = \varrho$ is interval nonnegative with a degree- D interval certificate. Telescoping $1 - \prod_{\ell \neq i} \omega(x_\ell)$, with SOS multipliers consisting of preceding ω factors, gives an ordinary-module certificate. Hence

$$c_s - j_s(x_i) \prod_{\ell \neq i} \omega(x_\ell) = c_s(1 - \prod_{\ell \neq i} \omega(x_\ell)) + (c_s - j_s(x_i)) \prod_{\ell \neq i} \omega(x_\ell)$$

is certified through degree $v_b D + 2 \leq r \leq 2r$. The product subtracted from c_s is SOS at the same degree, proving both inequalities. $\square$

Theorem 8.8 (Constrained ordinary-module rounding). *Assume [Assumption 8.1](#), $s \geq \max\{2, d_\infty\}$, even $N \geq 2$, and $r \geq wD + 2d$. Every feasible family for (8.6) admits a probability law ν on K with*

$$\int f d\nu \leq \sum_b L_b(f_b) + E + L_f H U^{\alpha/2}. \quad (8.9)$$

Consequently $0 \leq f^ - \rho_r^{c, \text{mod}} \leq E + L_f H U^{\alpha/2}$. For fixed data and width, parameters can be chosen for all sufficiently large r so that*

$$E = O(\log^3 r / r^2), \quad U = O(\log^2 r / r^2), \quad f^* - \rho_r^{c, \text{mod}} = O((\log r / r)^\alpha).$$

Proof. Let $\Psi_b(x, u) = \prod_{i \in B_b} \Psi(x_i, u_i)$ and $h_b(u) = L_b(\Psi_b(x, u)) \geq 0$. Since Ψ_b is SOS of degree at most $v_b D$, its conditional functional is positive on $(a + cg)^2$ and nonnegative at g , for $g = g_{bj}$. Indeed the required degrees are at most $v_b D + 2d \leq r$ and $v_b D + d \leq 2r$. The proof of (8.5) therefore applies to Ψ_b . Multiply the certificate in [Lemma 8.6](#) by the SOS Ψ_b and apply L_b for each fixed u . Its degree is at most $v_b D + 2d \leq r$. Integrating the resulting scalar inequality and using [Lemma 8.7](#) gives

$$\int (-g(u))_+^2 h_b(u) d\mu^{B_b}(u) \leq A(g) \sum_i A_i(g) L_b\left(j_s(x_i) \prod_{\ell \neq i} \omega(x_\ell)\right) \leq c_s A(g)^2. \quad (8.10)$$

No measurable choice of SOS certificates is required: certificates justify a scalar inequality at each fixed output before integration.

The source density has mass $Z_b = L_b(\prod_i \omega(x_i))$ with $a_0^{v_b} \leq Z_b \leq 1$. For completeness, the upper bound follows from the telescope in [Lemma 8.7](#). For the lower bound telescope $\prod_i \omega(x_i) - a_0^{v_b}$ into terms $(\omega(x_i) - a_0)$ times preceding SOS mass factors and nonnegative constant powers of a_0 . Each $\omega - a_0 \geq 0$ has an interval certificate through degree D . Both telescopes have degree at most $v_b D$, within the available order.

By [Lemma 4.5](#), the expansion of the signed density $\bar{h}_b = L_b(\prod_i \bar{\Psi}(x_i, u_i))$ has zero-residual term h_b , nonnegative singleton-residual terms, and higher terms whose sum is at least $-\Delta_w$. Therefore its common correction satisfies

$$h_b^+ = \frac{\bar{h}_b + \Delta_w}{1 + \Delta_w} \geq \frac{h_b}{1 + \Delta_w}.$$

The nonnegative remainder $e_b = h_b^+ - h_b/(1 + \Delta_w)$ has mass $(1 + \Delta_w - Z_b)/(1 + \Delta_w)$. Since $(-g)_+^2 \leq \|g\|_{\text{ch}}^2$, [\(8.10\)](#) gives

$$\int (-g)_+^2 h_b^+ d\mu^{B_b} \leq \frac{c_s \mathbf{A}(g)^2 + \|g\|_{\text{ch}}^2 (1 + \Delta_w - a_0^{v_b})}{1 + \Delta_w}.$$

The corrected bag laws have exactly consistent separators and glue to a single box law σ by [Theorem 4.6](#). That theorem gives objective error at most E , and summing the last display gives $\int V d\sigma \leq U$. Apply [Lemma 8.2](#), then take the hierarchy infimum.

To verify the rates, put $c_w = \max\{3, (w+1)/2\}$ and, for sufficiently large r , choose

$$s = \left\lfloor \frac{r - 2d}{2w(c_w + 3) \log_2(r + 2)} \right\rfloor, \quad N = 2 \left\lceil \frac{c_w}{2} \log_2 s \right\rceil.$$

Then $s \geq \max\{2, d_\infty\}$, $N \geq 2$ is even, and $wD \leq 2ws(N+1) \leq r - 2d$. These are the parameters of [Corollary 4.7](#) with the additive reserve. They give $\delta \leq \frac{1}{2}s^{-c_w}$, $\eta = O_{d_\infty}(\log s/s^2)$, and $\Delta_w = O_w((\log s)^{\max\{w-2, 0\}}/s^3) = o_w(s^{-2})$ (the correction is zero for $w = 1$). Thus $1 - a_0^{v_b} \leq v_b \delta$, $U = O(s^{-2})$, and $E = O(\log s/s^2)$. As $s = \Theta_w(r/\log r)$, the stated objective and squared-violation estimates follow. For $0 < \alpha \leq 1$ the objective term is of smaller order than $(\log r/r)^\alpha$. $\square$

8.4 Certificates with positive slack

The constrained moment problems can fail strict feasibility, for example when K has empty interior. The full box blocks nevertheless imply equality of the moment value and the certificate supremum.

Let $\mathcal{C}_r^{\text{c,mod}}$ be the sum over bags of the cones of polynomials

$$\sigma_{b0} + \sum_{i \in B_b} (1 - x_i^2) \sigma_{bi} + \sum_j g_{bj} \sigma_{bj}^g,$$

where each σ is SOS in x_{B_b} and each summand has degree at most $2r$. Define $\mathcal{C}_r^{\text{c,pre}}$ in the same way, using the summands $q_I \sigma_{bI}$ and $g_{bj} q_I \sigma_{bjI}$ for all $I \subseteq B_b$. All Gram bases are full; blocks with no allowed monomials are omitted. These cones correspond exactly to [\(8.6\)](#) and [\(8.3\)](#).

Proposition 8.9 (Constrained certificate supremum). *Suppose $K \neq \emptyset$, $r \geq 1$, and $\deg f_b \leq 2r$ for every bag. For $\mathcal{C}_r = \mathcal{C}_r^{\text{c,mod}}$ or $\mathcal{C}_r^{\text{c,pre}}$, let ρ_r be its corresponding moment value. The moment minimum is attained, and*

$$\sup\{\lambda : f - \lambda \in \mathcal{C}_r\} = \rho_r, \quad f - \lambda \in \mathcal{C}_r \quad \text{for every } \lambda < \rho_r.$$

No strict feasibility assumption is needed. The proposition does not assert membership at $\lambda = \rho_r$.

Proof. Work in the finite-dimensional coefficient space $W_r = \sum_b \mathbb{R}[x_{B_b}]_{\leq 2r}$. The ordinary sparse box cone is contained in either $\mathcal{C}_r$. Every bag monomial x^β of degree at most $2r$ splits as $x^\gamma x^\zeta$, with $|\gamma|, |\zeta| \leq r$. The identity

$$1 \pm x^\beta = \frac{1}{2}[(1 - x^{2\gamma}) + (1 - x^{2\zeta}) + (x^\gamma \pm x^\zeta)^2]$$

belongs to the ordinary box module at order r : each complement telescopes into square multiples of individual box generators through degree $2r$. Thus 1 plus any coefficient perturbation whose absolute coefficients sum to less than one is in $\mathcal{C}_r$, by taking nonnegative combinations of 1 and these identities. It follows that $1 \in \text{int}_{W_r} \mathcal{C}_r$.

By Lemma 2.2, normalized linear functionals on W_r that are nonnegative on $\mathcal{C}_r$ correspond exactly to the local moment families: separator equalities identify the different bag representations of a polynomial. Evaluation at any point of K makes this set nonempty. It is closed and bounded by the box moment bounds in Lemma 2.9, so its objective minimum ρ_r is attained.

Fix $\lambda < \rho_r$ and suppose $f - \lambda \notin \mathcal{C}_r$. Finite-dimensional separation from the interior of the convex cone gives a nonzero linear functional F with $F(q) \geq 0$ for every $q \in \mathcal{C}_r$ and $F(f - \lambda) \leq 0$. This separation does not require closedness of the cone. Since 1 is an interior point, $F(1) > 0$. Normalizing F therefore gives a feasible moment functional with objective at most λ , contradicting its strict inequality to ρ_r . Hence every strict level is certified. Weak duality gives the reverse inequality between the certificate supremum and ρ_r . $\square$

In particular, let E_r denote the last two terms of (8.4), or $E + L_f H U^{\alpha/2}$ in (8.9). The corresponding theorem and Proposition 8.9 give

$$f - f^* + E_r + \varepsilon \in \mathcal{C}_r \quad \text{for every } \varepsilon > 0.$$

Thus the same rates apply to real certificates with arbitrarily small positive slack, including problems with equality constraints or empty interior. Boundary attainment remains a separate question.

The constants retain the sums of generator budgets, H , and L_f . The exact-density proof introduces no additional tree disagreement term. Feasible repair is an existence result and can involve a global nonconvex projection problem. In the linear-error-bound class, the actual-measure example of Theorem 6.17 gives an inverse-order sparse gap even with stronger local positivity; the power one is therefore sharp for this class. It does not prove that the ordinary-module logarithm is necessary.

Remark 8.10 (Prior methods and the geometric assumption). Moment approximation followed by projection onto a feasible set is an established method. Tran and Toh [36] give dense preordering bounds with the same global Hölder exponent. Heijmans-Kuryatnikova, Vera and Zuluaga [13] transfer box certificates to general semialgebraic sets by a polynomial lift. For a dense formulation of the present global error-bound setting, after a fixed normalization of the generators, combining their lift with the dense ordinary box rate gives a bound $O((\log^{3/2} r/r)^\alpha)$ for fixed dimension and polynomial data; the direct displacement estimate above improves this particular logarithmic composition. This comparison concerns a dense certificate cone. These comparisons do not establish priority for the refinement. Korda, Magron and Ríos-Zertuche [17] use local geometric assumptions in their sparse general-domain theorem. Example 8.4 shows that local error bounds need not preserve the global exponent, so these rates do not imply uniform improvement for all such domains.

8.5 Finite-state labels

Now let each bag have continuous variables X_b and finite-state variables Z_b , so $B_b = X_b \sqcup Z_b$. Every finite-state variable has an explicitly listed finite domain, and running intersection holds for every

variable. Let $\mathcal{A}_b$ be the permitted assignments to Z_b , and assume

$$\mathcal{A} = \{a : a|_{Z_b} \in \mathcal{A}_b \text{ for every } b\} \neq \emptyset.$$

All labels have the same continuous box. There are no additional continuous constraints in this model. Its objective is $f(a, x) = \sum_b f_{b,a|Z_b}(x_{X_b})$, with polynomial local costs, and f^* is its minimum over $\mathcal{A} \times [-1, 1]^n$. In this subsection $v_b = |X_b|$ and $w = \max_b v_b$ are continuous sizes; empty X_b and $w = 0$ are allowed. This explicitly overrides the standing $w \geq 1, r \geq 1$ convention of [Section 2](#) when all variables are finite-state. An empty product kernel is one, and $\mu^\emptyset$ is the unit measure on the one-point empty product.

For each $a \in \mathcal{A}_b$, introduce $L_{b,a} : \mathbb{R}[x_{X_b}]_{\leq 2r} \rightarrow \mathbb{R}$, positive on either $P_r(X_b)$ or $M_r(X_b)$, with $\tau_{b,a} = L_{b,a}(1) \geq 0$ and $\sum_a \tau_{b,a} = 1$. For an edge $e = bc$ put $S_e^x = X_b \cap X_c$ and $S_e^z = Z_b \cap Z_c$. For every assignment s to S_e^z and every $p \in \mathbb{R}[x_{S_e^x}]_{\leq 2r}$ require

$$\sum_{a \in \mathcal{A}_b : a|_{S_e^z} = s} L_{b,a}(p) = \sum_{a \in \mathcal{A}_c : a|_{S_e^z} = s} L_{c,a}(p). \quad (8.11)$$

An empty fiber has sum zero; an empty discrete separator has one empty assignment. Equalities must include assignments whose fiber is absent on one side. Equating individual full-bag labels would impose the wrong condition. Denote the two moment values by $\rho_r^{\text{mix,pre}}$ and $\rho_r^{\text{mix,mod}}$.

Lemma 8.11 (Mass-scaled moment control). *If L is ordinary-box-module-positive through degree $2r$ with mass $\tau \geq 0$, then $|L(T_\beta)| \leq \tau$ for $|\beta| \leq r$. If $\tau = 0$, the entire functional through degree $2r$ vanishes. Both conclusions also hold under preordering positivity.*

Proof. The proof of [Lemma 2.8](#) is homogeneous: a certificate for $1 - T_\beta^2$ through degree $2|\beta|$ gives $0 \leq L(T_\beta^2) \leq \tau$, and the PSD form on $1, T_\beta$ gives $|L(T_\beta)|^2 \leq \tau L(T_\beta^2) \leq \tau^2$. If $\tau = 0$, all these diagonal values are zero. The PSD moment form on polynomials of degree at most r has zero diagonal in its Chebyshev basis, so every entry is zero by Cauchy–Schwarz. It follows that $L(pq) = 0$ for all $\deg p, \deg q \leq r$. Every monomial through degree $2r$ is such a product: split its exponent into two exponents of total degree at most r . Thus all moments vanish. For an empty continuous bag the only moment is the mass, so this conclusion also applies at order zero. $\square$

Set

$$A_{\text{mix}} = \sum_b \max_{a \in \mathcal{A}_b} A(f_{b,a}), \quad C_{b,a} = \|f_{b,a} - c_{b,a,0}\|_{\text{ch}}, \quad C_{\text{mix}} = \sum_b \max_{a \in \mathcal{A}_b} C_{b,a},$$

where $c_{b,a,0}$ is the constant Chebyshev coefficient. Let d and d_∞ bound the total and coordinate degrees of these costs, taking both zero for constant costs. Set $\Gamma_0 = \Delta_0 = 0$.

Theorem 8.12 (Finite-state preordering rounding). *For $w \geq 1$ and $r \geq \max\{w, d\}$, put $m = \lceil r/w \rceil + 1$. Every feasible labelled preordering family has a probability law ν supported on $\mathcal{A} \times [-1, 1]^n$ such that*

$$\left| \int f d\nu - \sum_{b,a} L_{b,a}(f_{b,a}) \right| \leq \frac{3}{D_m} \sum_{b,a} \tau_{b,a} A(f_{b,a}) \leq \frac{3A_{\text{mix}}}{D_m}.$$

Hence $0 \leq f^* - \rho_r^{\text{mix,pre}} \leq 3A_{\text{mix}}/D_m$. Empty continuous bags are permitted. If $w = 0$, the hierarchy is exact at order zero; this case uses neither m nor a division by w .

Proof. For each label define $h_{b,a}(u) = L_{b,a}(K_{m,X_b}(x, u))$. The kernel's degree is at most $2v_b(m-1) \leq 2r$, and its preordering certificate gives $h_{b,a} \geq 0$. Normalization of the kernel gives $\int h_{b,a} d\mu^{X_b} = \tau_{b,a}$.

After continuous marginalization and summation over the labels extending s , the separator density is

$$\sum_{a: a|_{S_e^z}=s} L_{b,a}(K_{m,S_e^x}(x,u)). \quad (8.12)$$

It agrees at b, c by (8.11). Thus these are consistent mixed bag probability laws. Apply Lemma 2.12 on the products of finite spaces and intervals. Equivalently, use the explicit conditional-density construction of Remark 3.5, summing over new finite labels and integrating over new continuous coordinates. At zero separator density any conditional law can be chosen, since the event is null. Every bag law assigns zero probability to forbidden local labels; a finite union of these null events shows that the global law has finite-state support in $\mathcal{A}$.

The Chebyshev multiplier calculation of Theorem 3.4, with Lemma 8.11, gives error at most $3\tau_{b,a}\mathbf{A}(f_{b,a})/D_m$ for each label. Sum and use bag mass normalization. Constants depending on labels are preserved exactly. Empty continuous bags have density $\tau_{b,a}$ and zero error. If $w = 0$, all functionals are scalar masses, (8.11) gives consistent finite bag probabilities already at order zero, and their exact gluing preserves the whole objective. Feasible Dirac families give weak duality and therefore exactness. $\square$

Theorem 8.13 (Finite-state ordinary-module rounding). *Suppose $w \geq 1$, $s \geq \max\{2, d_\infty\}$, even $N \geq 2$, and $r \geq wD$. Every feasible labelled ordinary-module family admits a law ν supported on $\mathcal{A} \times [-1, 1]^n$ whose local labelled densities are*

$$h_{b,a}^+(u) = \frac{\bar{h}_{b,a}(u) + \Delta_w \tau_{b,a}}{1 + \Delta_w}, \quad \bar{h}_{b,a} = L_{b,a}(\bar{\Psi}_{X_b}(x, u)), \quad (8.13)$$

and whose objective satisfies

$$\left| \int f d\nu - \sum_{b,a} L_{b,a}(f_{b,a}) \right| \leq \sum_{b,a} \tau_{b,a} [C_{b,a} \Gamma_{v_b} + \Delta_w \text{osc}(f_{b,a})] \quad (8.14)$$

$$\leq C_{\text{mix}}(\Gamma_w + 2\Delta_w). \quad (8.15)$$

The data-only bound (8.15) bounds $0 \leq f^* - \rho_r^{\text{mix,mod}}$. For fixed w, d_∞ it gives $C_{\text{mix}} O_{w,d_\infty}(\log^3 r/r^2)$ using the parameters of Corollary 4.7. Zero-mass labels and empty continuous bags require no normalization. If $w = 0$, the ordinary labelled hierarchy is exact at order zero, with the same conventions as Theorem 8.12.

Proof. The proof of Lemma 4.4 works with mass $\tau_{b,a}$: the signed density has this mass, and marginalization deletes the removed kernel factors exactly. Its coefficient error is at most $\tau_{b,a} C_{b,a} \Gamma_{v_b}$ by Lemma 8.11, since the transformed and original polynomials have degree at most $v_b D \leq r$. Indeed $\deg f_{b,a} \leq v_b d_\infty \leq v_b s \leq v_b D$ for nonempty bags; empty bags have constant costs.

In the residual expansion, for a subset of size j put $G_J = \prod_{i \notin J} \Psi(x_i, u_i)$ and $H_J = \prod_{i \in J} \varrho(x_i)$. The certificate and weighted Cauchy–Schwarz proof of Lemma 4.5 gives

$$|L_{b,a}(G_J H_J)| \leq \delta^j L_{b,a}(G_J) \leq \tau_{b,a} \delta^j \Lambda^{v_b-j}.$$

This proof never divides by the mass: the residual-square certificates have degree at most $(v_b + j)D \leq 2r$, and the coefficient bound for G_J uses degree $(v_b - j)D \leq r$. The terms with $j = 0, 1$ are nonnegative. Hence $\bar{h}_{b,a} \geq -\tau_{b,a} \Delta_{v_b} \geq -\tau_{b,a} \Delta_w$. The correction (8.13) is nonnegative and retains mass $\tau_{b,a}$. A zero-mass functional vanishes by Lemma 8.11; an empty bag has $\bar{h}_{b,a} = h_{b,a}^+ = \tau_{b,a}$.

For a separator label s , its corrected continuous density is exactly

$$\frac{\sum_{a: a|_{S_e^z}=s} L_{b,a}(\bar{\Psi}_{S_e^z}(x, u)) + \Delta_w \sum_{a: a|_{S_e^z}=s} \tau_{b,a}}{1 + \Delta_w}.$$

Both sums agree at the endpoints by (8.11), the second using $p = 1$. Absent fibers remain zero. Thus the mixed laws glue exactly as in Theorem 8.12, including null separators and permitted support. Adding a bare constant to each label would generally alter separator probabilities when the fibers have different sizes; the mass factor in (8.13) is essential.

Let $I_{b,a}^+ = \int f_{b,a} h_{b,a}^+ d\mu^{X_b}$ and define $\bar{I}_{b,a}$ with $\bar{h}_{b,a}$. Rearranging the correction gives

$$I_{b,a}^+ - \bar{I}_{b,a} = \Delta_w (\tau_{b,a} \int f_{b,a} d\mu^{X_b} - I_{b,a}^+).$$

The two terms in parentheses integrate $f_{b,a}$ against nonnegative measures of the same mass, so their difference is bounded by $\tau_{b,a} \text{osc}(f_{b,a})$. Combine this with the coefficient error and sum to obtain (8.14). Since $\sum_a \tau_{b,a} = 1$, $\text{osc}(f_{b,a}) \leq 2C_{b,a}$, and $\Gamma_{v_b} \leq \Gamma_w$, the data-only bound follows. It is uniform over feasible families, so it can be used before taking the infimum. The parameter proof of Corollary 4.7 applies unchanged. For $w = 0$ there is no kernel parameter choice; the scalar-mass gluing proof gives exactness at order zero directly. $\square$

If $C_{\text{mix}} = 0$, all local costs depend only on their finite labels, and the hierarchy is exact whenever their constant moments are defined. There is no extra label-count factor in these errors; explicit label lists and their SDP blocks can nevertheless be large. The results require a common continuous box for every label. They do not cover label-dependent boxes or additional label-dependent continuous constraints.

Proposition 8.14 (Real certificates after label pruning). *Delete every local label that has no extension in $\mathcal{A}$. This preserves the labelled primal values and feasible objectives. For the remaining finite SDP, at any order containing the objective, both labelled box hierarchies have an attained primal value and an attained dual of the same value. The certificate is an identity of labelled local polynomials on the permitted assignment space.*

Proof. The scalar masses of any feasible family satisfy consistent discrete bag probabilities, so finite tree gluing gives a law on $\mathcal{A}$. A label without a global extension therefore has zero mass. Its whole functional vanishes by Lemma 8.11; deleting its blocks and zero terms in the equalities preserves feasibility and objectives, in both directions.

Choose a distribution positive on every element of the finite nonempty set $\mathcal{A}$, independently of continuous product uniform measure. Every retained label has positive mass. A nonzero local polynomial square, weighted by any retained box-generator product, has strictly positive integral on the open box. Thus every retained labelled PSD block is positive definite; scalar blocks of empty continuous bags are positive as well. All separator and normalization equalities hold. This is primal Slater in the pruned finite SDP. Feasible moments are bounded by their masses: for monomials of degree at most r , a telescope for $1 - x^{2\beta}$ gives $0 \leq L_{b,a}(x^{2\beta}) \leq \tau_{b,a}$; splitting any exponent through degree $2r$ and applying Cauchy–Schwarz bounds its moment by $\tau_{b,a} \leq 1$. Consequently the feasible set is closed and bounded, so the primal optimum is finite and attained. Finite SDP Slater duality, as used in Lemma 2.10, gives dual attainment and equality.

To describe the identity, orient every edge and write $\epsilon_{b,e} = +1$ at its first endpoint, -1 at its second. Dual multipliers are scalars θ_b for normalization and separator polynomials $\phi_{e,s}(x_{S_e^z})$ of

degree at most $2r$ for each separator label s . Write ρ_r^{mix} for the value of the chosen labelled hierarchy. At the optimum they satisfy, for every retained label,

$$f_{b,a} - \theta_b - \sum_{e \ni b} \epsilon_{b,e} \phi_{e,a|_{S_e^z}} \in \text{Pr}(X_b) \quad \text{or} \quad \text{Mr}(X_b), \quad \sum_b \theta_b = \rho_r^{\text{mix}}.$$

For each global permitted assignment the separator terms cancel when these identities are summed. This is the asserted labelled certificate; it is not an unlabelled polynomial identity in surrogate integer variables. No attainment for the unpruned SDP is needed. $\square$

Finite-state tree gluing and labelled formulations are established tools; our use of them is a quantitative consequence of the kernels and the mass-scaled common correction. Pruning justifies the real finite-SDP certificate statement. It supplies no rational bit-size bound for the labelled cone, and neither labelled theorem is a general MINLP runtime claim.

9 Exact rational certificates with objective slack

The box hierarchy bounds can supply exact certificates for rational data. The hierarchy bounds first establish real finite-order membership with a positive constant margin. An explicit rational interior direction then allows rounding and exact coefficient correction. We prove a polynomial bound on the encoding length of the resulting rational Gram matrices in the expanded SDP dimensions and rational input length, including the slack. This is a specialization of established strict-feasibility rational SOS recovery methods, rather than a new general rationalization method.

9.1 The finite cone and its encoding

Fix $r \geq 1$ and nonempty continuous bags. Empty bags can have their constant terms assigned to another bag. Let $\mathcal{C}_r$ be either $\sum_b \text{Pr}(B_b)$ or $\sum_b \text{Mr}(B_b)$. For the preordering take $\mathcal{I}_b = \{I \subseteq B_b : |I| \leq r\}$; for the ordinary module take $\mathcal{I}_b = \{\emptyset\} \cup \{\{i\} : i \in B_b\}$. For each retained block define

$$q_{b,I} = \prod_{i \in I} (1 - x_i^2), \quad v_{b,I} = (x^\beta : |\beta| \leq r - |I|), \quad s_{b,I} = \binom{v_b + r - |I|}{v_b}.$$

These are full monomial bases, including every allowed monomial. Let

$$D_{\mathcal{C}} = \sum_{b,I} s_{b,I}, \quad V_{\mathcal{C}} = \sum_{b,I} \frac{s_{b,I}(s_{b,I} + 1)}{2}, \quad S_{\mathcal{C}} = \max_{b,I} s_{b,I}.$$

Let $M_{\mathcal{C}}$ be the number of distinct global monomials supported in some bag and of total degree at most $2r$. Coefficients of coincident monomials from overlapping bags are collected in the same output row. For a tuple of symmetric matrices the coefficient map is

$$\mathcal{G}_{\mathcal{C}}(Q) = \sum_b \sum_{I \in \mathcal{I}_b} q_{b,I} v_{b,I}^T Q_{b,I} v_{b,I}. \quad (9.1)$$

We use its monomial coefficient vector when taking a coefficient norm. The independent coordinates are the $V_{\mathcal{C}}$ upper-triangular entries. Its matrix has integer entries of modulus at most two. Each column expands into at most 2^w monomials; for the ordinary module the bound is two. Matrices $Q_{b,I} \succeq 0$ represent exactly $\mathcal{C}_r$.

Theorem 9.1 (Short exact rational Gram certificates). *Let p have rational coefficients and let $\tau \in \mathbb{Q}_{>0}$. Suppose*

$$p - \tau \in \mathcal{C}_r \quad (9.2)$$

with real Gram matrices in the full bases above. Then there are rational symmetric matrices with

$$\mathcal{G}_{\mathcal{C}}(Q) = p, \quad Q_{b,I} \succeq \frac{\tau}{2D_{\mathcal{C}}} I \quad \text{for every retained block.} \quad (9.3)$$

Their total binary encoding length is polynomial in $V_{\mathcal{C}}, M_{\mathcal{C}}$, the rational input length (including τ and the bags), and $\log^+(1/\tau)$, where $\log^+ t = \max\{0, \log_2 t\}$. The coefficient identity and positive semidefiniteness of a supplied tuple can be checked by exact rational arithmetic in polynomial time in its expanded encoding length. Running intersection is unnecessary for this rationalization statement once (9.2) is supplied.

The real membership prerequisite is stronger than pointwise positivity at a prescribed order. The theorem gives a polynomial-size witness; rounding an unspecified real witness in its proof does not assert a polynomial-time method for locating that witness.

9.2 An interior certificate of one

Lemma 9.2 (A diagonal rational interior direction). *There is a diagonal rational tuple H in the chosen full bases such that*

$$\mathcal{G}_{\mathcal{C}}(H) = 1, \quad H_{b,I} \succeq D_{\mathcal{C}}^{-1} I, \quad \sum_{b,I} \text{tr } H_{b,I} \leq r + 1. \quad (9.4)$$

Its entries have polynomial encoding length in the expanded dimensions.

Proof. In one bag order the coordinates $1, \dots, v$ and a generator set $I = \{i_1, \dots, i_k\}$. For each diagonal position indexed by β , with $|\beta| + k \leq r$, the following complement identity holds:

$$\begin{aligned} 1 - x^{2\beta} q_I &= \sum_{i=1}^v \sum_{j=0}^{\beta_i-1} (x_1^{\beta_1} \cdots x_{i-1}^{\beta_{i-1}} x_i^j)^2 (1 - x_i^2) \\ &\quad + \sum_{a=1}^k (x^\beta x_{i_a})^2 \prod_{h < a} (1 - x_{i_h}^2). \end{aligned} \quad (9.5)$$

The first telescope is $1 - x^{2\beta}$, and the second is $x^{2\beta}(1 - q_I)$. The first sum has total degree at most $2|\beta|$. A term in the second uses monomial degree $|\beta| + 1$ and $a - 1$ box generators, so its total degree is $2(|\beta| + a) \leq 2r$. All terms therefore use retained preordering blocks. If I is empty or singleton, the first sum uses only singleton blocks and the second uses only an empty-generator block; hence the identity also stays within the ordinary module.

Write $1 = x^{2\beta} q_I + (1 - x^{2\beta} q_I)$ for every one of the $D_{\mathcal{C}}$ diagonal positions across all bags, sum the displayed identities, and divide by $D_{\mathcal{C}}$. The resulting tuple is rational and diagonal. Each position receives its own baseline $1/D_{\mathcal{C}}$; all other contributions are nonnegative. Each source identity has $1 + |\beta| + |I| \leq r + 1$ monomial-square terms counted with multiplicity, proving the trace bound. Before division the entries are nonnegative integers at most $D_{\mathcal{C}}(r + 1)$, which proves the encoding claim. $\square$

9.3 A rational bound on the feasible Grams

Let ℓ be expectation under the global product uniform probability measure on the box and put

$$W_{b,I} = \ell(q_{b,I} v_{b,I} v_{b,I}^T).$$

The same global measure is used for every bag, so overlaps create no additional multiplicity in a polynomial identity.

Lemma 9.3 (Explicit outer bound). *Write $S = S_{\mathcal{C}}$ and set*

$$q_0 = ((2r+3)!)^{2w}, \quad \kappa_0 = q_0^{-S} S^{-(S-1)},$$

with the last factor one for $S = 1$. Then $W_{b,I} \succeq \kappa_0 I$. For C_p the sum of the absolute monomial coefficients of p , every PSD tuple representing p satisfies

$$\sum_{b,I} \text{tr } Q_{b,I} \leq C_p / \kappa_0, \quad \|Q\|_F \leq C_p / \kappa_0. \quad (9.6)$$

Here $\|Q\|_F$ is the aggregate Frobenius norm. The rational number $R_{\text{out}} = 1 + C_p / \kappa_0$ has polynomial encoding length in the expanded dimensions and input length.

Proof. A nonzero polynomial cannot vanish almost everywhere on an open box, and $q_{b,I}$ is strictly positive there. Thus $W_{b,I}$ is positive definite. Its entries are products of univariate uniform integrals, which for even a are

$$\ell(x^a) = \frac{1}{a+1}, \quad \ell(x^a(1-x^2)) = \frac{2}{(a+1)(a+3)};$$

for odd a both are zero. All exponents here are at most $2r$, and q_0 clears all denominators. A block of size s consequently has positive determinant at least q_0^{-s} , since $q_0 W_{b,I}$ is an integer matrix with positive integer determinant. Each diagonal is at most one, so $\text{tr } W_{b,I} \leq s$ and every eigenvalue is at most s . Dividing its determinant by the product of the other eigenvalues gives $\lambda_{\min}(W_{b,I}) \geq q_0^{-s} s^{-(s-1)} \geq \kappa_0$. For PSD matrices, trace pairing with $W_{b,I} \succeq \kappa_0 I$ gives

$$\kappa_0 \sum_{b,I} \text{tr } Q_{b,I} \leq \sum_{b,I} \text{tr}(W_{b,I} Q_{b,I}) = \ell(p) \leq C_p.$$

The aggregate Frobenius norm of a PSD tuple is at most its total trace. Finally $\log_2(1/\kappa_0) = O(Swr \log(r+2) + S \log(S+1))$. A nonempty bag's empty-generator block contains $1, x_i, \dots, x_i^r$, so $r+1 \leq S$; the width is at most the explicit bag-list length. These observations prove polynomial bit growth in the stated expanded parameters. The bound can be numerically very conservative. $\square$

9.4 Exact correction and proof of the theorem

Proof of Theorem 9.1. To simplify the formulas write $D_* = D_{\mathcal{C}}$, $V_* = V_{\mathcal{C}}$ and $\mathcal{G} = \mathcal{G}_{\mathcal{C}}$. By (9.2) there is a PSD tuple G with $\mathcal{G}(G) = p - \tau$. Add the interior direction of Lemma 9.2: $Q^* = G + \tau H$ satisfies

$$\mathcal{G}(Q^*) = p, \quad Q_{b,I}^* \succeq \varepsilon_* I, \quad \varepsilon_* = \tau / D_*.$$

Lemma 9.3 bounds all its entries by R_{out} .

The coefficient map has a simple rational right inverse. For each global output monomial x^β , choose a containing bag and split $\beta = \alpha + \zeta$ with $|\alpha|, |\zeta| \leq r$. Such a split exists because $|\beta| \leq 2r$.

In its empty-generator Gram block put one in the diagonal position if $\alpha = \zeta$, or $1/2$ in the two symmetric off-diagonal positions otherwise. Call this tuple Z_β . Then $\mathcal{G}(Z_\beta) = x^\beta$ and $\|Z_\beta\|_F \leq 1$. Distinct monomials cannot share a matrix position, since a position has a unique exponent sum. Thus the choices give simultaneous pivots. Their linear extension Z satisfies

$$\mathcal{G}Z = \text{id}, \quad \|Z(c)\|_F \leq \|c\|_1. \quad (9.7)$$

This inverse acts on global collected coefficients, rather than correcting each bag separately.

Round each upper-triangular entry of Q^* to a multiple of $h = 2^{-B}$ with absolute error at most h , preserving symmetry, and call the tuple Q^0 . Counting coordinates and generator expansions gives

$$\|Q^0 - Q^*\|_F \leq 2V_*h, \quad \|\mathcal{G}(Q^0 - Q^*)\|_1 \leq 2^{w+1}V_*h.$$

Define the rational tuple

$$Q = Q^0 + Z(p - \mathcal{G}(Q^0)). \quad (9.8)$$

Its coefficient identity is exact by (9.7), and

$$\|Q - Q^*\|_F \leq (2 + 2^{w+1})V_*h \leq 2^{w+2}V_*h.$$

Choose an integer $B \geq 0$ so that $2^{-B} \leq \varepsilon_*/(2^{w+3}V_*)$. Then each block perturbation has spectral norm at most $\varepsilon_*/2$, so every $Q_{b,I} \succeq (\varepsilon_*/2)I$, which proves (9.3).

The initial rounded entries have numerator length $O(B + \log_2(R_{\text{out}} + 1))$. If L is the least common multiple of 2^B and all denominators of coefficients of p , then coefficients of $p - \mathcal{G}(Q^0)$ have denominators dividing L . The correction Z introduces at most a factor of two. Thus $2L$ clears all corrected Gram entries, including an off-diagonal pivot that halves an input coefficient whose denominator has a larger power of two than 2^B . Its encoding length is at most a constant plus B and the sum of the input denominator lengths. Coefficient formation uses polynomially many rational additions and multiplications by integers of modulus at most two in the expanded map. The correction numerators therefore have polynomial bit length. Since $B = O(w + \log(V_* + 1) + \log(D_* + 1) + \log^+(1/\tau))$, Lemma 9.3 proves the total witness-size claim.

A supplied tuple is checked by expanding (9.1) and comparing all rational coefficients, then testing each rational symmetric matrix for PSD by exact elimination. Positive pivots permit Schur complements; a negative diagonal disproves PSD, and a zero diagonal with a nonzero off-diagonal entry also disproves PSD. An all-zero residual block is PSD. The elimination arithmetic has polynomial bit growth by determinant bounds. Thus both identity matching and PSD testing take polynomial time in the expanded certificate encoding length. $\square$

Formula (9.8) also describes a practical conversion step for an approximate Gram tuple: round, enforce coefficients exactly, and check the resulting PSD matrices. For a quantitative sufficient condition, if an approximate tuple is within aggregate Frobenius distance e of a tuple Q^* with margin ε_* and exact identity, its rounded-and-corrected tuple differs from Q^* by at most

$$(1 + 2^{w+1}\sqrt{V_C})e + 2^{w+2}V_C h.$$

Indeed the upper-triangular entrywise error sum is at most $\sqrt{V_C}e$, and each column of the map has coefficient one-norm at most 2^{w+1} . If this displayed bound is at most $\varepsilon_*/2$, the remaining Gram margin is at least $\varepsilon_*/2$. An arbitrary numerical SDP output need not satisfy these conditions. An exact identity and a successful rational PSD check are a certificate regardless of how the tuple was obtained.

9.5 From the two hierarchy rates to rational witnesses

Corollary 9.4 (Rational box certificates at the quantitative orders). *Let every supplied local objective polynomial f_b have rational coefficients, and let $\lambda \in \mathbb{Q}$ and $\tau \in \mathbb{Q}_{>0}$. Choose either of the following cones and rational gap budgets:*

(a) For $\mathcal{C}_r = \sum_b P_r(B_b)$, assume $r \geq \max\{w, d\}$ and set

$$E_{\text{pre}} = \frac{3A}{2(\lfloor r/w \rfloor + 1)^2 + 1}.$$

(b) For $\mathcal{C}_r = \sum_b M_r(B_b)$, use $s \geq \max\{2, d_\infty\}$, even $N \geq 2$, and $r \geq wD$ as in [Theorem 4.6](#). Define the rational majorant

$$\eta_{\mathbb{Q}} = \frac{N+1}{C_s} \left(\frac{d_\infty^2}{s^2} + \frac{3d_\infty^2}{2s} \right) + 2(D+1)\delta, \quad E_{\text{mod}} = C_f((1 + \eta_{\mathbb{Q}})^w - 1 + 2\Delta_w).$$

If $f^* - \lambda \geq E_C + \tau$, then $f - \lambda$ has the exact rational Gram certificate of [Theorem 9.1](#) in that same cone. The budget is $O(r^{-2})$ in (a) and $C_f O_{w,d_\infty}(\log^3 r/r^2)$ in (b), for the stated fixed-data parameter choices. Their encoding length is polynomial in the expanded finite cone dimensions and rational input length, including the slack.

Proof. All Chebyshev coefficients and budgets are rational for rational monomial data. In (b), $\sqrt{2(D+1)} \leq 2(D+1)$ gives $\eta_{\mathbb{Q}} \geq \eta$. Monotonicity and [Theorem 4.6](#) show that E_{mod} is a valid rational gap bound. In either case [Lemma 2.10](#) gives real membership $f - \rho_r \in \mathcal{C}_r$ with ρ_r the corresponding moment value. The gap bound and promise imply $\rho_r \geq \lambda + \tau$. Adding this nonnegative constant to a square block gives $f - \lambda - \tau \in \mathcal{C}_r$, precisely the prerequisite of [Theorem 9.1](#).

For (a), the rate is [Theorem 3.4](#). For (b), the parameter choice of [Corollary 4.7](#) has $D = O_w(s \log s)$ and $\delta = O_w(s^{-3})$. The replacement term $2(D+1)\delta$ is therefore $O_w(\log s/s^2)$, the same order as the other terms in η . The correction estimates are unchanged, so the rational majorant retains $O_{w,d_\infty}(\log s/s^2) = O_{w,d_\infty}(\log^3 r/r^2)$. $\square$

If the unlabelled problem has no continuous variables, its rational constant lower-bound certificate can be written directly; no width or kernel formula is needed. Finite-state certificates of [Proposition 8.14](#) are a separate real statement. The rational theorem above covers box-only unlabelled cones with full bases. It does not extend automatically to extra polynomial generators, labelled cones, or reduced Gram bases, which need their own interior directions and bounds.

Peyrl and Parrilo [\[30\]](#) already give numerical-symbolic rational SOS recovery by rounding and projection under strict feasibility. Davis and Papp [\[6\]](#) give quantitative rational certificate bounds and rational algorithms for weighted SOS, including discussion of sparse structure. Our contribution here is to connect the proved sparse hierarchy budgets to real finite-order membership and to supply elementary box-specific interior, outer, and coefficient correction bounds. The ordinary-module restriction uses the same proof; it is not a separate rationalization method. The encoding bound is in expanded order-dependent Gram dimensions, including all retained generator products. It gives no polynomial bound in the original optimization input size and no guarantee about a particular numerical solver's accuracy or running time.

10 Discussion

10.1 Separator information and the rate

The upper bounds share a construction: a common kernel converts local truncated functionals into laws whose separator marginals agree exactly. For the preordering, positivity and normalization

hold directly. For the ordinary module, products of globally SOS kernels give positive densities but have a mass defect. We restore exact normalization at the level of signed densities and use one common correction to make them positive (Theorem 4.6). Private conditional means, recourse repair, and label-dependent corrections act inside bags while retaining this scalar separator consistency. Additional polynomial constraints require a further global feasible repair.

The examples show that finite separator information can obstruct convergence even when local positivity is exact. In the paired two-bag models of Theorems 5.4 and 6.17, the local minima are $-h$ and h , and the exact local-measure gap is $2E_{2r}(h)$. This is a lower bound for the finite SDP gap, rather than an identity for every sparse relaxation. For the quadratic example, the order-one SDP gap is $1/4$, while the local-measure gap is $3 - 2\sqrt{2}$; at order two they agree (Proposition 5.5). A Lipschitz derivative gives an inverse-square approximation upper bound, attained in order by our piecewise quadratic example. The function $|x|$ has approximation error of exact order $1/r$. Smoothness alone does not determine an exact rate: smoother functions and polynomials can be approximated faster.

An active recourse constraint can create a convex corner, but activation alone is not sufficient. For example, minimizing y^2 over $y \in [-1, 1]$, $y \geq x$, and $y \geq 0$, with $x \in [-1, 1]$, gives the value x_+^2 , whose derivative is continuous despite the active-set change at zero. In the inverse-order example, by contrast, the value $|x|$ has a corner and the projected optimal multiplier jumps. The conditions of Theorems 7.2 and 7.3 control the projected multipliers and thereby improve the recourse repair estimate; they are not an equivalence between constraint activation and nonsmoothness.

For the sharp inverse-order instance, merging bags or splitting the shared domain at zero gives a low-order exact full-preordering certificate (Remark 6.18). Adding the separator statistic $|x|$ to the exact local-measure model also removes its gap. These instance-specific remedies suggest a question for sparse optimization: how should a method choose between increasing polynomial degree, enlarging bags, and exchanging additional separator information?

10.2 Computational interpretation and scope

The exponent of r in the recourse hierarchy’s block sizes depends only on the number of shared variables; at fixed shared width, its SDP dimensions grow polynomially in the private dimension. This is a useful structural distinction from a joint total-degree hierarchy. The constants in the accuracy bounds still depend on objective coefficients, private dimension, fiber conditioning, and multiplier regularity. The ordinary box bound is uniform in bag count and tree shape after normalization by C_f ; its total SDP size grows with the number of bags. A full preordering has up to 2^w blocks per bag, whereas the ordinary module has $w + 1$ in a bag of width w .

Discretization is a relevant comparator. For box objectives and fixed private domains, Chebyshev grids and tree dynamic programming give inverse-square error bounds. With exact local QP values and general affine complete recourse, the corresponding bound has order $1/r$ (Corollary 3.7 and Remark 6.20). A grid optimum G_r with proved error ε_r brackets the true optimum as $G_r - \varepsilon_r \leq f^* \leq G_r$. Thus discretization can provide lower bounds as well as feasible solutions. Our results control the particular sparse SDP, every feasible moment point, and its algebraic certificate cone. The convergence bounds do not establish a generic numerical runtime or a practical advantage over grid optimization. Solving local QPs, constructing SDP tables, numerical conditioning, and certified extraction from inexact moment data require their own analysis.

The recourse theorems assume continuous private variables belonging to one bag, pointwise private convexity on the shared box, and fixed polytopes or affine complete recourse with a fixed private constraint matrix. The regularity theorem’s inverse-square case uses weighted Chebyshev summability or coordinatewise Lipschitz projected multipliers; its coordinatewise Hölder case has

rate $r^{-(1+\beta)}$ for $0 < \beta \leq 1$. The constrained extension requires a global distance bound, whose exponent may be smaller than those of separate local geometric bounds. Its projection argument proves existence of a feasible repaired law, rather than an efficient projection algorithm.

Certificate conclusions depend on the model. The finite box SDPs have attained real dual certificates. The private-degree-two hierarchy has the same moment and certificate supremum values, with certificates at every strict level below the moment optimum ([Proposition 6.9](#)); boundary attainment is not established. The polynomial-constraint hierarchies also have equality of moment and certificate supremum values, and the proved rates yield real certificates with any further positive slack ([Proposition 8.9](#)). This does not require strict feasibility or imply boundary attainment. Exact rational box certificates need positive slack beyond the finite-order error, and their encoding length is polynomial in the expanded SDP dimensions and rational input length, including the slack ([Section 9](#)). This size statement does not make those dimensions polynomial in the original polynomial input.

10.3 Open problems

1. *Logarithms for the ordinary module.* The general sparse upper bound is $C_f O(\log^3 r/r^2)$, while the quadratic example gives a lower bound of order r^{-2} . The necessity of these logarithms is unresolved by the present bounds.
2. *Exact values beyond order two.* In the quadratic example of [Section 5](#), the sparse SDP and exact local-measure values agree at order two. We conjecture agreement for every $r \geq 2$. Retained floating-point computations are consistent with this conjecture but do not prove it. A proof would need a certificate for the relevant best separator approximation with a controlled degree.
3. *Width dependence.* The displayed rates hold at fixed width. The ordinary-module correction has substantial width dependence; the correct dependence of the accuracy order on w remains to be determined.
4. *Ordinary modules with private recourse.* The recourse results use the shared box preordering. It remains to combine ordinary-module exact-consistency correction with conditional private means and recourse repair at comparable rates.
5. *Multivariate multiplier regularity.* The proved hypotheses are weighted Chebyshev summability or coordinatewise dependence. Whether multivariate Lipschitz regularity suffices, including piecewise-affine projected multipliers with oblique region boundaries, remains open here.
6. *Leading constants.* Bernstein's approximation constant for $|x|$ determines the leading local-measure gap in the inverse-order example. We do not determine whether the finite SDP gaps have that same leading constant.
7. *Boundary certificates.* The order-unit argument gives real recourse certificates with arbitrary strict slack. It does not settle membership in the certificate cone at the optimal moment level.

The sparse preordering inverse-square rate was stated in lecture slides [\[23, 25\]](#) before this work. Our contribution for that cone is the complete compatible-density proof and its explicit finite-order consequences. The ordinary-module transfer and the recourse rate classification are separate theorem-specific contributions, with the prior-work boundaries stated in the relevant sections.

A Failure without private quadratic bounds

We prove [Proposition 6.11](#), including the nonmembership and closedness facts needed for separation. This argument concerns the scalar shared-box preordering; it does not assert closedness of the sparse recourse certificate cone of [Proposition 6.9](#).

Proof of Proposition 6.11. The arithmetic–geometric mean inequality gives

$$x_1^4 x_2^2 + x_1^2 x_2^4 + x_3^6 \geq 3x_1^2 x_2^2 x_3^2,$$

so $q \geq 0$, the objective $q(x)y^2$ is convex in y , and its true minimum is zero, attained at $y = 0$. Fix $r \geq 3$ and put $C = P_r(\{1, 2, 3\})$ on the space $\mathbb{R}[x]_{\leq 2r}$, which contains q .

First, $q \notin C$. Suppose a representation $q = \sum_{I,l} w_I a_{I,l}^2$ existed. Since $w_I(0) = 1$ and $q(0) = 0$, every $a_{I,l}$ vanishes at the origin. Let k be the smallest order of vanishing among the nonzero $a_{I,l}$. The degree- $2k$ homogeneous part of the representation is the sum of the squares of their degree- k parts. This sum cannot vanish: a sum of real polynomial squares is zero only if each square is zero. As q is homogeneous of degree six, $k = 3$. Every $a_{I,l}$ thus has no term of degree less than three, and comparison of degree-six parts expresses q as a sum of squares of homogeneous cubics.

Here is a direct contradiction. In any such sum, the zero coefficients of x_1^6 and x_2^6 force the coefficients of x_1^3 and x_2^3 in every cubic to vanish. A remaining cubic has the form

$$ax_1^2 x_2 + bx_1 x_2^2 + cx_1^2 x_3 + dx_2^2 x_3 + ex_1 x_3^2 + fx_2 x_3^2 + gx_1 x_2 x_3 + hx_3^3.$$

The coefficient of $x_1^4 x_3^2$ in the sum of squares is now $\sum c^2$, and that of $x_2^4 x_3^2$ is $\sum d^2$. Both vanish in q , so $c = d = 0$ in every cubic. The coefficients of $x_1^2 x_3^4$ and $x_2^2 x_3^4$ then give $\sum e^2 = \sum f^2 = 0$. With these coefficients removed, the coefficient of $x_1^2 x_2^2 x_3^2$ is $\sum g^2 \geq 0$, whereas it is -3 in q . Thus no representation exists, at any order.

Next, C is closed. For each I with $|I| \leq r$, let v_I be the vector of shared monomials of degree at most $r - |I|$. A Gram representation of an element of C is

$$p(x) = \sum_I w_I(x) v_I(x)^\top G_I v_I(x), \quad G_I \succeq 0.$$

For uniform probability measure ω on the cube, the matrix $B_I = \int w_I v_I v_I^\top d\omega$ is positive definite: a nonzero polynomial square has positive integral against the weight w_I , which is strictly positive in the open cube. If $p_j \in C$ converges coefficientwise, its integrals are bounded and, for any chosen Gram representations,

$$\int p_j d\omega = \sum_I \text{tr}(G_{I,j} B_I) \geq \sum_I \lambda_{\min}(B_I) \text{tr}(G_{I,j}).$$

All summands are nonnegative. Hence all Gram matrices are bounded, and a subsequence of the finite collection converges to positive semidefinite matrices giving a representation of the limit. This proves closedness.

Finite-dimensional separation now gives a linear functional $\mathcal{L}$ on $\mathbb{R}[x]_{\leq 2r}$ such that $\mathcal{L}(C) \geq 0$ and $\mathcal{L}(q) < 0$. Its mass is positive. Indeed, the unnormalized version of the proof of Lemma 2.9 gives $|\mathcal{L}(x^\alpha)| \leq \mathcal{L}(1)$ for all $|\alpha| \leq 2r$: shared generator tests bound even moments by $\mathcal{L}(1)$, and Cauchy–Schwarz bounds the remaining moments. If $\mathcal{L}(1) = 0$, the functional vanishes on a monomial basis, contradicting $\mathcal{L}(q) < 0$. Scale it so that $\mathcal{L}(1) = 1$. This normalization uses only $r \geq 3$, so it does not require q^2 to lie in the truncated domain.

For $R \geq 0$ define on the rectangular space

$$L_R(a(x) + b(x)y + c(x)y^2) = a(0) + R\mathcal{L}(c).$$

It has $L_R(1) = 1$. For every allowed I and every shared coefficient pair s_0, s_1 with degree at most $r - |I|$,

$$L_R(w_I(s_0 + s_1 y)^2) = w_I(0)s_0(0)^2 + R\mathcal{L}(w_I s_1^2) \geq 0.$$

Thus (R1) holds. The two affine private localizers, with the fixed-domain degree allowance, give $L_R(w_I s^2(1 \pm y)) = w_I(0)s(0)^2 \geq 0$, so (R2) holds. There are no separator equations in a single bag, and (R3) has been removed. Hence L_R is feasible for every R , while $L_R(q(x)y^2) = R\mathcal{L}(q) \rightarrow -\infty$. $\square$

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